Postgraduate (Master's) Programme Design for the Faculty of Geoscience and Technology

Benchmark research on graduate programmes, proposed 7-semester / 150-credit structure, semester plans for six courses, and a catalogue of every subject with credits, syllabus, passing requirements, open textbooks and similar subjects at other universities

Draft for discussion | 13 November 2026 | companion to the undergraduate curriculum and syllabus reports

Summary

- The programme has 3 taught semesters and 4 thesis semesters, 150 credits in total: 20 credits of common subjects, at least 36 credits of electives (12 subjects of 3 credits) and 94 credits of research milestones, practice, publication, thesis and defence.
- The 5 common subjects are Academic English; Applied Mathematics (numerical analysis and matrix methods); Applied Statistics, Geostatistics and Data Science; Research Methodology, Scientific Writing and Ethics; and Numerical Modelling and Process Simulation.
- Each of the six courses (Mineral Exploration; Civil Engineering; Mineral and Rock Engineering; Geophysics; Petroleum Exploration and Engineering; Geographic Information and Space) has 13-15 electives arranged by semester. Shared subjects are described once.
- The thesis is assessed through 9 milestones: literature review, proposal defence, practice, academic activities, thesis research, mid-term evaluation, publication, pre-defence, and review plus final defence.
- The proposals draw on the published rules of Colorado School of Mines, MIT, Tsinghua University, China University of Geosciences, China University of Petroleum, TU Bergakademie Freiberg and the ECTS Users' Guide (Appendix B).

How to read Part 4. Every subject appears once, with a catalogue code (GC = common, GM = research milestone, GE = elective). The 'Used in' line shows which courses use it and whether it is recommended (R) or optional (O). Open-access textbooks are listed first with a link to the free PDF or online edition and a table of contents in Appendix A; commercial textbooks have no legal free PDF, so a WorldCat library link is given. Mines and MIT publish subject descriptions, which are summarised; Tsinghua, CUG, CUP and Freiberg are cited by course or module title and credits from their published plans.

Contents

Contents 2

Part 1. Research findings on postgraduate programmes 5

1.1 What subjects do similar postgraduate programmes teach? . . . 5

1.2 Which assessments should be used from literature review and topic selection to graduation? . . . 5

1.3 How should the credit of each subject be designed and evaluated? . . . 6

1.4 Criteria for passing subjects and completing the programme . . . 6

1.5 Mathematics subjects in postgraduate programmes . . . 7

1.6 Subjects similar to 'process simulation analysis' . . . 7

1.7 Methods for gaining credit . . . 7

1.8 How to evaluate graduate students' academic activities . . . 8

1.9 Foreign language proficiency criteria . . . 8

Part 2. Proposed programme structure and rules 9

2.1 Structure of the 7-semester, 150-credit programme . . . 9

2.2 Credit rule . . . 9

2.3 Elective rules . . . 9

2.4 Milestone timeline . . . 9

2.5 Pass and completion criteria . . . 10

2.6 Foreign language criteria (proposal) . . . 10

2.7 Academic activities and research milestones . . . 11

Part 3. Subjects by semester for each course 12

Mineral Exploration course . . . 12

Civil Engineering course . . . 13

Mineral and Rock Engineering course . . . 14

Geophysics course . . . 15

Petroleum Exploration and Engineering course . . . 16

Geographic Information and Space course . . . 17

Part 4. Subject catalogue (each subject described once) 18

GC01 Academic English for Geoscience and Engineering . . . 18

GC02 Applied Mathematics: Numerical Analysis and Matrix Methods . . . 19

GC03 Research Methodology, Scientific Writing and Research Ethics . . . 20

GC04 Applied Statistics, Geostatistics and Geoscience Data Science . . . 21

GC05 Numerical Modelling and Process Simulation . . . 22

GM01 Literature Review and Research Topic Selection . . . 23

GM02 Research Proposal and Proposal Defence . . . 24

GM03 Professional, Field or Teaching Practice . . . 24

GM04 Academic Activities (seminars, conferences, presentations) . . . 25

GM05 Thesis Research I, II and III . . . 25

GM06 Mid-term Evaluation . . . 26

GM07 Research Publication and Conference Presentation . . . 26

GM08 Thesis Writing, Similarity Check and Pre-defence . . . 27

GM09 Thesis Review and Final Defence . . . 27

GE01 Advanced Ore Deposit Geology and Mineral Systems . . . 28

GE02 Geochemical Exploration and Geochemical Data Analysis . . . 28

GE03 Mineral Exploration Geophysics . . . 29
GE04 Advanced Structural Geology and Structural Controls on Ore . . . 30
GE05 Advanced GIS and Spatial Analysis . . . 30
GE06 Ore-forming Fluids, Fluid Inclusions and Isotope Geochemistry . . . 31
GE07 Analytical Methods in Mineralogy and Geochemistry (SEM, EPMA, LA-ICP-MS, XRD) . . . 32
GE08 3D Geological Modelling . . . 32
GE09 Advanced Optical and Hyperspectral Remote Sensing . . . 33
GE10 Mineral Resource Estimation and Geostatistical Simulation . . . 34
GE11 Exploration Project Design and Mineral Economics . . . 34
GE12 Field and Ore Deposit Geology (field school) . . . 35
GE13 Mineral Prospectivity Modelling and Targeting . . . 35
GE14 Critical and Strategic Minerals . . . 36
GE15 Machine Learning and AI for Geoscience and Engineering . . . 37
GE16 Advanced Soil Mechanics and Constitutive Modelling . . . 38
GE17 Advanced Structural Analysis (matrix and nonlinear methods) . . . 38
GE18 Finite Element Method in Engineering and Geomechanics . . . 39
GE19 Structural Dynamics and Earthquake Engineering . . . 40
GE20 Advanced Foundation Engineering and Deep Excavations . . . 40
GE21 Geotechnical Site Investigation, In-situ Testing and Monitoring . . . 41
GE22 Advanced Design of Concrete, Steel and Composite Structures . . . 42
GE23 SAR, InSAR and Deformation Monitoring . . . 42
GE24 Underground Construction and Tunnelling . . . 43
GE25 Slope Stability, Landslides and Geohazard Risk . . . 44
GE26 Reliability and Risk Analysis in Geotechnical and Structural Engineering . . . 44
GE27 Construction Project Management and BIM . . . 45
GE28 Ground Improvement and Geosynthetics . . . 45
GE29 Earthquake Seismology and Seismic Hazard Analysis . . . 46
GE30 Advanced Rock Mechanics . . . 47
GE31 Mine Planning and Optimisation . . . 47
GE32 Numerical Methods in Geomechanics (FDM, BEM, DEM) . . . 48
GE33 Advanced Mineral Processing and Plant Simulation . . . 49
GE34 Rock Fragmentation and Blasting Engineering . . . 49
GE35 Underground Mine Design and Ground Control . . . 50
GE36 Mine Ventilation, Health and Safety . . . 50
GE37 Tailings and Mine Waste Engineering . . . 51
GE38 Geophysical Inverse Theory . . . 51
GE39 Advanced Seismology and Wave Propagation . . . 52
GE40 Advanced Gravity and Magnetic Methods . . . 53
GE41 Advanced Formation Evaluation and Borehole Geophysics . . . 53
GE42 Seismic Data Processing and Imaging . . . 54
GE43 Electrical and Electromagnetic Methods . . . 54
GE44 Computational Geophysics (numerical simulation of geophysical fields) . . . 55
GE45 Rock Physics and Petrophysics . . . 56
GE46 Seismic Interpretation and Reservoir Characterisation . . . 56
GE47 Engineering, Environmental and Hydrogeophysics . . . 57
GE48 Geophysical Field School and Instrumentation . . . 57

GE49 Petroleum Systems Analysis and Basin Modelling . . . 58
GE50 Advanced Reservoir Engineering . . . 59
GE51 Reservoir Simulation . . . 59
GE52 Advanced Drilling Engineering and Well Design . . . 60
GE53 Reservoir Geomechanics . . . 61
GE54 Enhanced Oil Recovery and CO2 Storage . . . 61
GE55 Advanced Production Engineering . . . 62
GE56 Well Test and Rate-Transient Analysis . . . 62
GE57 Petroleum Economics, Risk and Decision Analysis . . . 63
GE58 Unconventional Oil and Gas Resources . . . 63
GE59 Satellite Geodesy and GNSS Positioning . . . 64
GE60 Orbital Mechanics and Space Mission Analysis . . . 64
GE61 Spatial Databases and WebGIS Development . . . 65
GE62 Photogrammetry, LiDAR and UAV Mapping . . . 65
GE63 Spacecraft Systems Engineering and Small Satellites . . . 66
GE64 Digital Terrain Analysis and Geohazard Modelling . . . 67
GE65 Planetary Geology and Space Resources . . . 67
Appendix A. Open-access textbooks: links and tables of contents 69
Appendix B. Sources 73

Part 1. Research findings on postgraduate programmes

1.1 What subjects do similar postgraduate programmes teach?

Graduate geoscience and engineering programmes at the benchmark universities share the same three layers: (1) a small common core of language, mathematics/statistics and research skills; (2) advanced discipline subjects at '500 level' (Mines) or 'degree course' level (Chinese universities); and (3) a large share of computational subjects - inversion, numerical modelling, geostatistics and machine learning - that did not exist in programmes 15 years ago. The table lists representative graduate subjects by field; full descriptions are given under each subject in Part 4.

Field	Typical graduate subjects	Examples at benchmark universities
Mineral exploration	Advanced ore deposits and mineral systems; hydrothermal geochemistry and isotopes; ore microscopy and microanalysis; exploration geochemistry; mineral exploration geophysics; resource estimation; field ore-deposit school	Mines GEOL515, GEOL524, GEOL513, GEOL540, GEOL523, GEOL521, GPGN543, MNGN538
Civil / geotechnical	Advanced soil mechanics; advanced structural analysis; FEM; structural dynamics; site investigation; underground construction; risk and reliability	Mines CEEN505, 506, 510, 513, 519, 523, 530, 531, 546; MIT 1.361, 1.322
Mining and rock engineering	Advanced rock mechanics; numerical methods in geomechanics; surface and underground mine design; blasting; ventilation; mineral processing and plant simulation; geostatistics	Mines MNGN504-508, 512, 516, 518, 524-526, 532, 533, 536, 538, 545; GEGN563
Geophysics	Inverse theory; seismology; seismic processing and interpretation; gravity/magnetic and EM methods; rock physics; computational geophysics; hydrogeophysics	Mines GPGN509-511, 519, 520, 536, 547, 552/553, 555, 558, 561, 574; MIT 12.510; CUP Fundamentals of Geophysical Inversion, Seismic Simulation and Imaging
Petroleum	Petroleum systems and basin modelling; reservoir engineering and simulation; geomechanics; drilling and well control; production; EOR; economics	Mines GEOL550, 551, 565, 575; PEGN501, 506, 513, 515, 517-519, 523, 541, 550, 590, 591, 594, 596; CUP Reservoir Characterization, Digital Rock
Geographic information and space	Advanced GIS and spatial statistics; remote sensing and SAR/InSAR; GNSS and geodesy; terrain analysis; space systems engineering; orbital mechanics; planetary geology	Mines GEGN542, 575, 579, 580, 588; GPGN570; SPRS501-507; MIT 12.540, 16.851, 16.346
Common to all	Academic English and scientific communication; numerical analysis; statistics and geostatistics; research methods and ethics; machine learning	Tsinghua First Foreign Language; CUG Scientific English Writing and Communication, Literature Review; Freiberg Multivariate Statistics and Geostatistics (9 LP), Scientific Communication (4 LP); MIT 18.335J, 1.151, 6.867

1.2 Which assessments should be used from literature review and topic selection to graduation?

All benchmark universities break the thesis into a chain of assessed milestones, so that problems are found early rather than at the final defence. The Chinese universities give these milestones credits; Mines and MIT treat them as requirements without credits; Freiberg grades the thesis and the colloquium separately. The recommended chain for the faculty is shown in the last column (details in Part 2 and in the milestone entries GM01-GM09).

Stage	Benchmark practice	Recommended for the faculty
Supervisor and committee	Mines: thesis committee of at least 3 approved by the end of semester 2. MIT EAPS: advisor + 2 faculty by the 3rd term.	Supervisor in week 4 of Semester 1; advisory committee of 3 by the end of Semester 1.
Literature review	Tsinghua: Literature Review and Proposal (1 cr) by end of semester 2, panel of at least 3. CUG Wuhan: Literature Review degree course 3 cr / 48 h. CUGB: review of at least 50 papers (30 foreign) from the last 10 years.	Literature Review report (4 cr) in Semester 2: at least 60 sources, 30 international, half from the last 10 years; 6,000-8,000 words; presentation.
Research proposal	Mines: written proposal approved at least one semester before the defence. CUGB: proposal in semesters 3-4; a failed proposal may be redone once within 3 months.	Proposal and public defence (4 cr) in Semester 3, panel 3-5; one re-defence within 3 months.
Progress checks	Tsinghua: mid-term check by a panel of 3-5 that must be passed.	Semester progress reports in Thesis Research I-III; Mid-term Evaluation (4 cr) in Semester 5.
Publication	Common in Chinese universities as a degree condition; not required at Mines, MIT or Freiberg.	At least one journal manuscript submitted (6 cr) in Semester 6; acceptance recommended, not required.
Pre-defence and review	Mines: thesis to committee at least 1 week before; MIT: 2 weeks. Freiberg: two reviewers, a third if grades differ by more than 1.7.	Pre-defence and similarity check (Semester 7), then two reviewers (one external).

Stage	Benchmark practice	Recommended for the faculty
Final defence	Mines: public defence, two negative votes = fail, one re-examination at least 1 week later, second failure ends the programme. MIT: oral presentation of at least 30 min plus questions. Freiberg: colloquium of about 30 min; thesis weight 2, colloquium 1.	Public defence: 30 min talk + at least 30 min questions; grade = (2 x thesis + 1 x defence)/3; one second defence allowed.

1.3 How should the credit of each subject be designed and evaluated?

The European Credit Transfer and Accumulation System (ECTS Users' Guide 2015) gives the most explicit method, and it fits the faculty's 150-credit system well. Credits are based on the total student workload needed to achieve the learning outcomes - lectures, laboratories, self-study, projects and examinations - not on contact hours alone. 60 ECTS correspond to one full-time year of about 1,500-1,800 hours, so 1 ECTS is about 25-30 hours. Credits are awarded only when the learning outcomes have been assessed and achieved, and master's programmes normally carry 90-120 ECTS. Chinese universities count about 16 contact hours per credit, and US universities about 15 contact hours (one weekly class hour over a semester) per credit.

System	Unit	Workload per credit	Master's total
ECTS (Europe, e.g. Freiberg)	1 ECTS / LP	25-30 h total workload	90-120 ECTS (Freiberg MSc Applied Geoscience: 120 LP)
US semester credit (Mines, MIT units)	1 semester hour (MIT: 1 unit = 1 h/week)	About 15 contact h + 30 h out of class	Mines MS: 30-36 credits; MIT EAPS SM: at least 66 units excluding thesis
Chinese graduate credit (Tsinghua, CUG, CUP)	1 credit	About 16 contact h	Coursework 23-32 credits; thesis not counted in credits
Proposed for the faculty	1 credit	About 45 h total workload (15 lecture h, or 30 lab/computer h, or 1 full-time week of research/field)	150 credits over 7 semesters (about 6,300-6,800 h, comparable to 210 ECTS over 3.5 years)

Recommended procedure for designing and checking a subject's credit:

- Write 4-6 measurable learning outcomes for the subject (what the student can do after it).
- List every learning activity and estimate its hours: lectures, laboratories/computer sessions, field days, guided reading, assignments, project work, examination preparation and examinations.
- Credit = total workload hours / 45, rounded to a whole number (3-credit elective = about 135 h; 4-credit common subject = about 180 h).
- Check that the assessment covers all learning outcomes and that the weight of each assessment component matches the time spent on it.
- After the first offering, collect a student workload survey and the pass rate. If the measured workload differs from the credit by more than about 20%, adjust the content or the credit.
- Review all subjects every 3 years in the curriculum committee, using student feedback, examination results and employer or industry advice.

1.4 Criteria for passing subjects and completing the programme

University	Subject pass / grade rule	Completion requirements
Colorado School of Mines (MS thesis)	Cumulative GPA at least 3.0; transfer courses only with grade C or better	At least 30 credits (36 incl. 12 research for MS Geology/Geological Engineering), at least 6 research credits, at most 9 transfer credits; committee of at least 3; written proposal approved at least one semester before defence; public defence; time limit 5 years
MIT EAPS (SM)	GPA at least 3.5 expected; A or B in programme subjects	At least 66 subject units excluding thesis; thesis committee (advisor + 2) by the 3rd term; thesis to committee 2 weeks before defence; oral presentation at least 30 min; usually 2 years
Tsinghua University (academic master's)	Degree courses must be passed (100-point scale)	At least 23 credits: public courses at least 5 (Foreign Language 2), professional courses at least 15, literacy at least 1; literature review and proposal 1 cr; academic activities 1 cr; mid-term check passed; at least 1 year of thesis research
CUG Wuhan (academic / professional master's)	Degree courses must be passed	Academic: at least 25 credits (public at least 7 incl. Scientific English 4; degree courses at least 18 incl. Literature Review 3 and a cross-discipline elective 2); 2 academic presentations; proposal; 3 years standard, 5 maximum; thesis at least 30,000 characters. Professional: 32 credits incl. applied mathematics 3, engineering ethics and 6 credits of practice (at least half a year); thesis at least 20,000 characters
China University of Petroleum	Degree courses must be passed	25-32 credits incl. Numerical Analysis 3; academic activities (at least 10 seminars + 1 presentation, or 8 lectures + 1 report); 3 years
TU Bergakademie Freiberg (MSc)	German scale, 4.0 is the lowest pass	120 LP in 4 semesters; at least 80 LP before the thesis can start; thesis 30 LP in 6 months (+3 months extension); colloquium; one repeat of a failed thesis within 1 year

The proposed criteria for the faculty, which combine these practices, are given in Part 2.5.

1.5 Mathematics subjects in postgraduate programmes

Mathematics subject	Main content	Where it is taught
Numerical analysis / numerical methods	Linear and nonlinear systems, interpolation, quadrature, ODE and PDE solvers, error and stability	CUP Numerical Analysis (3 cr); CUG applied mathematics (3 cr); Mines CEEN505, PEGN501; MIT 18.335J; Freiberg Numerical Analysis of Differential Equations
Matrix theory and applied linear algebra	Decompositions (LU, QR, SVD), eigenproblems, least squares, sparse matrices	MIT 18.335J, 18.085; Chinese 'Matrix Theory' courses
Applied statistics, multivariate statistics and geostatistics	Estimation, regression, multivariate analysis, variograms, kriging, simulation	Freiberg Multivariate Statistics and Geostatistics (9 LP, compulsory); Mines GEGN532, GPGN533, MNGN538; MIT 1.151
Inverse theory and optimisation	Least squares, regularisation, Bayesian inversion, MCMC	Mines GPGN509, GPGN510; Freiberg Inverse Problems in Geophysics (6 LP); CUP Fundamentals of Geophysical Inversion
Partial differential equations and numerical simulation	FD, FV, FE and spectral methods for flow, heat, waves and mechanics	Mines GPGN536, CEEN506, GEGN583; Freiberg Numerical Simulation Methods in Geophysics, Numerical Simulation with FE, PDEs
Probability, uncertainty quantification and data assimilation	Monte Carlo, Bayesian methods, ensemble methods	Freiberg Uncertainty Quantification, Probabilistic Forecasting and Data Assimilation, Bayesian Methods with R
Mathematics of machine learning	Linear algebra, optimisation and probability behind ML	Freiberg Mathematics of Machine Learning (6 LP); MIT 6.867

In the proposed programme, numerical analysis and matrix methods (GC02) and statistics/geostatistics (GC03) are compulsory for all students; numerical PDE methods are compulsory through GC05; inverse theory, FEM, reliability and machine learning are electives.

1.6 Subjects similar to 'process simulation analysis'

Yes. Every benchmark programme has simulation subjects, but they are named after the process being simulated rather than 'process simulation'. The closest general subject is MIT 10.34 Numerical Methods Applied to Chemical Engineering (process modelling of reactors and separations). In geoscience and engineering the equivalents are:

Process simulated	Benchmark subjects	Subject in this proposal
General numerical modelling of physical processes	MIT 10.34, 2.29, 18.085; Freiberg Numerical Simulation Methods in Geophysics, Numerical Simulation with FE	GC05 Numerical Modelling and Process Simulation (common)
Groundwater flow and transport	Mines GEGN583 Mathematical Modeling of Groundwater Systems	GC05 (project option)
Rock and soil mechanics (FEM/FDM/DEM/BEM)	Mines GEGN563, MNGN525, MNGN526, CEEN506	Finite Element Method; Numerical Methods in Geomechanics
Mineral processing plants	Mines MNGN532, MNGN533 (plant simulation software)	Advanced Mineral Processing and Plant Simulation
Reservoir flow	Mines PEGN513, PEGN541, PEGN550	Reservoir Simulation
Basin and petroleum-system processes	Mines GEOL550 Integrated Basin Modeling, GEOL575	Petroleum Systems Analysis and Basin Modelling
Seismic and EM wave fields	Mines GPGN536; CUP Seismic Wave Simulation and Imaging	Computational Geophysics
Geodynamics	Freiberg Numerical Modelling in Geodynamics	GC05 (project option)

1.7 Methods for gaining credit

- **Taught subjects** - examination, laboratory reports, projects and presentations (all universities).
- **Research credits** - Mines awards graded research credits each semester (at least 6 in a thesis MS); Freiberg awards 30 LP for the thesis; this proposal awards 46 credits for Thesis Research I-III.
- **Research milestones** - literature review, proposal and academic activities carry credits at Tsinghua (1 + 1 cr) and CUG (Literature Review 3 cr).
- **Practice** - CUG professional master's: 6 credits for at least half a year of professional practice; field schools (Mines GEOL521) and teaching assistantships.
- **Seminars and academic activities** - Mines GPGN581 Graduate Seminar; CUP seminar and presentation requirements.
- **Independent study and projects** - for example Mines SPRS599 Independent Study (0.5-6 cr) and SPRS591/592 team and individual projects.

- **Transfer and recognition** - Mines accepts at most 9 transfer credits with grade C or better; ECTS recommends recognising prior formal, non-formal and informal learning when the learning outcomes are equivalent.
- **Exemption by proficiency** - Tsinghua grants English course exemption for test scores or other evidence (Part 1.9).
- **Publications and conferences** - counted as a credit element in many Chinese programmes and in this proposal (GM07, 6 cr).
- **Online and summer courses** - MOOCs or summer schools from recognised universities can be recognised as electives if the workload and assessment meet the subject's requirements (recommended limit: 6 credits).

1.8 How to evaluate graduate students' academic activities

Benchmark universities set minimum counts and require evidence: China University of Petroleum asks for at least 10 seminars plus 1 presentation (or 8 lectures plus 1 report); CUG Wuhan asks for 2 academic presentations; Tsinghua gives 1 credit for academic activities; Mines geophysics requires attendance at the weekly distinguished lecture in every semester in residence (GPGN581). A points-based logbook is easy to administer and rewards active participation. The proposed system (GM04) is:

Activity	Evidence	Points
Attending a faculty seminar, distinguished lecture or thesis defence	Signature of chair; 1-page summary for 2 of them	1 each (at least 16 required)
Giving a talk in a faculty seminar or journal club	Programme and supervisor signature	4 each (at least 2 required)
Attending a national or international conference	Registration certificate	4 (at least 1 required)
Oral presentation at a conference	Programme listing	8
Poster at a conference	Programme listing	5
Completed summer or winter school	Certificate	6
Organising a seminar or workshop	Programme	4
Reviewing assistance for a journal (with supervisor)	Supervisor confirmation	3
Research report to an industry partner	Report	4
Public outreach (school visit, open day)	Organiser confirmation	2
Total required by the final defence	Checked at mid-term (at least 24) and before defence	At least 40 points = 4 credits

1.9 Foreign language proficiency criteria

University	Requirement
MIT (admission, international applicants)	Recommended minimum IELTS 7, TOEFL iBT 100, Duolingo 135 or Cambridge 190; some departments set higher (for example CEE requires TOEFL 100 or IELTS 7.5); an English Evaluation Test for international students after arrival
Tsinghua University (exemption from graduate English, 2026 rules)	TOEFL iBT 100, IELTS 7.0, GRE 310, GMAT 650/595, Duolingo 135, TEM-8, CET-6 600; or a degree taught in English abroad; or at least 6 months of study abroad; or a first-author English paper plus an oral interview; scores must be within the last 3 years
CUG Wuhan	Scientific English Writing and Communication (4 credits) is a compulsory public degree course; international students may write the thesis in English with a Chinese abstract
Tsinghua and CUP master's plans	First Foreign Language is a compulsory public course (Tsinghua: 2 credits)

Recommendation. Use the CEFR (Common European Framework of Reference) as the faculty's reference scale, accept several tests, and use two thresholds: an entry level that allows the student to follow Academic English (GC01), and a graduation level that the student must reach before the mid-term evaluation. The thresholds in Part 2.6 are proposals for the faculty to adjust to local conditions; CEFR equivalences of test scores are approximate and should be checked against the test publishers' current tables.

Part 2. Proposed programme structure and rules

2.1 Structure of the 7-semester, 150-credit programme

Semesters 1-3 are taught semesters; Semesters 4-7 are thesis semesters. Every student takes the 5 common subjects (20 credits) and at least 12 elective subjects of 3 credits (36 credits): 3 in Semester 1, 4 in Semester 2 and 5 in Semester 3. Research milestones, practice and academic activities carry the remaining 94 credits.

Semester	Common subjects	Electives	Research milestones and practice	Credits
1 (taught)	Academic English (4); Applied Mathematics (4); Research Methodology and Ethics (4)	3 x 3 = 9	Supervisor and committee chosen; Academic Activities logbook opened	21
2 (taught)	Statistics and Geostatistics (4); Numerical Modelling and Process Simulation (4)	4 x 3 = 12	Literature Review and Topic Selection (4)	24
3 (taught)	-	5 x 3 = 15	Research Proposal and Defence (4); Professional/Field/Teaching Practice (4)	23
4 (thesis)	-	-	Thesis Research I (18)	18
5 (thesis)	-	-	Thesis Research II (14); Mid-term Evaluation (4)	18
6 (thesis)	-	-	Thesis Research III (14); Research Publication (6); Academic Activities completed (4)	24
7 (thesis)	-	-	Thesis Writing and Pre-defence (12); Thesis Review and Final Defence (10)	22
Total	**20**	**36**	**94**	**150**

Lecture-based credits = 56 (37%); research, practice and presentation credits = 94 (63%). This balance is close to Mines and Freiberg, where research and the thesis make up about a fifth to a third of the credits of a 2-year programme, adjusted for the faculty's longer 4-semester thesis period.

2.2 Credit rule

- 1 credit = about 45 hours of total student workload.
- Taught subjects: 1 credit = 15 lecture/seminar hours (+ about 30 h self-study), or 30 laboratory/computer hours (+ about 15 h), or 1 field week.
- Research and practice: 1 credit = about 1 full-time week (40-45 h) of supervised research, practice or thesis work.
- Common subjects: 4 credits = 45 h lectures + 30 h laboratory/computer + about 105 h self-study.
- Electives: 3 credits = 135 h, delivered as lecture + laboratory (LL), lecture/seminar (L), lecture + field (F) or project-based (P) formats.

2.3 Elective rules

- At least 12 electives (36 credits); more may be taken but extra credits do not replace research credits.
- At least 8 electives from the student's own course list; 'R' subjects are recommended core electives, 'O' subjects are optional alternatives.
- At least 1 elective from another course list (cross-discipline elective, following CUG practice); up to 4 may be taken from other lists with the supervisor's approval.
- Up to 6 credits may be recognised from MOOCs, summer schools or exchange study (see Part 1.7) if workload and assessment are equivalent.
- An elective runs when at least 5 students enrol; shared subjects (for example FEM, Machine Learning, InSAR) are taught once for all courses.

2.4 Milestone timeline

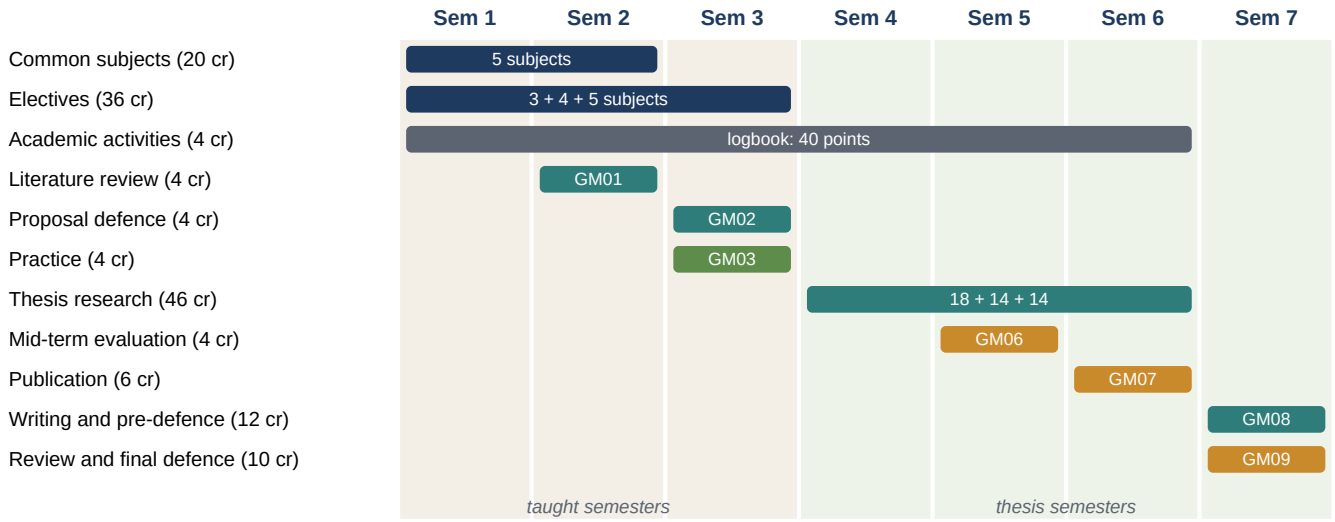


Figure 1. Milestones of the 7-semester programme. Bars show when each element is taken; credits in brackets. Amber items are formal checkpoints that must be passed before the student continues.

2.5 Pass and completion criteria

Mark	Letter	Grade points	Meaning
90-100	A	4.0	Excellent
85-89	A-	3.7	Very good
80-84	B+	3.3	Good
75-79	B	3.0	Good
70-74	B-	2.7	Satisfactory (minimum for common subjects)
65-69	C+	2.3	Pass (electives)
60-64	C	2.0	Pass (electives)
below 60	F	0	Fail

- **Subject pass:** attendance at least 80% of contact sessions; all compulsory laboratory, field or project work submitted; final mark at least 70 in common subjects and at least 60 in electives; where there is a final examination, at least 50 in the examination. Each subject's own requirements are listed in Part 4.
- **Resit:** one resit or resubmission in the following semester; a twice-failed elective must be repeated or replaced; a twice-failed common subject must be repeated.
- **Progression to thesis semesters (start of Semester 4):** all 56 coursework credits passed; coursework GPA at least 2.7; Literature Review and Proposal passed.
- **Mid-term (Semester 5):** Mid-term Evaluation passed; at least 24 academic-activity points; language requirement met.
- **Pre-defence (Semester 7):** manuscript submitted to a peer-reviewed journal (or accepted conference paper); 40 academic-activity points; thesis similarity below 15%.
- **Graduation:** 150 credits; thesis approved by two reviewers; final defence passed by majority vote with a final grade of at least 60; thesis deposited in the repository.
- **Time limits:** standard 7 semesters; maximum 10 semesters (5 years, as at Mines and CUG Wuhan); leave of absence counted separately.

2.6 Foreign language criteria (proposal)

Level	Purpose	Accepted evidence (any one; within the last 3 years)
Entry: CEFR B1	Admission; student takes Academic English (GC01)	IELTS 5.5; TOEFL iBT 60; Duolingo 95; CET-4 at least 425; or the faculty's own placement test
Graduation: CEFR B2	Required by the mid-term evaluation	GC01 passed with at least 70; or IELTS 6.0; TOEFL iBT 80; Duolingo 110; CET-6 at least 425
Exemption: CEFR C1	GC01 credits recognised; student takes Advanced Scientific Communication or a second foreign language instead	IELTS 7.0; TOEFL iBT 100; Duolingo 135 (the Tsinghua/MIT levels); CET-6 at least 550; previous degree taught in English; or a first-author paper in an international English-language journal plus a short oral interview

The thesis may be written in English or the national language, with an abstract in both. An English-language thesis is recommended for students who aim at international publication.

2.7 Academic activities and research milestones

The academic-activity points table in Part 1.8 applies. Detailed requirements for the literature review, proposal, practice, thesis research, mid-term evaluation, publication, pre-defence and final defence are given as catalogue entries GM01-GM09 in Part 4, in the same format as the taught subjects (credits, stages, assessment and passing requirements).

Part 3. Subjects by semester for each course

Each table shows the full 7-semester plan of one course. 'R' marks recommended electives (the standard choice); 'O' marks optional alternatives that may replace an R subject with the supervisor's approval. The student needs 3 electives in Semester 1, 4 in Semester 2 and 5 in Semester 3.

Mineral Exploration course

Lead unit: Institute of mineral exploration / geochemistry. **Graduate profile:** Researchers and exploration geologists who can model mineral systems, design and run exploration programmes, and estimate mineral resources to international reporting codes.

Semester	Code	Subject	Cr	Type
Semester 1	GC01	Academic English for Geoscience and Engineering	4	Common
	GC02	Applied Mathematics: Numerical Analysis and Matrix Methods	4	Common
	GC03	Research Methodology, Scientific Writing and Research Ethics	4	Common
	GE01	Advanced Ore Deposit Geology and Mineral Systems	3	Elective R
	GE02	Geochemical Exploration and Geochemical Data Analysis	3	Elective R
	GE03	Mineral Exploration Geophysics	3	Elective R
	GE04	*Advanced Structural Geology and Structural Controls on Ore*	*3*	*Elective O*
	GE05	*Advanced GIS and Spatial Analysis*	*3*	*Elective O*
		Semester 1 credits (electives counted as 3 x 3)	**21**	
Semester 2	GC04	Applied Statistics, Geostatistics and Geoscience Data Science	4	Common
	GC05	Numerical Modelling and Process Simulation	4	Common
	GE06	Ore-forming Fluids, Fluid Inclusions and Isotope Geochemistry	3	Elective R
	GE07	Analytical Methods in Mineralogy and Geochemistry (SEM, EPMA, LA-ICP-MS, XRD)	3	Elective R
	GE08	3D Geological Modelling	3	Elective R
	GE09	Advanced Optical and Hyperspectral Remote Sensing	3	Elective R
	GM01	Literature Review and Research Topic Selection	4	Milestone
		Semester 2 credits (electives counted as 4 x 3)	**24**	
Semester 3	GE10	Mineral Resource Estimation and Geostatistical Simulation	3	Elective R
	GE11	Exploration Project Design and Mineral Economics	3	Elective R
	GE12	Field and Ore Deposit Geology (field school)	3	Elective R
	GE13	Mineral Prospectivity Modelling and Targeting	3	Elective R
	GE14	Critical and Strategic Minerals	3	Elective R
	GE15	*Machine Learning and AI for Geoscience and Engineering*	*3*	*Elective O*
	GM02	Research Proposal and Proposal Defence	4	Milestone
	GM03	Professional, Field or Teaching Practice	4	Milestone
		Semester 3 credits (electives counted as 5 x 3)	**23**	
Semester 4	GM05	Thesis Research I	18	Milestone
		Semester 4 credits	**18**	
Semester 5	GM05	Thesis Research II	14	Milestone
	GM06	Mid-term Evaluation	4	Milestone
		Semester 5 credits	**18**	
Semester 6	GM05	Thesis Research III	14	Milestone
	GM07	Research Publication and Conference Presentation	6	Milestone
	GM04	Academic Activities (seminars, conferences, presentations) (accumulated Sem 1-6, credited here)	4	Milestone
		Semester 6 credits	**24**	
Semester 7	GM08	Thesis Writing, Similarity Check and Pre-defence	12	Milestone
	GM09	Thesis Review and Final Defence	10	Milestone
		Semester 7 credits	**22**	
		Programme total	**150**	

Civil Engineering course

Lead unit: Institute of civil engineering. **Graduate profile:** Geotechnical and structural engineers able to analyse, design and monitor foundations, underground works, slopes and structures with advanced numerical and reliability methods.

Semester	Code	Subject	Cr	Type
Semester 1	GC01	Academic English for Geoscience and Engineering	4	Common
	GC02	Applied Mathematics: Numerical Analysis and Matrix Methods	4	Common
	GC03	Research Methodology, Scientific Writing and Research Ethics	4	Common
	GE16	Advanced Soil Mechanics and Constitutive Modelling	3	Elective R
	GE17	Advanced Structural Analysis (matrix and nonlinear methods)	3	Elective R
	GE18	Finite Element Method in Engineering and Geomechanics	3	Elective R
		Semester 1 credits (electives counted as 3 x 3)	**21**	
Semester 2	GC04	Applied Statistics, Geostatistics and Geoscience Data Science	4	Common
	GC05	Numerical Modelling and Process Simulation	4	Common
	GE19	Structural Dynamics and Earthquake Engineering	3	Elective R
	GE20	Advanced Foundation Engineering and Deep Excavations	3	Elective R
	GE21	Geotechnical Site Investigation, In-situ Testing and Monitoring	3	Elective R
	GE22	Advanced Design of Concrete, Steel and Composite Structures	3	Elective R
	GE23	*SAR, InSAR and Deformation Monitoring*	*3*	*Elective O*
	GM01	Literature Review and Research Topic Selection	4	Milestone
		Semester 2 credits (electives counted as 4 x 3)	**24**	
Semester 3	GE24	Underground Construction and Tunnelling	3	Elective R
	GE25	Slope Stability, Landslides and Geohazard Risk	3	Elective R
	GE26	Reliability and Risk Analysis in Geotechnical and Structural Engineering	3	Elective R
	GE27	Construction Project Management and BIM	3	Elective R
	GE28	Ground Improvement and Geosynthetics	3	Elective R
	GE29	*Earthquake Seismology and Seismic Hazard Analysis*	*3*	*Elective O*
	GE15	*Machine Learning and AI for Geoscience and Engineering*	*3*	*Elective O*
	GM02	Research Proposal and Proposal Defence	4	Milestone
	GM03	Professional, Field or Teaching Practice	4	Milestone
		Semester 3 credits (electives counted as 5 x 3)	**23**	
Semester 4	GM05	Thesis Research I	18	Milestone
		Semester 4 credits	**18**	
Semester 5	GM05	Thesis Research II	14	Milestone
	GM06	Mid-term Evaluation	4	Milestone
		Semester 5 credits	**18**	
Semester 6	GM05	Thesis Research III	14	Milestone
	GM07	Research Publication and Conference Presentation	6	Milestone
	GM04	Academic Activities (seminars, conferences, presentations) (accumulated Sem 1-6, credited here)	4	Milestone
		Semester 6 credits	**24**	
Semester 7	GM08	Thesis Writing, Similarity Check and Pre-defence	12	Milestone
	GM09	Thesis Review and Final Defence	10	Milestone
		Semester 7 credits	**22**	
		Programme total	**150**	

Mineral and Rock Engineering course

Lead unit: Institute of mining / mineral engineering. **Graduate profile:** Mining and rock engineers who can characterise rock masses, design and optimise surface and underground mines, simulate processing plants and manage mine safety and waste.

Semester	Code	Subject	Cr	Type
Semester 1	GC01	Academic English for Geoscience and Engineering	4	Common
	GC02	Applied Mathematics: Numerical Analysis and Matrix Methods	4	Common
	GC03	Research Methodology, Scientific Writing and Research Ethics	4	Common
	GE30	Advanced Rock Mechanics	3	Elective R
	GE18	Finite Element Method in Engineering and Geomechanics	3	Elective R
	GE31	Mine Planning and Optimisation	3	Elective R
	GE16	*Advanced Soil Mechanics and Constitutive Modelling*	*3*	*Elective O*
		Semester 1 credits (electives counted as 3 x 3)	**21**	
Semester 2	GC04	Applied Statistics, Geostatistics and Geoscience Data Science	4	Common
	GC05	Numerical Modelling and Process Simulation	4	Common
	GE32	Numerical Methods in Geomechanics (FDM, BEM, DEM)	3	Elective R
	GE33	Advanced Mineral Processing and Plant Simulation	3	Elective R
	GE34	Rock Fragmentation and Blasting Engineering	3	Elective R
	GE21	Geotechnical Site Investigation, In-situ Testing and Monitoring	3	Elective R
	GE08	*3D Geological Modelling*	*3*	*Elective O*
	GM01	Literature Review and Research Topic Selection	4	Milestone
		Semester 2 credits (electives counted as 4 x 3)	**24**	
Semester 3	GE35	Underground Mine Design and Ground Control	3	Elective R
	GE25	Slope Stability, Landslides and Geohazard Risk	3	Elective R
	GE10	Mineral Resource Estimation and Geostatistical Simulation	3	Elective R
	GE36	Mine Ventilation, Health and Safety	3	Elective R
	GE37	Tailings and Mine Waste Engineering	3	Elective R
	GE24	*Underground Construction and Tunnelling*	*3*	*Elective O*
	GM02	Research Proposal and Proposal Defence	4	Milestone
	GM03	Professional, Field or Teaching Practice	4	Milestone
		Semester 3 credits (electives counted as 5 x 3)	**23**	
Semester 4	GM05	Thesis Research I	18	Milestone
		Semester 4 credits	**18**	
Semester 5	GM05	Thesis Research II	14	Milestone
	GM06	Mid-term Evaluation	4	Milestone
		Semester 5 credits	**18**	
Semester 6	GM05	Thesis Research III	14	Milestone
	GM07	Research Publication and Conference Presentation	6	Milestone
	GM04	Academic Activities (seminars, conferences, presentations) (accumulated Sem 1-6, credited here)	4	Milestone
		Semester 6 credits	**24**	
Semester 7	GM08	Thesis Writing, Similarity Check and Pre-defence	12	Milestone
	GM09	Thesis Review and Final Defence	10	Milestone
		Semester 7 credits	**22**	
		Programme total	**150**	

Geophysics course

Lead unit: Institute of geophysics. **Graduate profile:** Geophysicists able to acquire, process, model and invert seismic, potential-field and electromagnetic data for resources, engineering and hazards.

Semester	Code	Subject	Cr	Type
Semester 1	GC01	Academic English for Geoscience and Engineering	4	Common
	GC02	Applied Mathematics: Numerical Analysis and Matrix Methods	4	Common
	GC03	Research Methodology, Scientific Writing and Research Ethics	4	Common
	GE38	Geophysical Inverse Theory	3	Elective R
	GE39	Advanced Seismology and Wave Propagation	3	Elective R
	GE40	Advanced Gravity and Magnetic Methods	3	Elective R
	GE03	*Mineral Exploration Geophysics*	*3*	*Elective O*
	GE41	*Advanced Formation Evaluation and Borehole Geophysics*	*3*	*Elective O*
		Semester 1 credits (electives counted as 3 x 3)	**21**	
Semester 2	GC04	Applied Statistics, Geostatistics and Geoscience Data Science	4	Common
	GC05	Numerical Modelling and Process Simulation	4	Common
	GE42	Seismic Data Processing and Imaging	3	Elective R
	GE43	Electrical and Electromagnetic Methods	3	Elective R
	GE44	Computational Geophysics (numerical simulation of geophysical fields)	3	Elective R
	GE45	Rock Physics and Petrophysics	3	Elective R
	GE23	*SAR, InSAR and Deformation Monitoring*	*3*	*Elective O*
	GM01	Literature Review and Research Topic Selection	4	Milestone
		Semester 2 credits (electives counted as 4 x 3)	**24**	
Semester 3	GE46	Seismic Interpretation and Reservoir Characterisation	3	Elective R
	GE15	Machine Learning and AI for Geoscience and Engineering	3	Elective R
	GE47	Engineering, Environmental and Hydrogeophysics	3	Elective R
	GE48	Geophysical Field School and Instrumentation	3	Elective R
	GE29	Earthquake Seismology and Seismic Hazard Analysis	3	Elective R
	GM02	Research Proposal and Proposal Defence	4	Milestone
	GM03	Professional, Field or Teaching Practice	4	Milestone
		Semester 3 credits (electives counted as 5 x 3)	**23**	
Semester 4	GM05	Thesis Research I	18	Milestone
		Semester 4 credits	**18**	
Semester 5	GM05	Thesis Research II	14	Milestone
	GM06	Mid-term Evaluation	4	Milestone
		Semester 5 credits	**18**	
Semester 6	GM05	Thesis Research III	14	Milestone
	GM07	Research Publication and Conference Presentation	6	Milestone
	GM04	Academic Activities (seminars, conferences, presentations) (accumulated Sem 1-6, credited here)	4	Milestone
		Semester 6 credits	**24**	
Semester 7	GM08	Thesis Writing, Similarity Check and Pre-defence	12	Milestone
	GM09	Thesis Review and Final Defence	10	Milestone
		Semester 7 credits	**22**	
		Programme total	**150**	

Petroleum Exploration and Engineering course

Lead unit: Institute of petroleum / drilling engineering. **Graduate profile:** Petroleum geoscientists and engineers who can evaluate petroleum systems, characterise and simulate reservoirs, design wells and plan recovery and CO2 storage.

Semester	Code	Subject	Cr	Type
Semester 1	GC01	Academic English for Geoscience and Engineering	4	Common
	GC02	Applied Mathematics: Numerical Analysis and Matrix Methods	4	Common
	GC03	Research Methodology, Scientific Writing and Research Ethics	4	Common
	GE49	Petroleum Systems Analysis and Basin Modelling	3	Elective R
	GE50	Advanced Reservoir Engineering	3	Elective R
	GE41	Advanced Formation Evaluation and Borehole Geophysics	3	Elective R
		Semester 1 credits (electives counted as 3 x 3)	**21**	
Semester 2	GC04	Applied Statistics, Geostatistics and Geoscience Data Science	4	Common
	GC05	Numerical Modelling and Process Simulation	4	Common
	GE45	Rock Physics and Petrophysics	3	Elective R
	GE51	Reservoir Simulation	3	Elective R
	GE52	Advanced Drilling Engineering and Well Design	3	Elective R
	GE53	Reservoir Geomechanics	3	Elective R
	GE42	*Seismic Data Processing and Imaging*	*3*	*Elective O*
	GE08	*3D Geological Modelling*	*3*	*Elective O*
	GM01	Literature Review and Research Topic Selection	4	Milestone
		Semester 2 credits (electives counted as 4 x 3)	**24**	
Semester 3	GE46	Seismic Interpretation and Reservoir Characterisation	3	Elective R
	GE54	Enhanced Oil Recovery and CO2 Storage	3	Elective R
	GE55	Advanced Production Engineering	3	Elective R
	GE56	Well Test and Rate-Transient Analysis	3	Elective R
	GE57	Petroleum Economics, Risk and Decision Analysis	3	Elective R
	GE58	*Unconventional Oil and Gas Resources*	*3*	*Elective O*
	GM02	Research Proposal and Proposal Defence	4	Milestone
	GM03	Professional, Field or Teaching Practice	4	Milestone
		Semester 3 credits (electives counted as 5 x 3)	**23**	
Semester 4	GM05	Thesis Research I	18	Milestone
		Semester 4 credits	**18**	
Semester 5	GM05	Thesis Research II	14	Milestone
	GM06	Mid-term Evaluation	4	Milestone
		Semester 5 credits	**18**	
Semester 6	GM05	Thesis Research III	14	Milestone
	GM07	Research Publication and Conference Presentation	6	Milestone
	GM04	Academic Activities (seminars, conferences, presentations) (accumulated Sem 1-6, credited here)	4	Milestone
		Semester 6 credits	**24**	
Semester 7	GM08	Thesis Writing, Similarity Check and Pre-defence	12	Milestone
	GM09	Thesis Review and Final Defence	10	Milestone
		Semester 7 credits	**22**	
		Programme total	**150**	

Geographic Information and Space course

Lead unit: Institute of geographic information / space technology. **Graduate profile:** Geospatial and space engineers who can build spatial databases and analyses, process satellite, SAR, GNSS and LiDAR data, and design small Earth-observation missions.

Semester	Code	Subject	Cr	Type
Semester 1	GC01	Academic English for Geoscience and Engineering	4	Common
	GC02	Applied Mathematics: Numerical Analysis and Matrix Methods	4	Common
	GC03	Research Methodology, Scientific Writing and Research Ethics	4	Common
	GE05	Advanced GIS and Spatial Analysis	3	Elective R
	GE59	Satellite Geodesy and GNSS Positioning	3	Elective R
	GE60	Orbital Mechanics and Space Mission Analysis	3	Elective R
		Semester 1 credits (electives counted as 3 x 3)	**21**	
Semester 2	GC04	Applied Statistics, Geostatistics and Geoscience Data Science	4	Common
	GC05	Numerical Modelling and Process Simulation	4	Common
	GE09	Advanced Optical and Hyperspectral Remote Sensing	3	Elective R
	GE23	SAR, InSAR and Deformation Monitoring	3	Elective R
	GE61	Spatial Databases and WebGIS Development	3	Elective R
	GE62	Photogrammetry, LiDAR and UAV Mapping	3	Elective R
	GE08	*3D Geological Modelling*	*3*	*Elective O*
	GM01	Literature Review and Research Topic Selection	4	Milestone
		Semester 2 credits (electives counted as 4 x 3)	**24**	
Semester 3	GE63	Spacecraft Systems Engineering and Small Satellites	3	Elective R
	GE15	Machine Learning and AI for Geoscience and Engineering	3	Elective R
	GE64	Digital Terrain Analysis and Geohazard Modelling	3	Elective R
	GE65	Planetary Geology and Space Resources	3	Elective R
	GE13	Mineral Prospectivity Modelling and Targeting	3	Elective R
	GM02	Research Proposal and Proposal Defence	4	Milestone
	GM03	Professional, Field or Teaching Practice	4	Milestone
		Semester 3 credits (electives counted as 5 x 3)	**23**	
Semester 4	GM05	Thesis Research I	18	Milestone
		Semester 4 credits	**18**	
Semester 5	GM05	Thesis Research II	14	Milestone
	GM06	Mid-term Evaluation	4	Milestone
		Semester 5 credits	**18**	
Semester 6	GM05	Thesis Research III	14	Milestone
	GM07	Research Publication and Conference Presentation	6	Milestone
	GM04	Academic Activities (seminars, conferences, presentations) (accumulated Sem 1-6, credited here)	4	Milestone
		Semester 6 credits	**24**	
Semester 7	GM08	Thesis Writing, Similarity Check and Pre-defence	12	Milestone
	GM09	Thesis Review and Final Defence	10	Milestone
		Semester 7 credits	**22**	
		Programme total	**150**	

Part 4. Subject catalogue (each subject described once)

GC01 Academic English for Geoscience and Engineering

Credits: 4 | **Type:** Common compulsory subject (degree subject) | **Offered:** Semester 1

Workload: 45 h lectures + 30 h laboratory/computer + about 105 h self-study (= 180 h workload)

Used in: All six courses

Syllabus

Chapter 1. Academic reading - **1.1** Reading research articles: *skimming, scanning, reading for argument* **1.2** Critical reading: *evaluating claims, evidence and limitations* **1.3** Note-making and synthesis matrices

Chapter 2. Language of science - **2.1** Academic vocabulary: *Academic Word List, discipline terms in geoscience and engineering* **2.2** Grammar for research writing: *nominalisation, passive and active voice, tense in methods and results* **2.3** Hedging and boosting claims

Chapter 3. Listening and note-taking - **3.1** Lectures and seminars **3.2** Conference talks and panel discussion **3.3** Asking and answering questions

Chapter 4. Writing the research article - **4.1** Structure of IMRaD papers **4.2** Abstract and title: *structured abstracts, keywords* **4.3** Introduction: *CARS model, research gap* **4.4** Methods, results and discussion: *describing data, comparing with literature, limitations* **4.5** Conclusions and highlights

Chapter 5. Using sources - **5.1** Paraphrase and summary **5.2** Citation and referencing styles: *APA, Harvard, IEEE, GB/T 7714* **5.3** Reference managers: *Zotero* **5.4** Plagiarism and similarity checking

Chapter 6. Figures, tables and numbers - **6.1** Describing graphs, maps and cross-sections **6.2** Captions **6.3** Units and significant figures

Chapter 7. Presentations and posters - **7.1** Structure and timing of a 12-minute talk **7.2** Slide design **7.3** Poster design **7.4** Handling questions

Chapter 8. Professional correspondence - **8.1** Emails to supervisors and collaborators **8.2** Cover letter to a journal editor **8.3** Response to reviewers **8.4** Conference abstracts and CVs

Laboratory practice:

- Weekly speaking and listening lab (30 h)
- Mock conference session with recorded talks
- Peer review of a draft paragraph each week

Assessment and passing requirements

Assessment: Writing portfolio (abstract, introduction, figure description, response letter) 40%; oral presentation 20%; final reading-listening-writing examination 40%.

To pass: Final mark at least 70/100 and at least 50/100 in each of writing and oral components; students who meet the exemption criteria of Part 2.6 receive the credits by recognition and instead take Advanced Scientific Communication or a second foreign language (Chinese, Russian, German or Japanese) as a non-credit or elective subject.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Academic Phrasebank** - J. Morley (University of Manchester). *Free web resource.* Open PDF/online: https://www.phrasebank.manchester.ac.uk/. Contents: Appendix A, A01.
- **Technical Writing Essentials** - S. Last (University of Victoria; BCcampus). *CC BY 4.0; free web and PDF.* Open PDF/online: https://pressbooks.bccampus.ca/technicalwriting/. Contents: Appendix A, A02.
- Swales & Feak, Academic Writing for Graduate Students (3rd ed., Univ. of Michigan Press). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Glasman-Deal, Science Research Writing for Non-Native Speakers of English (Imperial College Press). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Tsinghua University (graduate training regulations) - First Foreign Language (2 cr, public course).** Compulsory public course in master's plans; exemption by test scores (see Part 1.9). Source: https://www.tsinghua.edu.cn/yjsy/fujian/pygz.pdf
- **China University of Geosciences (Wuhan) (graduate programme rules) - Scientific English Writing and Communication (4 cr, public course).** Public degree course for academic master's students. Source: https://graduate.cug.edu.cn/__local/C/BB/44/40D6AD23FA7CDCBB4769049FEAF_14DF9605_12E76F.pdf?e=.pdf
- **TU Bergakademie Freiberg (MSc Applied Geoscience) - Scientific Communication (4 LP, compulsory).** Compulsory module of the programme; see the module handbook for content. Source: https://tu-freiberg.de/sites/default/files/2025-12/english-taught-modules_applied_geoscience.pdf

GC02 Applied Mathematics: Numerical Analysis and Matrix Methods

Credits: 4 | **Type:** Common compulsory subject (degree subject) | **Offered:** Semester 1

Workload: 45 h lectures + 30 h laboratory/computer + about 105 h self-study (= 180 h workload)

Used in: All six courses

Syllabus

Chapter 1. Computer arithmetic and errors - **1.1** Floating-point representation **1.2** Round-off and truncation error **1.3** Conditioning and stability of algorithms

Chapter 2. Linear systems - **2.1** Gaussian elimination and LU factorisation: *pivoting, Cholesky* **2.2** Iterative solvers: *Jacobi, Gauss-Seidel, SOR, conjugate gradient, GMRES* **2.3** Sparse matrices and preconditioning

Chapter 3. Matrix theory and decompositions - **3.1** Vector and matrix norms **3.2** Eigenvalues and eigenvectors: *power method, QR algorithm* **3.3** Singular value decomposition **3.4** Least squares and pseudo-inverse **3.5** Low-rank approximation

Chapter 4. Nonlinear equations and optimisation - **4.1** Bisection, Newton and secant methods **4.2** Newton for systems **4.3** Gauss-Newton and Levenberg-Marquardt **4.4** Gradient, quasi-Newton (BFGS) and constrained optimisation **4.5** Linear programming

Chapter 5. Interpolation and approximation - **5.1** Polynomial and spline interpolation **5.2** Orthogonal polynomials **5.3** Fourier series, DFT and FFT **5.4** Introduction to wavelets

Chapter 6. Numerical differentiation and integration - **6.1** Finite-difference formulas **6.2** Newton-Cotes and Gaussian quadrature **6.3** Multidimensional integration and Monte Carlo

Chapter 7. Ordinary differential equations - **7.1** Euler and Runge-Kutta methods **7.2** Stiff systems and implicit methods **7.3** Boundary-value problems: *shooting, finite differences*

Chapter 8. Introduction to numerical PDEs - **8.1** Classification: *elliptic, parabolic, hyperbolic* **8.2** Finite-difference schemes **8.3** Consistency, stability (von Neumann, CFL) and convergence

Computer practice:

- Eight Python (NumPy/SciPy) laboratories: floating-point experiments
- LU and CG solvers
- SVD image and data compression
- Newton and Levenberg-Marquardt curve fitting
- spline and FFT analysis of a geophysical signal
- quadrature
- RK4 and stiff ODE
- explicit and implicit heat equation

Assessment and passing requirements

Assessment: Laboratory reports 30%; mid-term test 20%; final written examination 50%.

To pass: Final mark at least 70/100 (degree subject); final examination at least 50/100; all eight laboratory reports submitted.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Python Programming and Numerical Methods: A Guide for Engineers and Scientists** - Q. Kong, T. Siauw, A. Bayen (UC Berkeley; Elsevier). *Free online edition from the authors (print edition commercial).* Open PDF/online: https://bayen.berkeley.edu/python-programming-and-numerical-methods. Contents: Appendix A, A03.
- **Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares** - S. Boyd, L. Vandenberghe (Cambridge University Press). *Free PDF from the authors' site.* Open PDF/online: https://web.stanford.edu/~boyd/vmls/vmls.pdf. Contents: Appendix A, A04.
- **Convex Optimization** - S. Boyd, L. Vandenberghe (Cambridge University Press). *Free PDF from the authors' site.* Open PDF/online: https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf. Contents: Appendix A, A05.
- **Notes on Diffy Qs: Differential Equations for Engineers** - Jiri Lebl. *CC BY-SA / CC BY-NC-SA; free PDF.* Open PDF/online: https://jirka.org/diffyqs/diffyqs.pdf. Contents: Appendix A, A06.
- Burden, Faires & Burden, Numerical Analysis (10th ed., Cengage). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Golub & Van Loan, Matrix Computations (4th ed., Johns Hopkins). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Strang, Computational Science and Engineering (Wellesley-Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Massachusetts Institute of Technology (MIT OpenCourseWare) - 18.335J Introduction to Numerical Methods (graduate, Spring 2019).** Floating-point arithmetic, numerical linear algebra (direct and iterative), eigenproblems, optimisation and stability of algorithms. Source: https://ocw.mit.edu/courses/18-335j-introduction-to-numerical-methods-spring-2019/pages/syllabus/
- **Massachusetts Institute of Technology (MIT OpenCourseWare) - 18.085 Computational Science and Engineering I (graduate, Fall 2008).** Applied linear algebra, differential equations, finite differences and finite elements, Fourier methods and optimisation for engineering. Source: https://www.ocw.mit.edu/courses/18-085-computational-science-and-engineering-i-fall-2008/pages/syllabus

- **Colorado School of Mines - CEEN505 Numerical Methods for Engineers (3 cr).** Linear systems, nonlinear roots, eigenvalue problems, curve fitting, integration and boundary-value problems. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf
- **China University of Petroleum (master's training plans) - Numerical Analysis (3 cr).** Degree mathematics course in geoscience master's plans. Source: https://cie.upc.edu.cn/pdf/Master/dizhiziyuanyudizhigongcheng.pdf
- **China University of Geosciences (Wuhan) (graduate programme rules) - Applied Mathematics (3 cr: Numerical Analysis or Applied Statistics, professional master's).** Required mathematics course in professional master's plans. Source: https://graduate.cug.edu.cn/__local/C/BB/44/40D6AD23FA7CDCBB4769049FEAF_14DF9605_12E76F.pdf?e=.pdf
- **TU Bergakademie Freiberg (MSc Applied Geoscience) - Numerical Analysis of Differential Equations (4 LP).** Elective module in the computational and mathematical geoscience area. Source: https://tu-freiberg.de/sites/default/files/2025-12/english-taught-modules_applied_geoscience.pdf

GC03 Research Methodology, Scientific Writing and Research Ethics

Credits: 4 | **Type:** Common compulsory subject (degree subject) | **Offered:** Semester 1

Workload: 45 h lectures + 30 h laboratory/computer + about 105 h self-study (= 180 h workload)

Used in: All six courses

Syllabus

Chapter 1. Research in science and engineering - **1.1** Basic, applied and design research **1.2** Research questions, hypotheses and objectives **1.3** From problem to research gap

Chapter 2. Literature search and review - **2.1** Bibliographic databases: *Web of Science, Scopus, GeoRef, CNKI, Google Scholar* **2.2** Search strategies and Boolean logic **2.3** Narrative, critical and systematic reviews: *PRISMA* **2.4** Reference managers

Chapter 3. Research design - **3.1** Field, laboratory and computational study design **3.2** Sampling design and representativeness **3.3** Measurement uncertainty and error propagation **3.4** Reproducibility and replication

Chapter 4. Research data management - **4.1** FAIR principles **4.2** Data management plans **4.3** Metadata, version control (Git) and archiving

Chapter 5. Writing a research proposal - **5.1** Aims, objectives and hypotheses **5.2** Methods and work packages **5.3** Gantt chart and milestones **5.4** Budget, resources and risk register **5.5** Expected results and impact

Chapter 6. Publishing research - **6.1** Choosing a journal **6.2** Peer review process **6.3** Open access and licences **6.4** Citation metrics and their limits **6.5** Predatory journals and conferences

Chapter 7. Research ethics and integrity - **7.1** Fabrication, falsification and plagiarism **7.2** Authorship and contributor roles **7.3** Conflicts of interest **7.4** Responsible use of AI tools **7.5** Field and laboratory safety **7.6** Ethics of community and environmental impact

Chapter 8. Intellectual property and research impact - **8.1** Patents, software licences and standards **8.2** Technology transfer **8.3** Communicating with industry, government and the public

Project work: Literature review outline (at least 40 sources) and a 4-page mini research proposal on the student's provisional thesis topic, presented in a 10-minute talk

Assessment and passing requirements

Assessment: Assignments 40%; mini-proposal and talk 40%; research-integrity test 20%.

To pass: Final mark at least 70/100; research-integrity test passed with at least 80% (compulsory before the proposal defence).

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **On Being a Scientist: A Guide to Responsible Conduct in Research (3rd ed.)** - National Academy of Sciences, NAE, IOM. *Free PDF from National Academies Press.* Open PDF/online: https://nap.nationalacademies.org/catalog/12192. Contents: Appendix A, A07.
- **Technical Writing Essentials** - S. Last (University of Victoria; BCcampus). *CC BY 4.0; free web and PDF.* Open PDF/online: https://pressbooks.bccampus.ca/technicalwriting/. Contents: Appendix A, A02.
- Booth, Colomb, Williams et al., The Craft of Research (4th ed., Univ. of Chicago Press). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Day & Gastel, How to Write and Publish a Scientific Paper (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **TU Bergakademie Freiberg (MSc Applied Geoscience) - Scientific Communication (4 LP, compulsory).** Compulsory module of the programme; see the module handbook for content. Source: https://tu-freiberg.de/sites/default/files/2025-12/english-taught-modules_applied_geoscience.pdf
- **China University of Geosciences (Wuhan) (graduate programme rules) - Engineering Ethics (professional master's).** Required course in professional master's plans. Source: https://graduate.cug.edu.cn/__local/C/BB/44/40D6AD23FA7CDCBB4769049FEAF_14DF9605_12E76F.pdf?e=.pdf
- **China University of Geosciences (Wuhan) (graduate programme rules) - Literature Review (3 cr / 48 h, degree course).** Degree course in the academic master's plan. Source: https://graduate.cug.edu.cn/__local/C/BB/44/40D6AD23FA7CDCBB4769049FEAF_14DF9605_12E76F.pdf?e=.pdf

GC04 Applied Statistics, Geostatistics and Geoscience Data Science

Credits: 4 | **Type:** Common compulsory subject (degree subject) | **Offered:** Semester 2

Workload: 45 h lectures + 30 h laboratory/computer + about 105 h self-study (= 180 h workload)

Used in: All six courses

Syllabus

Chapter 1. Probability and distributions - **1.1** Random variables **1.2** Normal, lognormal, binomial, Poisson, exponential and extreme-value distributions **1.3** Transformations and the central limit theorem

Chapter 2. Estimation and hypothesis testing - **2.1** Point and interval estimation **2.2** t, F and chi-square tests **2.3** Non-parametric tests **2.4** Bootstrap and resampling **2.5** Multiple testing

Chapter 3. Regression and experimental design - **3.1** Simple and multiple linear regression **3.2** Diagnostics and transformations **3.3** Generalised linear models **3.4** ANOVA and design of experiments

Chapter 4. Multivariate analysis - **4.1** Principal component and factor analysis **4.2** Cluster analysis **4.3** Discriminant analysis **4.4** Compositional data (log-ratio transforms) for geochemical data

Chapter 5. Time series and spectral analysis - **5.1** Trend and autocorrelation **5.2** Power spectra **5.3** Filtering and smoothing

Chapter 6. Spatial statistics and variography - **6.1** Regionalised variables **6.2** Experimental and model variograms **6.3** Anisotropy and nested structures

Chapter 7. Kriging and geostatistical simulation - **7.1** Simple, ordinary and universal kriging **7.2** Cokriging **7.3** Indicator kriging **7.4** Sequential Gaussian and indicator simulation **7.5** Change of support

Chapter 8. Bayesian inference and statistical learning - **8.1** Bayes' theorem and priors **8.2** Markov chain Monte Carlo: *basic ideas* **8.3** Cross-validation **8.4** Regularised regression, decision trees and random forests

Computer practice: Weekly Python/R laboratories with course-specific data sets (geochemical survey, borehole assays, soil test data, well logs, satellite image samples)

Project work: Individual data-analysis project on a data set from the student's thesis field, with a 6-page report

Assessment and passing requirements

Assessment: Laboratory work 30%; project 20%; final examination 50%.

To pass: Final mark at least 70/100 (degree subject); project report at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **An Introduction to Statistical Learning (2nd ed.; R and Python editions)** - James, Witten, Hastie, Tibshirani (Springer). *Free PDF from the authors' site.* Open PDF/online: https://www.statlearning.com. Contents: Appendix A, A08.
- **OpenIntro Statistics (4th ed.)** - Diez, Cetinkaya-Rundel, Barr. *CC BY-SA; free PDF (pay-what-you-want, $0 minimum).* Open PDF/online: https://leanpub.com/os. Contents: Appendix A, A09.
- **Geostatistics Lessons** - Centre for Computational Geostatistics, University of Alberta (C.V. Deutsch et al.). *Free peer-reviewed online lessons.* Open PDF/online: https://geostatisticslessons.com. Contents: Appendix A, A10.
- **Spatial Data Science: With Applications in R** - E. Pebesma, R. Bivand (CRC Press). *Free online edition from the authors.* Open PDF/online: https://r-spatial.org/book/. Contents: Appendix A, A11.
- Davis, Statistics and Data Analysis in Geology (3rd ed., Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Wackernagel, Multivariate Geostatistics (3rd ed., Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Montgomery & Runger, Applied Statistics and Probability for Engineers (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Massachusetts Institute of Technology (MIT OpenCourseWare) - 1.151 Probability and Statistics in Engineering (graduate, Spring 2005).** Probability, random variables and distributions, estimation, hypothesis testing and regression for engineering problems. Source: https://ocw.mit.edu/courses/1-151-probability-and-statistics-in-engineering-spring-2005/pages/syllabus/
- **Colorado School of Mines - GEGN532 Geological Data Analysis (3 cr).** Statistical, multivariate and geostatistical methods for geologic data with MATLAB coding. Source: https://catalog.mines.edu/coursesaz/geoleng/
- **Colorado School of Mines - GPGN533 Geophysical Data Integration and Geostatistics (3 cr).** Combining data of differing support and resolution into stochastic subsurface models with geostatistical methods. Source: https://catalog.mines.edu/coursesaz/geophyseng/
- **TU Bergakademie Freiberg (MSc Applied Geoscience) - Multivariate Statistics and Geostatistics (9 LP, compulsory).** Compulsory module of the programme; see the module handbook for content. Source: https://tu-freiberg.de/sites/default/files/2025-12/english-taught-modules_applied_geoscience.pdf
- **Colorado School of Mines - MNGN536 Operations Research Techniques in the Mineral Industry (3 cr).** Statistics, simulation, Monte Carlo methods and linear programming for mining systems. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf

GC05 Numerical Modelling and Process Simulation

Credits: 4 | **Type:** Common compulsory subject (degree subject) | **Offered:** Semester 2

Workload: 45 h lectures + 30 h laboratory/computer + about 105 h self-study (= 180 h workload)

Used in: All six courses

Syllabus

Chapter 1. The modelling workflow - **1.1** Conceptual model and assumptions **1.2** Governing equations and boundary conditions **1.3** Verification, validation and calibration **1.4** Model documentation

Chapter 2. Conservation laws and constitutive models - **2.1** Mass, momentum and energy balance **2.2** Diffusion-advection, Darcy flow, heat conduction **2.3** Linear elasticity and plasticity **2.4** Wave equation

Chapter 3. Finite difference methods - **3.1** Explicit and implicit schemes **3.2** Accuracy, stability and convergence **3.3** Grids and boundary conditions

Chapter 4. Finite volume and finite element methods - **4.1** Control-volume formulation **4.2** Weak form and Galerkin method **4.3** Element types, shape functions and assembly **4.4** Meshing and mesh quality

Chapter 5. Discrete and particle methods - **5.1** Discrete element method: *contact laws, time stepping* **5.2** Particle and lattice methods **5.3** Monte Carlo simulation of processes

Chapter 6. Coupled processes - **6.1** Thermo-hydro-mechanical (THM) coupling **6.2** Multiphase flow **6.3** Reactive transport **6.4** Sequential and fully coupled solution strategies

Chapter 7. Calibration, sensitivity and uncertainty - **7.1** Parameter estimation and history matching **7.2** Local and global sensitivity: *Sobol indices* **7.3** Monte Carlo uncertainty propagation **7.4** Surrogate and reduced-order models

Chapter 8. Simulation software and high-performance computing - **8.1** Open codes: *MODFLOW 6, OPM Flow/MRST, FEniCS, OpenFOAM, Yade* **8.2** Commercial codes: *overview* **8.3** Parallel computing basics **8.4** Visualisation: *ParaView*

Computer practice:

- Six laboratories: 1D heat conduction by FD
- 2D steady groundwater flow by FV
- 2D elasticity by FEM (FEniCS)
- DEM granular column collapse
- Monte Carlo uncertainty propagation
- calibration of a simple model to observations

Project work: Course-specific simulation project (for example aquifer pumping and drawdown; consolidation under an embankment; stress around a tunnel or stope; seismic wave propagation; reservoir depletion; flood or landslide run-out on a DEM), with a model report

Assessment and passing requirements

Assessment: Laboratories 30%; simulation project 40%; final examination 30%.

To pass: Final mark at least 70/100 (degree subject); simulation project at least 60/100 with a verified model (benchmark against an analytical solution).

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Finite Difference Computing with PDEs: A Modern Software Approach** - H.P. Langtangen, S. Linge (Springer, Texts in Computational Science and Engineering 16). *Open access (CC BY 4.0), free PDF from Springer.* Open PDF/online: https://link.springer.com/book/10.1007/978-3-319-55456-3. Contents: Appendix A, A12.
- **Solving PDEs in Python: The FEniCS Tutorial I** - H.P. Langtangen, A. Logg (Springer, Simula SpringerBriefs). *Open access (CC BY 4.0), free PDF from Springer.* Open PDF/online: https://link.springer.com/book/10.1007/978-3-319-52462-7. Contents: Appendix A, A13.
- **An Introduction to Reservoir Simulation Using MATLAB/GNU Octave (MRST user guide)** - K.-A. Lie (SINTEF; Cambridge University Press, 2019). *Open access (CC BY-NC-ND 4.0) on Cambridge Core.* Open PDF/online: https://www.cambridge.org/core/books/an-introduction-to-reservoir-simulation-using-matlabgnu-octave/introduction/C66C3A56440AA6B95D7F814CAD3CD971. Contents: Appendix A, A14.
- **Documentation for the MODFLOW 6 Groundwater Flow Model (USGS Techniques and Methods 6-A55)** - C.D. Langevin et al., USGS 2017. *Free PDF (U.S. Government).* Open PDF/online: https://pubs.usgs.gov/publication/tm6A55. Contents: Appendix A, A15.
- Chapra & Canale, Numerical Methods for Engineers (McGraw-Hill). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Zienkiewicz, Taylor & Zhu, The Finite Element Method: Its Basis and Fundamentals (Elsevier). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Anderson, Woessner & Hunt, Applied Groundwater Modeling (2nd ed., Academic Press). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Massachusetts Institute of Technology (MIT OpenCourseWare) - 10.34 Numerical Methods Applied to Chemical Engineering (graduate, Fall 2015).** Numerical linear algebra, nonlinear equations, optimisation, ODE/DAE and PDE solution, and stochastic simulation for process modelling. Source: https://ocw.mit.edu/courses/10-34-numerical-methods-applied-to-chemical-engineering-fall-2015/pages/syllabus

- **Massachusetts Institute of Technology (MIT OpenCourseWare) - 2.29 Numerical Fluid Mechanics (graduate, Spring 2015).** Numerical methods for fluid flow: finite differences and finite volumes, stability, linear solvers, and computational fluid dynamics projects. Source: https://ocw.mit.edu/courses/2-29-numerical-fluid-mechanics-spring-2015/pages/syllabus/
- **Colorado School of Mines - GEGN583 Mathematical Modeling of Groundwater Systems (3 cr).** Analytical and numerical (finite-difference and finite-element) solutions for groundwater flow and mass transport. Source: https://catalog.mines.edu/coursesaz/geoleng/
- **Colorado School of Mines - GEGN563 Applied Numerical Modelling for Geomechanics (3 cr).** FEM, FDM, DEM and BEM for design of rock and soil excavations and slopes. Source: https://catalog.mines.edu/coursesaz/geoleng/
- **Colorado School of Mines - PEGN513 Reservoir Simulation I (3 cr).** Flow equations, finite-difference and control-volume methods, solvers and multiphase models. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf
- **TU Bergakademie Freiberg (MSc Applied Geoscience) - Numerical Simulation Methods in Geophysics (6 LP).** Elective module in the computational and mathematical geoscience area. Source: https://tu-freiberg.de/sites/default/files/2025-12/english-taught-modules_applied_geoscience.pdf
- **TU Bergakademie Freiberg (MSc Applied Geoscience) - Numerical Simulation with Finite Elements (6 LP).** Elective module in the computational and mathematical geoscience area. Source: https://tu-freiberg.de/sites/default/files/2025-12/english-taught-modules_applied_geoscience.pdf
- **TU Bergakademie Freiberg (MSc Applied Geoscience) - Uncertainty Quantification (6 LP).** Elective module in the computational and mathematical geoscience area. Source: https://tu-freiberg.de/sites/default/files/2025-12/english-taught-modules_applied_geoscience.pdf
- **China University of Petroleum (master's training plans) - Seismic Wave Simulation and Imaging.** Elective course in geophysics master's plans. Source: https://cie.upc.edu.cn/pdf/Master/dizhiziyuanyudizhigongcheng.pdf

GM01 Literature Review and Research Topic Selection

Credits: 4 | **Type:** Research milestone | **Offered:** Semester 2

Workload: 180 h independent study, supervision and presentation

Used in: All six courses

Stages and requirements

- Semester 1, week 4: supervisor and provisional research area confirmed, thesis advisory committee of at least 3 members formed
- Semesters 1-2: systematic literature search with a recorded search strategy
- Read and annotate at least 60 sources, at least 30 in international peer-reviewed journals and at least half published in the last 10 years
- Write a 6,000-8,000-word critical review report (state of knowledge, methods, gaps, proposed research questions)
- Week 14 of Semester 2: 15-minute presentation and 10 minutes of questions before the committee

Assessment and passing requirements

Assessment: Report 70% (coverage, critical synthesis, identification of gaps, quality of writing and referencing); presentation and discussion 30%.

To pass: Committee mark at least 60/100 and similarity check below 15% (excluding references); a failed review may be revised and re-presented once within 6 weeks.

Recommended textbooks

- **On Being a Scientist: A Guide to Responsible Conduct in Research (3rd ed.)** - National Academy of Sciences, NAE, IOM. *Free PDF from National Academies Press.* Open PDF/online: https://nap.nationalacademies.org/catalog/12192. Contents: Appendix A, A07.
- Ridley, The Literature Review: A Step-by-Step Guide for Students (SAGE). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Tsinghua University (graduate training regulations) - Literature Review and Thesis Proposal (1 cr).** Required training element completed by the end of the second semester, assessed by a panel of at least 3. Source: https://www.tsinghua.edu.cn/yjsy/fujian/pygz.pdf
- **China University of Geosciences (Wuhan) (graduate programme rules) - Literature Review (3 cr / 48 h, degree course).** Degree course in the academic master's plan. Source: https://graduate.cug.edu.cn/__local/C/BB/44/40D6AD23FA7CDCBB4769049FEAF_14DF9605_12E76F.pdf?e=.pdf

GM02 Research Proposal and Proposal Defence

Credits: 4 | **Type:** Research milestone | **Offered:** Semester 3

Workload: 180 h independent study, supervision and defence

Used in: All six courses

Stages and requirements

- Proposal document of 15-25 pages: title
- background and problem statement
- aims, objectives and hypotheses
- literature summary
- study area, data and methods
- work plan with Gantt chart
- expected results and contribution
- risks and mitigation
- resources and budget
- ethics, safety and data management plan
- Submission to the panel at least 2 weeks before the defence
- Public proposal defence: 20-minute presentation and 20 minutes of questions
- Panel of 3-5 members including the supervisor, at least one member from another institute and, if possible, one from industry or another university

Assessment and passing requirements

Assessment: Proposal document 60%; presentation and defence 40%.

To pass: Approved by a majority of the panel (outcomes: approved; approved with minor revisions; major revision and re-defence; rejected); a student who does not pass may re-defend once within 3 months; a second failure leads to a change of topic or transfer to a coursework-only exit award.

Recommended textbooks

- **On Being a Scientist: A Guide to Responsible Conduct in Research (3rd ed.)** - National Academy of Sciences, NAE, IOM. *Free PDF from National Academies Press.* Open PDF/online: https://nap.nationalacademies.org/catalog/12192. Contents: Appendix A, A07.
- **Technical Writing Essentials** - S. Last (University of Victoria; BCcampus). *CC BY 4.0; free web and PDF.* Open PDF/online: https://pressbooks.bccampus.ca/technicalwriting/. Contents: Appendix A, A02.
- Booth, Colomb, Williams et al., The Craft of Research (4th ed., Univ. of Chicago Press). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Tsinghua University (graduate training regulations) - Literature Review and Thesis Proposal (1 cr).** Required training element completed by the end of the second semester, assessed by a panel of at least 3. Source: https://www.tsinghua.edu.cn/yjsy/fujian/pygz.pdf

GM03 Professional, Field or Teaching Practice

Credits: 4 | **Type:** Research milestone | **Offered:** Semester 3

Workload: 4 weeks full time (160 h) + report

Used in: All six courses

Stages and requirements

- Choose one track (4 weeks full time or 160 hours in total): industry or institute internship
- Field school or field mapping campaign
- Teaching practice (tutorials, laboratory demonstration and marking for an undergraduate subject)
- Laboratory or instrument training (for example EPMA, LA-ICP-MS, triaxial rig, seismograph array, GNSS network)
- Agree a work plan with the supervisor and the host
- Keep a daily logbook
- Submit a 10-15-page practice report and give a 10-minute presentation

Assessment and passing requirements

Assessment: Host or supervisor evaluation 40%; practice report 40%; presentation 20%.

To pass: At least 160 documented hours, logbook signed by the host, and overall mark at least 60/100.

Recommended textbooks

Faculty guidelines and templates; recent journal literature in the student's field.

Similar subjects at other universities

- **China University of Geosciences (Wuhan) (graduate programme rules) - Professional Practice (6 cr, at least half a year, professional master's).** Required practice element of professional master's plans. Source:

https://graduate.cug.edu.cn/__local/C/BB/44/40D6AD23FA7CDCBB4769049FEAF_14DF9605_12E76F.pdf?e=.pdf

- **Colorado School of Mines - GEOL521 Field and Ore Deposit Geology (3 cr).** Field study of major mineral districts through mine visits, regional traverses and an intensive summer field trip. Source: https://catalog.mines.edu/coursesaz/geology

GM04 Academic Activities (seminars, conferences, presentations)

Credits: 4 | **Type:** Research milestone | **Offered:** Semesters 1-6

Workload: Accumulated over Semesters 1-6 (about 180 h)

Used in: All six courses

Stages and requirements

- Recorded in an academic-activity logbook (paper or online) from Semester 1 to Semester 6 and signed by the speaker, chair or supervisor
- Required minimum: 40 activity points including at least 16 attended seminars or lectures (1 point each), 2 talks by the student in faculty seminars or journal clubs (4 points each), and 1 conference attendance (4 points)
- Other activities: oral presentation at a national or international conference (8)
- poster (5)
- summer or winter school completed (6)
- reviewing or editorial assistance for a journal (3)
- organising a seminar or workshop (4)
- public science outreach event (2)
- research report to industry partner (4)

Assessment and passing requirements

Assessment: Points and quality review of two seminar summaries (1 page each) by the supervisor.

To pass: At least 40 points with the required minima met, and the logbook approved at the mid-term evaluation (at least 24 points) and before the final defence.

Recommended textbooks

Faculty guidelines and templates; recent journal literature in the student's field.

Similar subjects at other universities

- **Tsinghua University (graduate training regulations) - Academic Activities (1 cr).** Required training element: attendance and presentation at academic activities. Source: https://www.tsinghua.edu.cn/yjsy/fujian/pygz.pdf
- **China University of Petroleum (master's training plans) - Academic Activities (at least 10 seminars + 1 presentation, or 8 lectures + 1 report).** Required element of master's plans. Source: https://cie.upc.edu.cn/pdf/Master/dizhiziyuanyudizhigongcheng.pdf
- **Colorado School of Mines - GPGN581 Graduate Seminar (0-1 cr).** Attendance at the weekly distinguished-lecture series in each semester in residence. Source: https://catalog.mines.edu/coursesaz/geophyseng/

GM05 Thesis Research I, II and III

Credits: 46 | **Type:** Research milestone | **Offered:** Semesters 4, 5 and 6

Workload: Full-time research: about 20 weeks x 40 h per semester (18 + 14 + 14 credits)

Used in: All six courses

Stages and requirements

- Thesis Research I (Semester 4): fieldwork, sampling, experiments or data acquisition according to the approved proposal
- development and testing of methods or models
- quarterly meeting with the supervisor and a written progress report at the end of the semester
- Thesis Research II (Semester 5): analysis and interpretation of first results
- model calibration or laboratory testing programme
- draft chapters on data and methods
- Thesis Research III (Semester 6): completion of analysis, synthesis, comparison with literature and uncertainty assessment
- draft results and discussion chapters
- preparation of the publication

Assessment and passing requirements

Assessment: Each semester: supervisor assessment of progress against the work plan (50%), semester progress report (30%) and laboratory or field notebook and data archive (20%).

To pass: Each semester graded Satisfactory/Unsatisfactory; an Unsatisfactory semester requires a corrective plan agreed by the committee and a re-assessment after 3 months; two consecutive Unsatisfactory semesters lead to a committee review of continuation.

Recommended textbooks

Faculty guidelines and templates; recent journal literature in the student's field.

Similar subjects at other universities

- Equivalent to thesis research credits: Mines requires at least 6 research credits in a 30-credit MS (12 in the 36-credit MS Geology/Geological Engineering); Freiberg awards 30 LP for a 6-month thesis.

GM06 Mid-term Evaluation

Credits: 4 | **Type:** Research milestone | **Offered:** Semester 5

Workload: Report preparation and panel review (about 180 h)

Used in: All six courses

Stages and requirements

- Written mid-term report (about 30 pages): progress against the proposal, results obtained (at least 30-40% of planned results), changes to scope, work plan for the remaining semesters, publication plan
- Check of credits (all coursework completed), academic-activity logbook (at least 24 points) and language requirement
- 20-minute presentation and 20 minutes of questions before a panel of 3-5 members

Assessment and passing requirements

Assessment: Report 50%; presentation and discussion 30%; completeness of requirements 20%.

To pass: Outcome Pass, Conditional pass (re-check of named items within 3 months) or Fail; a student who fails may repeat the mid-term evaluation once after 3-6 months; a second failure ends the thesis programme.

Recommended textbooks

Faculty guidelines and templates; recent journal literature in the student's field.

Similar subjects at other universities

- **Tsinghua University (graduate training regulations) - Mid-term Assessment.** Required check of progress by a panel of 3-5 experts; must be passed before the thesis defence. Source: https://www.tsinghua.edu.cn/yjsy/fujian/pygz.pdf

GM07 Research Publication and Conference Presentation

Credits: 6 | **Type:** Research milestone | **Offered:** Semester 6

Workload: About 270 h of writing, revision and presentation

Used in: All six courses

Stages and requirements

- Prepare at least one research paper from the thesis work with the supervisor
- Submit it to a peer-reviewed journal (Scopus or SCI/EI indexed, or a faculty-approved national journal) as first author or first student author
- Alternatively, publish an accepted full paper in the proceedings of a recognised international conference and give the oral presentation
- Additional credit-equivalent outputs accepted by the committee: granted patent or software registration, technical standard contribution, or industry report adopted by the partner

Assessment and passing requirements

Assessment: Evidence of submission or acceptance; supervisor assessment of the student's own contribution (contributor-role statement).

To pass: At least one journal manuscript submitted and under review (or accepted conference paper and talk) before the pre-defence; acceptance is recommended but not required for graduation.

Recommended textbooks

- **Academic Phrasebank** - J. Morley (University of Manchester). *Free web resource.* Open PDF/online: https://www.phrasebank.manchester.ac.uk/. Contents: Appendix A, A01.

Similar subjects at other universities

- Not a credit-bearing element at Mines, MIT or Freiberg; many Chinese universities set a publication requirement for degree award (see Part 1.4).

GM08 Thesis Writing, Similarity Check and Pre-defence

Credits: 12 | **Type:** Research milestone | **Offered:** Semester 7

Workload: About 12 weeks full time

Used in: All six courses

Stages and requirements

- Thesis structure: abstract (in English and the national language), introduction, literature review, study area or materials, methods, results, discussion, conclusions, references, appendices with data
- Recommended length 25,000-40,000 words (or a paper-based thesis with an introduction, at least two manuscripts and a synthesis)
- Similarity check with the university system (below 15% excluding references and the student's own publications)
- Pre-defence (internal) before the advisory committee 6-8 weeks before the final defence, with a written list of required changes

Assessment and passing requirements

Assessment: Supervisor and committee assessment of the draft thesis (completeness, scientific quality, writing) and the pre-defence.

To pass: Pre-defence approved by the committee and similarity requirement met; only then is the thesis sent to external reviewers.

Recommended textbooks

- **Technical Writing Essentials** - S. Last (University of Victoria; BCcampus). *CC BY 4.0; free web and PDF.* Open PDF/online: https://pressbooks.bccampus.ca/technicalwriting/. Contents: Appendix A, A02.
- **Academic Phrasebank** - J. Morley (University of Manchester). *Free web resource.* Open PDF/online: https://www.phrasebank.manchester.ac.uk/. Contents: Appendix A, A01.
- Day & Gastel, How to Write and Publish a Scientific Paper (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- Mines: thesis to the committee at least 1 week before the defence; MIT EAPS: thesis to the committee 2 weeks before the defence.

GM09 Thesis Review and Final Defence

Credits: 10 | **Type:** Research milestone | **Offered:** Semester 7

Workload: About 10 weeks: revisions after review, defence preparation and defence

Used in: All six courses

Stages and requirements

- Thesis submitted to at least two reviewers (at least one external to the university) 4 weeks before the defence
- each reviewer gives a written report and a grade
- if the two grades differ by more than one grade step a third reviewer is appointed
- Public defence: 30-minute presentation and at least 30 minutes of questions from the defence committee (at least 3, preferably 5 members, chaired by a senior academic who is not the supervisor) and from the audience
- Committee closed session and vote

Assessment and passing requirements

Assessment: Final thesis grade = (2 x thesis grade from reviewers + 1 x defence grade) / 3.

To pass: Majority vote of the committee and a final grade of at least 60/100 (or 'Pass'); two or more negative votes mean the defence is not passed; a second defence may be held once, at least 1 month and at most 12 months later; after successful defence the final thesis is deposited in the library repository.

Recommended textbooks

Faculty guidelines and templates; recent journal literature in the student's field.

Similar subjects at other universities

- Mines: public defence, two negative votes = fail, one re-examination at least 1 week later; MIT EAPS: oral presentation of at least 30 minutes; Freiberg: colloquium of about 30 minutes, thesis weighted 2 and colloquium 1, two reviewers plus a third if grades differ by more than 1.7.

GE01 Advanced Ore Deposit Geology and Mineral Systems

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Mineral Exploration (recommended)

Syllabus

Chapter 1. Mineral systems concept - **1.1** Sources of metals, fluids and energy **1.2** Pathways and traps **1.3** Scale: *craton to deposit*

Chapter 2. Magmatic deposits - **2.1** Orthomagmatic Ni-Cu-PGE and chromite **2.2** Rare-metal granites and pegmatites: *Li, Ta, Nb, Sn, W*

Chapter 3. Porphyry and epithermal systems - **3.1** Porphyry Cu-Mo-Au architecture and alteration zoning **3.2** High-, intermediate- and low-sulfidation epithermal deposits **3.3** Skarns

Chapter 4. Orogenic and intrusion-related gold - **4.1** Structural controls **4.2** Fluid sources **4.3** Exploration vectors

Chapter 5. Sediment-hosted and volcanogenic deposits - **5.1** VMS **5.2** SEDEX and Mississippi Valley-type Pb-Zn **5.3** Sediment-hosted Cu **5.4** Iron formations and manganese

Chapter 6. Supergene and surficial deposits - **6.1** Laterites: *Ni, bauxite* **6.2** Supergene enrichment **6.3** Placers and heavy-mineral sands **6.4** Regolith geochemistry

Chapter 7. Metallogeny and tectonic setting - **7.1** Deposits through time **7.2** Metallogenic provinces and belts **7.3** Mineral system targeting criteria

Laboratory practice:

- Hand-specimen and drill-core suites of major deposit types
- Alteration mapping on core
- Construction of a deposit-scale cross-section with alteration and grade zoning

Assessment and passing requirements

Assessment: Laboratory portfolio 30%; seminar on a deposit type 20%; final examination 50%.

To pass: Final mark at least 60/100 with at least 50/100 in the examination; all laboratory exercises submitted.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Mineral Deposit Models (USGS Bulletin 1693)** - D.P. Cox, D.A. Singer (eds), U.S. Geological Survey, 1986. *Public domain; free PDF.* Open PDF/online: https://pubs.usgs.gov/bul/b1693/. Contents: Appendix A, A16.
- Robb, Introduction to Ore-Forming Processes (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Ridley, Ore Deposit Geology (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Pirajno, Hydrothermal Processes and Mineral Systems (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEOL515 Advanced Mineral Deposits (2 cr).** Mineral systems at deposit, district and regional scales: metal sources, transport paths and traps. Source: https://catalog.mines.edu/coursesaz/geology
- **Colorado School of Mines - GEOL524 Ore Deposit Models and Exploration Strategies (3 cr).** Magmatic, hydrothermal and sedimentary metal deposits, exploration strategies and lab-based core logging. Source: https://catalog.mines.edu/coursesaz/geology

GE02 Geochemical Exploration and Geochemical Data Analysis

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 15 h lectures + 2 weeks of field work + about 40 h report writing (= 135 h)

Used in: Mineral Exploration (recommended)

Syllabus

Chapter 1. Geochemical cycle and dispersion - **1.1** Primary and secondary dispersion halos **1.2** Mobility of elements in surface environments

Chapter 2. Survey design - **2.1** Orientation surveys **2.2** Sample media: *stream sediment, soil, till, rock, water, biogeochemistry* **2.3** Sampling density and grid design

Chapter 3. Sampling and analysis - **3.1** Sample preparation **3.2** Partial and total digestion **3.3** ICP-MS, ICP-OES, XRF and portable XRF **3.4** Detection limits

Chapter 4. Quality assurance and control - **4.1** Standards, blanks and duplicates **4.2** Control charts **4.3** Data acceptance criteria

Chapter 5. Data processing - **5.1** Censored data and below-detection values **5.2** Compositional data and log-ratio transforms **5.3** Background and threshold: *EDA, concentration-area fractal* **5.4** Multivariate methods and element associations

Chapter 6. Anomaly interpretation - **6.1** Scavenging and false anomalies **6.2** Pathfinder elements by deposit type **6.3** Lithogeochemistry and alteration indices

Chapter 7. Integration and targeting - **7.1** Geochemical maps in GIS **7.2** Combination with geophysics and geology **7.3** Follow-up strategy

Field practice: One-week geochemical survey: stream-sediment and soil sampling, field pXRF measurements, sample documentation

Computer practice: Processing a regional geochemical data set (QA/QC, log-ratio PCA, threshold mapping) in Python or ioGAS-type software

Assessment and passing requirements

Assessment: Field report 30%; data-processing project 30%; final examination 40%.

To pass: Final mark at least 60/100; field report and QA/QC analysis both at least 50/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Analytical Chemistry 2.1** - David Harvey (DePauw University; LibreTexts). *Free; full-book PDF download.* Open PDF/online: https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Analytical_Chemistry_2.1_by_David_Harvey. Contents: Appendix A, A17.
- **OpenIntro Statistics (4th ed.)** - Diez, Cetinkaya-Rundel, Barr. *CC BY-SA; free PDF (pay-what-you-want, $0 minimum).* Open PDF/online: https://leanpub.com/os. Contents: Appendix A, A09.
- Rose, Hawkes & Webb, Geochemistry in Mineral Exploration (Academic Press). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEGN532 Geological Data Analysis (3 cr).** Statistical, multivariate and geostatistical methods for geologic data with MATLAB coding. Source: https://catalog.mines.edu/coursesaz/geoleng/

GE03 Mineral Exploration Geophysics

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Mineral Exploration (recommended); Geophysics (optional)

Syllabus

Chapter 1. Physical properties of ores and host rocks - **1.1** Density, magnetic susceptibility, conductivity, chargeability, radioactivity **1.2** Property databases and petrophysical logging of core

Chapter 2. Magnetic and gravity methods for minerals - **2.1** Survey design: *airborne, ground, drone* **2.2** Processing: *RTP, derivatives, tilt, upward continuation* **2.3** Interpretation of intrusions, structures and alteration

Chapter 3. Electrical and electromagnetic methods - **3.1** Induced polarisation and resistivity for disseminated sulfides **3.2** Time-domain and frequency-domain EM for massive sulfides **3.3** Magnetotellurics for mineral systems

Chapter 4. Radiometric and other methods - **4.1** Gamma-ray spectrometry (K, U, Th) for alteration **4.2** Seismic reflection in hard-rock terrains **4.3** Borehole geophysics

Chapter 5. Inversion and 3D modelling - **5.1** Constrained and joint inversion **5.2** Geological constraints and petrophysics **5.3** Non-uniqueness and depth of investigation

Chapter 6. Exploration strategy - **6.1** Geophysical signatures of deposit types **6.2** Integration with geology and geochemistry **6.3** Cost-effective survey sequences

Chapter 7. Case histories - **7.1** Porphyry, VMS, IOCG, kimberlite and orogenic-gold examples

Computer practice:

- Processing of airborne magnetic and radiometric data
- 2D IP inversion
- 3D magnetic inversion with geological constraints (for example SimPEG)

Assessment and passing requirements

Assessment: Computer exercises 40%; case-history seminar 20%; final examination 40%.

To pass: Final mark at least 60/100; all computer exercises submitted.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* Open PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Contents: Appendix A, A18.
- **SimPEG documentation and tutorials (Simulation and Parameter Estimation in Geophysics)** - SimPEG developers (open source). *Free web resource (MIT licence).* Open PDF/online: https://simpeg.xyz. Contents: Appendix A, A19.
- Dentith & Mudge, Geophysics for the Mineral Exploration Geoscientist (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GPGN543 Mineral Exploration Geophysics (3 cr).** Links mineral-deposit geology with geophysical methods, inversion and exploration strategy design. Source: https://catalog.mines.edu/coursesaz/geophyseng/

GE04 Advanced Structural Geology and Structural Controls on Ore

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 15 h lectures + 2 weeks of field work + about 40 h report writing (= 135 h)

Used in: Mineral Exploration (optional)

Syllabus

Chapter 1. Stress, strain and rheology - **1.1** Stress tensor and Mohr analysis **1.2** Finite and incremental strain **1.3** Rheology of crust and fault rocks

Chapter 2. Faults and shear zones - **2.1** Fault architecture and damage zones **2.2** Shear-zone kinematics and microstructures **2.3** Fault-valve behaviour and fluid flow

Chapter 3. Folds and fabrics - **3.1** Fold mechanics **3.2** Foliations and lineations **3.3** Superposed deformation

Chapter 4. Structural controls on mineralisation - **4.1** Dilational jogs and fault bends **4.2** Vein systems and stress inversion **4.3** Ore shoot geometry and plunge

Chapter 5. Structural analysis of drill core - **5.1** Oriented core **5.2** Alpha-beta measurements **5.3** Structural logging and 3D interpretation

Chapter 6. Regional tectonics and metallogeny - **6.1** Extensional, compressional and strike-slip settings **6.2** Inversion tectonics

Field practice: Five-day structural mapping of a mineralised district with vein and fault measurements

Computer practice:

- Stereonet and kinematic analysis
- Oriented-core calculations
- 3D ore-shoot modelling

Assessment and passing requirements

Assessment: Field map and report 40%; exercises 20%; final examination 40%.

To pass: Final mark at least 60/100; field map and report at least 50/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Geological Structures: a Practical Introduction** - John W.F. Waldron, Morgan Snyder (Univ. of Alberta). *CC BY-NC; free web and PDF.* Open PDF/online: https://pressbooks.openeducationalberta.ca/introductorystructuralgeology/. Contents: Appendix A, A20.
- Fossen, Structural Geology (2nd ed., Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEOL502 Structural Methods for Seismic Interpretation (3 cr).** Structural techniques for interpreting 2D/3D seismic data in complex areas, with a 4-day field trip. Source: https://catalog.mines.edu/coursesaz/geology

GE05 Advanced GIS and Spatial Analysis

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Mineral Exploration (optional); Geographic Information and Space (recommended)

Syllabus

Chapter 1. Geographic information science - **1.1** Spatial data models: *vector, raster, TIN, voxel* **1.2** Coordinate reference systems, datums and projections **1.3** Data quality and uncertainty

Chapter 2. Spatial data acquisition and management - **2.1** Geodatabases **2.2** Metadata and standards: *ISO 19115, OGC* **2.3** Data integration and topology

Chapter 3. Vector and raster analysis - **3.1** Overlay, buffering and network analysis **3.2** Map algebra **3.3** Terrain and hydrological analysis

Chapter 4. Spatial statistics - **4.1** Spatial autocorrelation: *Moran's I, LISA* **4.2** Point-pattern analysis **4.3** Geographically weighted regression **4.4** Spatial interpolation: *IDW, splines, kriging*

Chapter 5. Multi-criteria and suitability analysis - **5.1** Weighting methods: *AHP* **5.2** Fuzzy logic **5.3** Sensitivity of results

Chapter 6. Geoprocessing automation - **6.1** Python scripting: *GeoPandas, Rasterio, PyQGIS* **6.2** Model builder workflows **6.3** Reproducible spatial analysis

Chapter 7. Cartography and geovisualisation - **7.1** Map design **7.2** 3D and web visualisation **7.3** Story maps

Computer practice:

- Weekly QGIS and Python laboratories
- Final GIS project in the student's thesis area with metadata and a reproducible script

Assessment and passing requirements

Assessment: Laboratories 40%; project 40%; final test 20%.

To pass: Final mark at least 60/100; project at least 50/100 with documented, reproducible workflow.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Essentials of Geographic Information Systems** - Jonathan Campbell, Michael Shin. *CC BY-NC-SA; free (Open Textbook Library).* Open PDF/online: https://open.umn.edu/opentextbooks/textbooks/67. Contents: Appendix A, A21.
- **Spatial Data Science: With Applications in R** - E. Pebesma, R. Bivand (CRC Press). *Free online edition from the authors.* Open PDF/online: https://r-spatial.org/book/. Contents: Appendix A, A11.
- **Geographic Data Science with Python** - S. Rey, D. Arribas-Bel, L.J. Wolf (CRC Press). *Free online edition (CC BY-NC-ND).* Open PDF/online: https://geographicdata.science/book/. Contents: Appendix A, A22.
- Longley, Goodchild, Maguire & Rhind, Geographic Information Science and Systems (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat
- O'Sullivan & Unwin, Geographic Information Analysis (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEGN575 Applications of Geographic Information Systems (3 cr).** GIS principles, data structures, DEMs and spatial analysis for geology. Source: https://catalog.mines.edu/coursesaz/geoleng/
- **Colorado School of Mines - GEGN579 Python Scripting for GIS (3 cr).** Python with ArcGIS to automate geoprocessing and build standalone GIS scripts and tools. Source: https://catalog.mines.edu/coursesaz/geoleng/

GE06 Ore-forming Fluids, Fluid Inclusions and Isotope Geochemistry

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Mineral Exploration (recommended)

Syllabus

Chapter 1. Hydrothermal fluids - **1.1** Sources: *magmatic, meteoric, metamorphic, basinal* **1.2** Physical properties of H2O-NaCl-CO2 fluids **1.3** Phase separation and boiling

Chapter 2. Metal transport and deposition - **2.1** Complexing: *chloride, bisulfide* **2.2** Solubility controls: *T, P, pH, fO2* **2.3** Deposition mechanisms

Chapter 3. Fluid inclusions - **3.1** Petrography and fluid-inclusion assemblages **3.2** Microthermometry **3.3** Raman and LA-ICP-MS of inclusions **3.4** Pressure-temperature reconstruction

Chapter 4. Stable isotopes - **4.1** O, H, C and S isotopes **4.2** Fractionation and geothermometry **4.3** Fluid-source tracing

Chapter 5. Radiogenic isotopes and geochronology - **5.1** Rb-Sr, Sm-Nd, Pb isotopes **5.2** U-Pb zircon and Re-Os molybdenite dating **5.3** Ar-Ar alteration dating

Chapter 6. Alteration geochemistry - **6.1** Mass-balance: *isocon* **6.2** Alteration mineral chemistry (chlorite, white mica) as vectors

Chapter 7. Application to exploration - **7.1** Isotope and fluid vectors **7.2** Case studies

Laboratory practice:

- Fluid-inclusion petrography and heating-freezing measurements on vein quartz
- Interpretation of isotope data sets

Assessment and passing requirements

Assessment: Laboratory report 30%; literature seminar 20%; final examination 50%.

To pass: Final mark at least 60/100; laboratory report at least 50/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Mineral Deposit Models (USGS Bulletin 1693)** - D.P. Cox, D.A. Singer (eds), U.S. Geological Survey, 1986. *Public domain; free PDF.* Open PDF/online: https://pubs.usgs.gov/bul/b1693/. Contents: Appendix A, A16.
- Faure & Mensing, Isotopes: Principles and Applications (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Pirajno, Hydrothermal Processes and Mineral Systems (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEOL513 Hydrothermal Geochemistry (3 cr).** High-temperature aqueous systems, fluid-rock interaction, stable isotopes, transport and deposition of ore minerals. Source: https://catalog.mines.edu/coursesaz/geology
- **Colorado School of Mines - GEOL540 Isotope Geochemistry and Geochronology (3 cr).** Geochronology and stable-isotope principles applied to igneous petrology and ore-deposit case studies. Source: https://catalog.mines.edu/coursesaz/geology

GE07 Analytical Methods in Mineralogy and Geochemistry (SEM, EPMA, LA-ICP-MS, XRD)

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Mineral Exploration (recommended)

Syllabus

Chapter 1. Sample preparation - **1.1** Polished thin sections and blocks **1.2** Mineral separation **1.3** Contamination control

Chapter 2. Optical and electron microscopy - **2.1** Reflected-light ore microscopy **2.2** SEM imaging (SE, BSE) and EDS **2.3** Automated mineralogy: *QEMSCAN, MLA*

Chapter 3. Electron probe microanalysis - **3.1** WDS principles **3.2** Standards and matrix correction **3.3** Detection limits and mapping

Chapter 4. X-ray methods - **4.1** XRD: *qualitative and Rietveld quantitative* **4.2** XRF: *major and trace elements*

Chapter 5. Mass spectrometry - **5.1** ICP-MS and LA-ICP-MS: *trace elements, U-Pb dating* **5.2** Isotope-ratio MS overview

Chapter 6. Spectroscopy - **6.1** Raman **6.2** SWIR/VNIR reflectance spectroscopy of alteration minerals **6.3** Moessbauer and FTIR overview

Chapter 7. Data quality and interpretation - **7.1** Accuracy, precision, uncertainty **7.2** Reference materials **7.3** Reporting analytical data

Laboratory practice:

- Hands-on sessions on SEM-EDS, XRD and SWIR spectrometer
- Analysis plan and interpretation of EPMA and LA-ICP-MS data for the student's own samples

Assessment and passing requirements

Assessment: Laboratory reports 50%; analytical plan for thesis samples 20%; final examination 30%.

To pass: Final mark at least 60/100; instrument safety induction completed; all laboratory reports submitted.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Analytical Chemistry 2.1** - David Harvey (DePauw University; LibreTexts). *Free; full-book PDF download.* Open PDF/online: https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Analytical_Chemistry_2.1_by_David_Harvey. Contents: Appendix A, A17.
- **Mineralogy** - Dexter Perkins et al. (Univ. of North Dakota; LibreTexts). *CC BY-NC-SA; free web and PDF export.* Open PDF/online: https://geo.libretexts.org/Bookshelves/Geology/Mineralogy_(Perkins_et_al.). Contents: Appendix A, A23.
- Reed, Electron Microprobe Analysis and Scanning Electron Microscopy in Geology (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Goldstein et al., Scanning Electron Microscopy and X-Ray Microanalysis (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEOL523 Reflected Light and Electron Microscopy (2 cr).** Reflected-light and electron microscopy (SEM, QEMSCAN) for ore-mineral identification and paragenesis. Source: https://catalog.mines.edu/coursesaz/geology

GE08 3D Geological Modelling

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: Project-based: 15 h lectures + 30 h studio/computer + about 90 h project work (= 135 h)

Used in: Mineral Exploration (recommended); Mineral and Rock Engineering (optional); Petroleum Exploration and Engineering (optional); Geographic Information and Space (optional)

Syllabus

Chapter 1. Geological data for 3D models - **1.1** Maps, sections, drill holes, geophysics **1.2** Data validation and desurveying

Chapter 2. Explicit modelling - **2.1** Wireframes and sections **2.2** Triangulated surfaces

Chapter 3. Implicit modelling - **3.1** Radial basis functions and potential-field (co-kriging) methods **3.2** Fault networks **3.3** Stratigraphic and intrusive relationships

Chapter 4. Structural and stratigraphic grids - **4.1** Corner-point and unstructured grids **4.2** Property modelling preparation

Chapter 5. Model uncertainty - **5.1** Stochastic geological models **5.2** Topological uncertainty **5.3** Information entropy

Chapter 6. Model validation and use - **6.1** Section checks and drilling **6.2** Links to resource estimation, reservoir and geomechanical models

Computer practice:

- Implicit modelling of a faulted deposit or reservoir in GemPy (open source) and Leapfrog/Petrel-type software
- Uncertainty ensemble of 50 models

Assessment and passing requirements

Assessment: Modelling project 60%; final test 40%.

To pass: Final mark at least 60/100; model delivered with data, script and validation report.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **GemPy documentation and tutorials (implicit 3D geological modelling)** - GemPy developers (open source). *Free web resource.* Open PDF/online: https://docs.gempy.org. Contents: Appendix A, A24.
- Mallet, Geomodeling (Oxford University Press). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **TU Bergakademie Freiberg (MSc Applied Geoscience) - Applied Geomodelling (4 LP) / Geomodelling and Geostatistics (5 LP).** Elective modules in the programme. Source: https://tu-freiberg.de/sites/default/files/2025-12/english-taught-modules_applied_geoscience.pdf
- **Colorado School of Mines - PEGN542 Integrated Reservoir Characterization (3 cr).** Integration of diverse data into a comprehensive reservoir model. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf

GE09 Advanced Optical and Hyperspectral Remote Sensing

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Mineral Exploration (recommended); Geographic Information and Space (recommended)

Syllabus

Chapter 1. Radiometric principles - **1.1** Radiative transfer **1.2** Sensor calibration **1.3** Atmospheric correction

Chapter 2. Optical and thermal sensors - **2.1** Multispectral: *Landsat, Sentinel-2, ASTER* **2.2** Hyperspectral: *EnMAP, PRISMA* **2.3** Thermal infrared

Chapter 3. Spectral analysis - **3.1** Spectral libraries **3.2** Band ratios and indices **3.3** Spectral unmixing and spectral angle mapping **3.4** Mineral mapping: *alteration, iron oxides, clays*

Chapter 4. Image classification - **4.1** Object-based image analysis **4.2** Machine-learning classifiers: *random forest, SVM, CNN* **4.3** Accuracy assessment

Chapter 5. Change detection and time series - **5.1** Multi-temporal analysis **5.2** Data cubes **5.3** Google Earth Engine and openEO

Chapter 6. Applications - **6.1** Lithological and alteration mapping **6.2** Land cover and environment **6.3** Mining and reclamation monitoring

Computer practice:

- Atmospheric correction and alteration mapping with ASTER/Sentinel-2
- Hyperspectral unmixing
- Time-series analysis in a cloud platform

Assessment and passing requirements

Assessment: Laboratories 40%; project 30%; final examination 30%.

To pass: Final mark at least 60/100; project at least 50/100 with accuracy assessment.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Fundamentals of Remote Sensing (tutorial)** - Canada Centre for Remote Sensing / Natural Resources Canada. *Free web tutorial.* Open PDF/online: https://natural-resources.canada.ca/science-data/science-research/geomatics/remote-sensing/tutorial-fundamentals-remote-sensing. Contents: Appendix A, A25.
- Richards, Remote Sensing Digital Image Analysis (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Lillesand, Kiefer & Chipman, Remote Sensing and Image Interpretation (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GPGN570 Applications of Satellite Remote Sensing (3 cr).** Satellite remote sensing of Earth and planets across visible, thermal, microwave and geodetic methods, with term projects. Source: https://catalog.mines.edu/coursesaz/geophyseng/
- **Colorado School of Mines - GEGN580 Applied Remote Sensing for Geoengineering and Geosciences (3 cr).** SAR, InSAR and optical remote sensing and their geoengineering and geoscience applications. Source: https://catalog.mines.edu/coursesaz/geoleng/

GE10 Mineral Resource Estimation and Geostatistical Simulation

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: Project-based: 15 h lectures + 30 h studio/computer + about 90 h project work (= 135 h)

Used in: Mineral Exploration (recommended); Mineral and Rock Engineering (recommended)

Syllabus

Chapter 1. Data for resource estimation - **1.1** Drilling and sampling **1.2** QA/QC **1.3** Density **1.4** Compositing and domaining

Chapter 2. Exploratory data analysis - **2.1** Statistics by domain **2.2** Top cuts and outliers **2.3** Contact analysis

Chapter 3. Variography - **3.1** Experimental and model variograms **3.2** Anisotropy **3.3** Variograms of transformed data

Chapter 4. Estimation - **4.1** Ordinary and simple kriging **4.2** Kriging neighbourhood analysis **4.3** Inverse-distance and nearest-neighbour checks **4.4** Block models and change of support

Chapter 5. Simulation - **5.1** Sequential Gaussian and indicator simulation **5.2** Grade-tonnage uncertainty **5.3** Recoverable resources

Chapter 6. Validation and classification - **6.1** Swath plots and visual checks **6.2** Reconciliation **6.3** Classification: *measured, indicated, inferred*

Chapter 7. Reporting codes - **7.1** JORC, NI 43-101 and CRIRSCO template **7.2** Competent Person **7.3** Reasonable prospects for eventual economic extraction

Computer practice: Complete resource estimate of a drill-hole data set in open (for example GeostatsPy, pygslib) and commercial software

Assessment and passing requirements

Assessment: Estimation project with JORC-style report 60%; final examination 40%.

To pass: Final mark at least 60/100; project at least 50/100 and validation plots provided.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Geostatistics Lessons** - Centre for Computational Geostatistics, University of Alberta (C.V. Deutsch et al.). *Free peer-reviewed online lessons.* Open PDF/online: https://geostatisticslessons.com. Contents: Appendix A, A10.
- **The JORC Code (2012 edition)** - Joint Ore Reserves Committee. *Free PDF (Chinese translation also free).* Open PDF/online: https://jorc.org/docs/jorc_code2012.pdf. Contents: Appendix A, A26.
- Rossi & Deutsch, Mineral Resource Estimation (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Sinclair & Blackwell, Applied Mineral Inventory Estimation (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - MNGN538 Geostatistical Ore Reserve Estimation (3 cr).** Variograms, kriging, estimation variance and geostatistical simulation for reserve estimation. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf
- **Colorado School of Mines - GPGN533 Geophysical Data Integration and Geostatistics (3 cr).** Combining data of differing support and resolution into stochastic subsurface models with geostatistical methods. Source: https://catalog.mines.edu/coursesaz/geophyseng/

GE11 Exploration Project Design and Mineral Economics

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: Project-based: 15 h lectures + 30 h studio/computer + about 90 h project work (= 135 h)

Used in: Mineral Exploration (recommended)

Syllabus

Chapter 1. Exploration strategy - **1.1** Commodity and target selection **1.2** Mineral system models **1.3** Exploration stages and decision gates

Chapter 2. Planning and budgeting - **2.1** Work programmes **2.2** Cost estimates **2.3** Contractors and logistics

Chapter 3. Drilling programmes - **3.1** Drill-hole planning **3.2** RC and diamond drilling **3.3** Logging and sampling protocols

Chapter 4. Mineral economics - **4.1** Supply, demand and prices **4.2** Critical minerals **4.3** Cost curves

Chapter 5. Project evaluation - **5.1** Scoping studies **5.2** Discounted cash flow, NPV and IRR **5.3** Sensitivity and risk analysis **5.4** Real options

Chapter 6. Licensing, ESG and social licence - **6.1** Mining law and licences **6.2** Environmental and social impact assessment **6.3** Community engagement

Project work:

- Team exploration programme for a real or simulated licence: target generation, budget, drilling plan, preliminary economic assessment
- report and pitch to a panel

Assessment and passing requirements

Assessment: Team report 50%; pitch 20%; individual examination 30%.

To pass: Final mark at least 60/100; peer-assessed individual contribution at least satisfactory.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **The JORC Code (2012 edition)** - Joint Ore Reserves Committee. *Free PDF (Chinese translation also free).* Open PDF/online: https://jorc.org/docs/jorc_code2012.pdf. Contents: Appendix A, A26.
- **Mineral Deposit Models (USGS Bulletin 1693)** - D.P. Cox, D.A. Singer (eds), U.S. Geological Survey, 1986. *Public domain; free PDF.* Open PDF/online: https://pubs.usgs.gov/bul/b1693/. Contents: Appendix A, A16.
- Marjoribanks, Geological Methods in Mineral Exploration and Mining (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Moon, Whateley & Evans, Introduction to Mineral Exploration (Blackwell). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEOL524 Ore Deposit Models and Exploration Strategies (3 cr).** Magmatic, hydrothermal and sedimentary metal deposits, exploration strategies and lab-based core logging. Source: https://catalog.mines.edu/coursesaz/geology

GE12 Field and Ore Deposit Geology (field school)

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 15 h lectures + 2 weeks of field work + about 40 h report writing (= 135 h)

Used in: Mineral Exploration (recommended)

Syllabus

Chapter 1. Preparation - **1.1** Literature on the districts **1.2** Safety and personal protective equipment **1.3** Field notebook standards

Chapter 2. Mine geology - **2.1** Ore controls **2.2** Grade control and reconciliation **2.3** Geometallurgy

Chapter 3. Reporting - **3.1** Daily field notes **3.2** District synthesis report **3.3** Presentation

Field practice:

- Two-week field school in mineral districts: underground and open-pit mine visits
- Mapping of alteration and veining
- Core logging at the mine core shed
- Discussions with mine geologists on grade control and exploration

Assessment and passing requirements

Assessment: Field notebook 30%; field report 50%; presentation 20%.

To pass: Full participation in the field school (at least 90% of field days) and final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Mineral Deposit Models (USGS Bulletin 1693)** - D.P. Cox, D.A. Singer (eds), U.S. Geological Survey, 1986. *Public domain; free PDF.* Open PDF/online: https://pubs.usgs.gov/bul/b1693/. Contents: Appendix A, A16.
- Marjoribanks, Geological Methods in Mineral Exploration and Mining (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEOL521 Field and Ore Deposit Geology (3 cr).** Field study of major mineral districts through mine visits, regional traverses and an intensive summer field trip. Source: https://catalog.mines.edu/coursesaz/geology

GE13 Mineral Prospectivity Modelling and Targeting

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: Project-based: 15 h lectures + 30 h studio/computer + about 90 h project work (= 135 h)

Used in: Mineral Exploration (recommended); Geographic Information and Space (recommended)

Syllabus

Chapter 1. Mineral prospectivity concepts - **1.1** Knowledge-driven and data-driven approaches **1.2** Mineral system proxies

Chapter 2. Evidence layers - **2.1** Geology, structure, geochemistry, geophysics, remote sensing **2.2** Feature engineering and buffering

Chapter 3. Knowledge-driven methods - **3.1** Index overlay **3.2** Fuzzy logic **3.3** Multi-criteria decision analysis

Chapter 4. Data-driven methods - **4.1** Weights of evidence **4.2** Logistic regression **4.3** Random forest, gradient boosting and neural networks

Chapter 5. Handling imbalance and uncertainty - **5.1** Training and negative samples **5.2** Spatial cross-validation **5.3** Uncertainty and success-rate curves

Chapter 6. Target ranking - **6.1** Integration with economics **6.2** Reporting targets

Computer practice:

- Prospectivity mapping of a district with fuzzy logic and random forest in Python/QGIS
- Comparison of models with prediction-rate curves

Assessment and passing requirements

Assessment: Project 60%; final test 40%.

To pass: Final mark at least 60/100; project with spatial cross-validation and documented code.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **An Introduction to Statistical Learning (2nd ed.; R and Python editions)** - James, Witten, Hastie, Tibshirani (Springer). *Free PDF from the authors' site.* Open PDF/online: https://www.statlearning.com. Contents: Appendix A, A08.
- **Spatial Data Science: With Applications in R** - E. Pebesma, R. Bivand (CRC Press). *Free online edition from the authors.* Open PDF/online: https://r-spatial.org/book/. Contents: Appendix A, A11.
- Bonham-Carter, Geographic Information Systems for Geoscientists: Modelling with GIS (Pergamon). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEGN532 Geological Data Analysis (3 cr).** Statistical, multivariate and geostatistical methods for geologic data with MATLAB coding. Source: https://catalog.mines.edu/coursesaz/geoleng/
- **Colorado School of Mines - GEGN575 Applications of Geographic Information Systems (3 cr).** GIS principles, data structures, DEMs and spatial analysis for geology. Source: https://catalog.mines.edu/coursesaz/geoleng/

GE14 Critical and Strategic Minerals

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 45 h lectures/seminars + about 90 h self-study (= 135 h)

Used in: Mineral Exploration (recommended)

Syllabus

Chapter 1. Critical minerals - **1.1** Definitions and criticality assessment **1.2** Supply-chain risks **1.3** Energy-transition demand

Chapter 2. Battery and magnet metals - **2.1** Lithium: *brines, pegmatites, clays* **2.2** Cobalt, nickel and graphite **2.3** Rare-earth elements: *carbonatite, ionic clay*

Chapter 3. Other critical metals - **3.1** Gallium, germanium, indium, tellurium: *by-products* **3.2** Tungsten, tin, tantalum, niobium **3.3** Platinum-group metals

Chapter 4. Exploration and evaluation - **4.1** Deposit models and exploration methods **4.2** Mineralogy and processing constraints

Chapter 5. Secondary resources - **5.1** Tailings and waste reprocessing **5.2** Recycling and urban mining

Chapter 6. Policy and ESG - **6.1** Strategic reserves and trade **6.2** Responsible sourcing

Assessment and passing requirements

Assessment: Commodity report 40%; seminar 20%; final examination 40%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Critical Mineral Resources of the United States (USGS Professional Paper 1802)** - K.J. Schulz, J.H. DeYoung, R.R. Seal, D.C. Bradley (eds), USGS 2017. *Free PDF (U.S. Government); one chapter per commodity.* Open PDF/online: https://pubs.usgs.gov/publication/pp1802. Contents: Appendix A, A27.
- Gunn (ed.), Critical Metals Handbook (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEOL524 Ore Deposit Models and Exploration Strategies (3 cr).** Magmatic, hydrothermal and sedimentary metal deposits, exploration strategies and lab-based core logging. Source: https://catalog.mines.edu/coursesaz/geology
- **Colorado School of Mines - SPRS501 Space Resources Fundamentals (3 cr).** Known solar-system resources, extraction systems, feasibility studies, legal issues and resource-enabled exploration. Source: https://catalog.mines.edu/coursesaz/spaceresources/spaceresources.pdf

GE15 Machine Learning and AI for Geoscience and Engineering

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Mineral Exploration (optional); Civil Engineering (optional); Geophysics (recommended); Geographic Information and Space (recommended)

Syllabus

Chapter 1. Machine-learning workflow - **1.1** Problem framing **1.2** Data preparation and feature engineering **1.3** Training, validation and testing

Chapter 2. Supervised learning - **2.1** Linear and logistic regression **2.2** Decision trees, random forests and gradient boosting **2.3** Support vector machines

Chapter 3. Neural networks and deep learning - **3.1** Multilayer perceptrons **3.2** Convolutional networks for images and seismic data **3.3** Recurrent networks and transformers for sequences

Chapter 4. Unsupervised learning - **4.1** Clustering **4.2** Dimensionality reduction: *PCA, t-SNE, UMAP* **4.3** Anomaly detection

Chapter 5. Geoscience-specific issues - **5.1** Spatial autocorrelation and spatial cross-validation **5.2** Small and imbalanced data **5.3** Physics-informed machine learning

Chapter 6. Explainability and uncertainty - **6.1** Feature importance and SHAP **6.2** Uncertainty quantification **6.3** Responsible and reproducible AI

Computer practice:

- Weekly Python labs (scikit-learn, PyTorch): lithology prediction from logs
- image classification
- seismic fault detection
- geohazard susceptibility mapping

Project work: Machine-learning project on the student's own data

Assessment and passing requirements

Assessment: Laboratories 30%; project 50%; final test 20%.

To pass: Final mark at least 60/100; project code and report reproducible by the assessor.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **An Introduction to Statistical Learning (2nd ed.; R and Python editions)** - James, Witten, Hastie, Tibshirani (Springer). *Free PDF from the authors' site.* Open PDF/online: https://www.statlearning.com. Contents: Appendix A, A08.
- **Mathematics for Machine Learning** - M.P. Deisenroth, A.A. Faisal, C.S. Ong (Cambridge University Press). *Free PDF from the authors' site (published by CUP).* Open PDF/online: https://mml-book.github.io/book/mml-book.pdf. Contents: Appendix A, A28.
- **Dive into Deep Learning** - Zhang, Lipton, Li, Smola. *Free online book and PDF (arXiv).* Open PDF/online: https://arxiv.org/abs/2106.11342. Contents: Appendix A, A29.
- **Deep Learning** - I. Goodfellow, Y. Bengio, A. Courville (MIT Press). *Free online edition from the authors.* Open PDF/online: https://www.deeplearningbook.org. Contents: Appendix A, A30.
- Geron, Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow (O'Reilly). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Massachusetts Institute of Technology (MIT OpenCourseWare) - 6.867 Machine Learning (graduate, Fall 2006).** Supervised and unsupervised learning, regression, classification, kernel methods, graphical models and model selection. Source: https://ocw.mit.edu/courses/6-867-machine-learning-fall-2006/pages/syllabus/
- **Colorado School of Mines - GPGN510 Machine Learning Inversion in Applied Geoscience (3 cr).** Building geoscience models through Tikhonov regularisation, machine learning and generative AI, including uncertainty quantification. Source: https://catalog.mines.edu/coursesaz/geophyseng/
- **TU Bergakademie Freiberg (MSc Applied Geoscience) - Machine Learning Methods (5 LP) / Mathematics of Machine Learning (6 LP).** Elective modules in the computational and mathematical geoscience area. Source: https://tu-freiberg.de/sites/default/files/2025-12/english-taught-modules_applied_geoscience.pdf

GE16 Advanced Soil Mechanics and Constitutive Modelling

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Civil Engineering (recommended); Mineral and Rock Engineering (optional)

Syllabus

Chapter 1. Soil composition and behaviour - **1.1** Clay mineralogy and fabric **1.2** Effective stress principle **1.3** Partially saturated soils: *suction*

Chapter 2. Seepage and consolidation - **2.1** 2D seepage and flow nets **2.2** One-dimensional and 3D consolidation **2.3** Secondary compression

Chapter 3. Stress-strain and strength - **3.1** Stress paths **3.2** Drained and undrained shear strength **3.3** Critical state soil mechanics

Chapter 4. Constitutive models - **4.1** Elastic-plastic models **4.2** Cam Clay and Modified Cam Clay **4.3** Hardening soil models

Chapter 5. Laboratory testing - **5.1** Triaxial: *CU, CD, stress-path* **5.2** Oedometer **5.3** Direct simple shear

Chapter 6. Numerical analysis - **6.1** Finite-element analysis of consolidation and stability **6.2** Parameter determination

Laboratory practice:

- Triaxial CU test with pore pressure
- Oedometer test
- Interpretation of stress paths

Computer practice: FE analysis of an embankment on soft clay (PLAXIS-type or open code)

Assessment and passing requirements

Assessment: Laboratory reports 30%; computer project 20%; final examination 50%.

To pass: Final mark at least 60/100 with at least 50/100 in the examination.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **An Introduction to Soil Mechanics** - A. Verruijt (Springer, 2018). *Open access PDF from Springer.* Open PDF/online: https://link.springer.com/book/10.1007/978-3-319-61185-3. Contents: Appendix A, A31.
- Wood, Soil Behaviour and Critical State Soil Mechanics (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Mitchell & Soga, Fundamentals of Soil Behavior (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Massachusetts Institute of Technology (MIT OpenCourseWare) - 1.361 Advanced Soil Mechanics (graduate, Fall 2004).** Soil composition and behaviour, effective stress, consolidation, shear strength, stress-strain behaviour and critical-state concepts. Source: https://www.ocw.mit.edu/courses/1-361-advanced-soil-mechanics-fall-2004/pages/syllabus
- **Massachusetts Institute of Technology (MIT OpenCourseWare) - 1.322 Soil Behavior (graduate, Spring 2005).** Composition, fabric and physico-chemical controls on the engineering behaviour of soils. Source: https://www.ocw.mit.edu/courses/1-322-soil-behavior-spring-2005/pages/syllabus
- **Colorado School of Mines - CEEN510 Advanced Soil Mechanics (3 cr).** Seepage, consolidation, shear strength, failure criteria and constitutive models, solved numerically by FE and FD. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf
- **Colorado School of Mines - CEEN513 Advanced Geomaterial Mechanics (4 cr).** Advanced soil and rock mechanics: rock-mass properties, failure theories, shear strength, stresses around underground openings. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf

GE17 Advanced Structural Analysis (matrix and nonlinear methods)

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Civil Engineering (recommended)

Syllabus

Chapter 1. Matrix methods - **1.1** Stiffness method for trusses, beams and frames **1.2** Coordinate transformation **1.3** Assembly and boundary conditions

Chapter 2. Advanced members - **2.1** Non-prismatic members **2.2** Arches **2.3** Cables and cable-stayed systems

Chapter 3. Nonlinear analysis - **3.1** Geometric nonlinearity: *P-delta* **3.2** Material nonlinearity and plastic hinges **3.3** Incremental-iterative solution

Chapter 4. Stability - **4.1** Elastic buckling of frames **4.2** Effective length **4.3** Second-order analysis

Chapter 5. Plates and shells - **5.1** Plate bending theory **5.2** Introduction to shells

Chapter 6. Analysis software - **6.1** Modelling with commercial and open software: *OpenSees* **6.2** Verification of results

Computer practice:

- Writing a 2D frame stiffness program in Python

- Nonlinear pushover analysis in OpenSees

Assessment and passing requirements

Assessment: Assignments and program 40%; final examination 60%.

To pass: Final mark at least 60/100; working stiffness program submitted.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Matrix Structural Analysis (2nd ed.)** - W. McGuire, R.H. Gallagher, R.D. Ziemian. *Free PDF from Bucknell Digital Commons.* Open PDF/online: https://digitalcommons.bucknell.edu/books/7/. Contents: Appendix A, A32.
- Kassimali, Matrix Analysis of Structures (Cengage). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - CEEN530 Advanced Structural Analysis (3 cr).** Non-prismatic members, arches, suspension and cable-stayed bridges, computer methods, nonlinear materials, force redistribution. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf
- **Colorado School of Mines - CEEN533 Matrix Structural Analysis (3 cr).** Stiffness-based matrix methods for trusses and frames with MATLAB programming and commercial software. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf

GE18 Finite Element Method in Engineering and Geomechanics

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Civil Engineering (recommended); Mineral and Rock Engineering (recommended)

Syllabus

Chapter 1. Variational and weighted residual methods - **1.1** Strong and weak forms **1.2** Galerkin method **1.3** Principle of virtual work

Chapter 2. One-dimensional elements - **2.1** Bars and beams **2.2** Shape functions **2.3** Element stiffness and assembly

Chapter 3. Two- and three-dimensional elements - **3.1** Triangles and quadrilaterals **3.2** Isoparametric formulation **3.3** Numerical integration **3.4** Solid elements

Chapter 4. Elasticity and plasticity - **4.1** Plane stress and strain **4.2** Axisymmetry **4.3** Mohr-Coulomb and Drucker-Prager plasticity **4.4** Return mapping

Chapter 5. Dynamics and time-dependent problems - **5.1** Mass and damping **5.2** Modal analysis **5.3** Newmark time integration **5.4** Transient heat and consolidation

Chapter 6. Geomechanics applications - **6.1** Excavation and construction stages **6.2** Interface and joint elements **6.3** Mesh convergence and error estimation

Computer practice:

- Own FE code for 2D elasticity in Python
- Analyses with FEniCS and a commercial FE package
- Verification against analytical solutions

Assessment and passing requirements

Assessment: Programming assignments 40%; project 20%; final examination 40%.

To pass: Final mark at least 60/100; FE code verified against a benchmark.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Solving PDEs in Python: The FEniCS Tutorial I** - H.P. Langtangen, A. Logg (Springer, Simula SpringerBriefs). *Open access (CC BY 4.0), free PDF from Springer.* Open PDF/online: https://link.springer.com/book/10.1007/978-3-319-52462-7. Contents: Appendix A, A13.
- **Finite Difference Computing with PDEs: A Modern Software Approach** - H.P. Langtangen, S. Linge (Springer, Texts in Computational Science and Engineering 16). *Open access (CC BY 4.0), free PDF from Springer.* Open PDF/online: https://link.springer.com/book/10.1007/978-3-319-55456-3. Contents: Appendix A, A12.
- Zienkiewicz, Taylor & Zhu, The Finite Element Method: Its Basis and Fundamentals (Elsevier). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Potts & Zdravkovic, Finite Element Analysis in Geotechnical Engineering (Thomas Telford). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - CEEN506 Finite Element Methods for Engineers (3 cr).** FE theory and programming for structural elements, elastic foundations, solid elasticity, steady and transient analysis, with geomechanics applications. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf
- **Massachusetts Institute of Technology (MIT OpenCourseWare) - 18.085 Computational Science and Engineering I (graduate, Fall 2008).** Applied linear algebra, differential equations, finite differences and finite elements, Fourier methods and optimisation for engineering. Source: https://www.ocw.mit.edu/courses/18-085-computational-science-and-engineering-i-fall-2008/pages/syllabus

- **TU Bergakademie Freiberg (MSc Applied Geoscience) - Numerical Simulation with Finite Elements (6 LP).** Elective module in the computational and mathematical geoscience area. Source: https://tu-freiberg.de/sites/default/files/2025-12/english-taught-modules_applied_geoscience.pdf

GE19 Structural Dynamics and Earthquake Engineering

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Civil Engineering (recommended)

Syllabus

Chapter 1. Single-degree-of-freedom systems - **1.1** Free and forced vibration **1.2** Damping **1.3** Response to harmonic, impulse and general loading: *Duhamel, numerical methods*

Chapter 2. Multi-degree-of-freedom systems - **2.1** Equations of motion **2.2** Modal analysis **2.3** Modal superposition

Chapter 3. Earthquake ground motion - **3.1** Seismic sources and waves **3.2** Ground-motion parameters **3.3** Response spectra

Chapter 4. Seismic design - **4.1** Design spectra and building codes **4.2** Equivalent lateral force and response-spectrum analysis **4.3** Ductility and capacity design

Chapter 5. Nonlinear and performance-based analysis - **5.1** Nonlinear time-history analysis **5.2** Pushover **5.3** Performance-based earthquake engineering

Chapter 6. Soil-structure interaction and site effects - **6.1** Site amplification **6.2** Liquefaction overview **6.3** Base isolation and dampers

Computer practice:

- Response-spectrum and time-history analysis of a multi-storey frame
- Site response analysis (1D equivalent linear)

Assessment and passing requirements

Assessment: Assignments 30%; design project 20%; final examination 50%.

To pass: Final mark at least 60/100; design project at least 50/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Earthquake-Resistant Design Concepts (FEMA P-749)** - FEMA / NIBS Building Seismic Safety Council. *Free PDF (U.S. Government).* Open PDF/online: https://www.fema.gov/sites/default/files/2020-10/earthquake-resistant-design-concepts_nehrp-seismic-provisions.pdf. Contents: Appendix A, A33.
- Chopra, Dynamics of Structures (Pearson). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Kramer, Geotechnical Earthquake Engineering (Pearson). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - CEEN531 Structural Dynamics (3 cr).** Linear and nonlinear SDOF and MDOF dynamics linked to earthquake code design. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf

GE20 Advanced Foundation Engineering and Deep Excavations

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 45 h lectures/seminars + about 90 h self-study (= 135 h)

Used in: Civil Engineering (recommended)

Syllabus

Chapter 1. Site characterisation for foundations - **1.1** Design parameters from in-situ and laboratory tests

Chapter 2. Shallow foundations - **2.1** Bearing capacity **2.2** Settlement: *immediate, consolidation* **2.3** Raft foundations

Chapter 3. Deep foundations - **3.1** Axial capacity of driven and bored piles **3.2** Pile groups **3.3** Lateral loading: *p-y curves* **3.4** Pile load testing

Chapter 4. Retaining structures - **4.1** Earth pressure theories **4.2** Gravity and cantilever walls **4.3** Embedded walls and anchors

Chapter 5. Deep excavations - **5.1** Support systems **5.2** Basal heave and groundwater control **5.3** Ground movements and adjacent structures

Chapter 6. Limit-state and reliability-based design - **6.1** Eurocode 7 and LRFD approaches **6.2** Partial factors

Computer practice:

- Pile group and lateral pile analysis
- Excavation support design with FE

Assessment and passing requirements

Assessment: Design project 40%; final examination 60%.

To pass: Final mark at least 60/100 with at least 50/100 in the examination.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **An Introduction to Soil Mechanics** - A. Verruijt (Springer, 2018). *Open access PDF from Springer.* Open PDF/online: https://link.springer.com/book/10.1007/978-3-319-61185-3. Contents: Appendix A, A31.
- Das, Principles of Foundation Engineering (Cengage). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Tomlinson & Woodward, Pile Design and Construction Practice (CRC). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - CEEN510 Advanced Soil Mechanics (3 cr).** Seepage, consolidation, shear strength, failure criteria and constitutive models, solved numerically by FE and FD. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf
- **Colorado School of Mines - CEEN513 Advanced Geomaterial Mechanics (4 cr).** Advanced soil and rock mechanics: rock-mass properties, failure theories, shear strength, stresses around underground openings. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf

GE21 Geotechnical Site Investigation, In-situ Testing and Monitoring

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 15 h lectures + 2 weeks of field work + about 40 h report writing (= 135 h)

Used in: Civil Engineering (recommended); Mineral and Rock Engineering (recommended)

Syllabus

Chapter 1. Planning a site investigation - **1.1** Desk study and walk-over survey **1.2** Investigation stages and scope **1.3** Ground models

Chapter 2. Drilling and sampling - **2.1** Drilling methods **2.2** Sample quality classes **2.3** Core logging: *soil and rock description standards*

Chapter 3. In-situ testing - **3.1** SPT, CPTu and seismic CPT **3.2** Pressuremeter and dilatometer **3.3** Vane, plate-load and permeability tests

Chapter 4. Geophysical site investigation - **4.1** Seismic refraction and MASW **4.2** ERT and GPR **4.3** Borehole geophysics

Chapter 5. Rock mass characterisation - **5.1** Discontinuity surveys **5.2** RQD, RMR, Q and GSI **5.3** Hydrogeological testing: *packer tests*

Chapter 6. Instrumentation and monitoring - **6.1** Piezometers, inclinometers, extensometers **6.2** InSAR and fibre-optic monitoring **6.3** Data management

Chapter 7. Geotechnical reporting - **7.1** Factual and interpretative reports **7.2** Geotechnical baseline report

Field practice: One-week field investigation: borehole logging, CPTu, MASW and ERT survey, installation of a piezometer

Assessment and passing requirements

Assessment: Field report 50%; ground model and interpretative report 30%; test 20%.

To pass: Field participation and final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Practical Rock Engineering** - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).* Open PDF/online: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf. Contents: Appendix A, A34.
- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* Open PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Contents: Appendix A, A18.
- Clayton, Matthews & Simons, Site Investigation (Blackwell). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEGN573 Geological Engineering Site Investigation (3 cr).** Field investigation, testing and monitoring methods for geotechnical and hazardous-waste sites. Source: https://catalog.mines.edu/coursesaz/geoleng/
- **Colorado School of Mines - GEGN571 Advanced Engineering Geology (3 cr).** Engineering-geology mapping and geologic hazard assessment for site selection and evaluation. Source: https://catalog.mines.edu/coursesaz/geoleng/

GE22 Advanced Design of Concrete, Steel and Composite Structures

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: Project-based: 15 h lectures + 30 h studio/computer + about 90 h project work (= 135 h)

Used in: Civil Engineering (recommended)

Syllabus

Chapter 1. Advanced reinforced concrete - **1.1** Slender columns and biaxial bending **1.2** Two-way slabs **1.3** Strut-and-tie models **1.4** Shear walls

Chapter 2. Prestressed concrete - **2.1** Pre- and post-tensioning **2.2** Losses **2.3** Flexural and shear design

Chapter 3. Advanced steel design - **3.1** Beam-columns and frame stability **3.2** Connections **3.3** Plate girders

Chapter 4. Composite structures - **4.1** Composite beams and columns **4.2** Shear connectors

Chapter 5. Durability and serviceability - **5.1** Cracking and deflection **5.2** Fatigue **5.3** Fire resistance

Chapter 6. Design codes - **6.1** Eurocodes, ACI/AISC and national codes **6.2** Load combinations

Project work: Design of a multistorey building or bridge element: analysis model, design calculations and drawings

Assessment and passing requirements

Assessment: Design project 50%; final examination 50%.

To pass: Final mark at least 60/100; design project checked and signed off by the instructor.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- Wight, Reinforced Concrete: Mechanics and Design (Pearson). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Salmon, Johnson & Malhas, Steel Structures: Design and Behavior (Pearson). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - CEEN540 Advanced Design of Steel Structures (3 cr).** Slender columns, beam-columns, frame stability, bracing, connections and composite design. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf
- **Colorado School of Mines - CEEN541 Design of Reinforced Concrete Structures II (3 cr).** Slender columns, biaxial bending, two-way slabs, strut-and-tie models, multistorey lateral and vertical load analysis. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf

GE23 SAR, InSAR and Deformation Monitoring

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Civil Engineering (optional); Geophysics (optional); Geographic Information and Space (recommended)

Syllabus

Chapter 1. SAR principles - **1.1** Radar imaging geometry **1.2** Range-Doppler processing **1.3** Speckle **1.4** Polarimetry

Chapter 2. SAR sensors and data - **2.1** Sentinel-1, ALOS-2, TerraSAR-X, NISAR **2.2** Data formats and access

Chapter 3. Interferometry - **3.1** Phase and coherence **3.2** DEM generation **3.3** Phase unwrapping

Chapter 4. Differential InSAR - **4.1** Deformation mapping **4.2** Atmospheric and orbital errors

Chapter 5. Time-series InSAR - **5.1** Persistent scatterers **5.2** Small baseline subset: *SBAS* **5.3** Validation with GNSS and levelling

Chapter 6. Applications - **6.1** Subsidence from mining, groundwater and hydrocarbons **6.2** Landslides and slope monitoring **6.3** Earthquake and volcano deformation **6.4** Infrastructure monitoring

Computer practice:

- InSAR processing of Sentinel-1 data (SNAP, ISCE or cloud services)
- Time-series analysis of a subsidence area

Assessment and passing requirements

Assessment: Laboratories 40%; project 40%; test 20%.

To pass: Final mark at least 60/100; project at least 50/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **The SAR Handbook: Comprehensive Methodologies for Forest Monitoring and Biomass Estimation** - A. Flores-Anderson et al. (NASA SERVIR, 2019). *Free PDF (NASA Technical Reports Server).* Open PDF/online: https://ntrs.nasa.gov/citations/20190002563. Contents: Appendix A, A35.
- **Fundamentals of Remote Sensing (tutorial)** - Canada Centre for Remote Sensing / Natural Resources Canada. *Free web tutorial.* Open PDF/online: https://natural-resources.canada.ca/science-data/science-research/geomatics/remote-sensing/tutorial-fundamentals-remote-sensing. Contents: Appendix A, A25.
- Hanssen, Radar Interferometry: Data Interpretation and Error Analysis (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEGN580 Applied Remote Sensing for Geoengineering and Geosciences (3 cr).** SAR, InSAR and optical remote sensing and their geoengineering and geoscience applications. Source: https://catalog.mines.edu/coursesaz/geoleng/
- **Colorado School of Mines - GPGN570 Applications of Satellite Remote Sensing (3 cr).** Satellite remote sensing of Earth and planets across visible, thermal, microwave and geodetic methods, with term projects. Source: https://catalog.mines.edu/coursesaz/geophyseng/

GE24 Underground Construction and Tunnelling

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 45 h lectures/seminars + about 90 h self-study (= 135 h)

Used in: Civil Engineering (recommended); Mineral and Rock Engineering (optional)

Syllabus

Chapter 1. Planning of underground works - **1.1** Route selection **1.2** Ground investigation for tunnels **1.3** Geotechnical risk register

Chapter 2. Ground behaviour - **2.1** Squeezing, swelling, rock burst and running ground **2.2** Ground reaction and convergence-confinement method

Chapter 3. Excavation methods - **3.1** Drill and blast **3.2** Tunnel boring machines: *EPB, slurry, hard-rock TBM* **3.3** Sequential excavation: *NATM/SEM*

Chapter 4. Support and lining design - **4.1** Rock bolts, shotcrete, steel sets **4.2** Segmental linings **4.3** Empirical (Q, RMR) and numerical design

Chapter 5. Soft-ground tunnelling - **5.1** Face stability **5.2** Surface settlement and building damage assessment **5.3** Ground improvement and grouting

Chapter 6. Shafts, caverns and portals - **6.1** Design and construction **6.2** Groundwater control

Chapter 7. Monitoring and observational method - **7.1** Convergence and settlement monitoring **7.2** Trigger levels **7.3** Back analysis

Computer practice:

- Convergence-confinement and 2D FE analysis of a tunnel
- Settlement trough prediction

Assessment and passing requirements

Assessment: Design project 40%; final examination 60%.

To pass: Final mark at least 60/100; design project at least 50/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Technical Manual for Design and Construction of Road Tunnels - Civil Elements (FHWA-NHI-10-034)** - U.S. Federal Highway Administration. *Free PDF (U.S. Government).* Open PDF/online: https://www.fhwa.dot.gov/bridge/tunnel/pubs/nhi09010/tunnel_manual.pdf. Contents: Appendix A, A36.
- **Practical Rock Engineering** - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).* Open PDF/online: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf. Contents: Appendix A, A34.
- Chapman, Metje & Staerk, Introduction to Tunnel Construction (CRC). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Kolymbas, Tunnelling and Tunnel Mechanics (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - CEEN523 Underground Construction Engineering in Soft Ground (4 cr).** Design and construction of tunnels, shafts and excavations in soft ground with temporary and permanent support. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf
- **Colorado School of Mines - MNGN504 Underground Construction Engineering in Hard Rock (3 cr).** Rock engineering, tunnelling methods, excavation support and risk assessment for underground projects. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf
- **Colorado School of Mines - MNGN506 Design and Support of Underground Excavations (3 cr).** Stress and deformation analysis, rock-mass rating, support selection and numerical models for shafts and tunnels. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf

GE25 Slope Stability, Landslides and Geohazard Risk

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Civil Engineering (recommended); Mineral and Rock Engineering (recommended)

Syllabus

Chapter 1. Slope failure mechanisms - **1.1** Soil and rock slopes **1.2** Planar, wedge, toppling and circular failures **1.3** Debris flows and rockfalls

Chapter 2. Limit-equilibrium analysis - **2.1** Method of slices: *Bishop, Janbu, Spencer, Morgenstern-Price* **2.2** Kinematic analysis of rock slopes

Chapter 3. Numerical analysis - **3.1** Strength reduction FE/FD **3.2** Discrete element analysis of jointed rock

Chapter 4. Hydrology and triggering - **4.1** Rainfall infiltration and pore pressure **4.2** Seismic loading **4.3** Rapid drawdown

Chapter 5. Probabilistic analysis - **5.1** Probability of failure **5.2** Monte Carlo **5.3** Reliability index

Chapter 6. Hazard and risk - **6.1** Landslide susceptibility mapping **6.2** Run-out analysis **6.3** Quantitative risk assessment

Chapter 7. Stabilisation and monitoring - **7.1** Drainage, anchors, retaining structures, rockfall barriers **7.2** Monitoring (inclinometers, radar, InSAR) and early warning

Field practice: Two-day slope mapping and kinematic data collection

Computer practice:

- Limit-equilibrium and strength-reduction analyses
- Probabilistic slope analysis
- Susceptibility map in GIS

Assessment and passing requirements

Assessment: Field and computer project 50%; final examination 50%.

To pass: Final mark at least 60/100; project at least 50/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Practical Rock Engineering** - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).* Open PDF/online: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf. Contents: Appendix A, A34.
- **The Landslide Handbook - A Guide to Understanding Landslides (USGS Circular 1325)** - L.M. Highland, P. Bobrowsky (2008). *Public domain; free PDF.* Open PDF/online: https://pubs.usgs.gov/publication/cir1325. Contents: Appendix A, A37.
- Wyllie, Rock Slope Engineering (5th ed., CRC). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Duncan, Wright & Brandon, Soil Strength and Slope Stability (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - CEEN515 Hillslope Hydrology and Stability (3 cr).** Shallow landslide mechanics with unsaturated effective stress; computer modelling of slope flow and stress. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf
- **Colorado School of Mines - MNGN545 Rock Slope Engineering (3 cr).** Rock-mass classification, slope stability analysis, wedge failure and remedial measures. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf
- **Colorado School of Mines - GEGN571 Advanced Engineering Geology (3 cr).** Engineering-geology mapping and geologic hazard assessment for site selection and evaluation. Source: https://catalog.mines.edu/coursesaz/geoleng/

GE26 Reliability and Risk Analysis in Geotechnical and Structural Engineering

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 45 h lectures/seminars + about 90 h self-study (= 135 h)

Used in: Civil Engineering (recommended)

Syllabus

Chapter 1. Uncertainty in engineering - **1.1** Aleatory and epistemic uncertainty **1.2** Spatial variability of ground properties

Chapter 2. Probability models - **2.1** Random variables and distributions of soil and rock properties **2.2** Random fields

Chapter 3. Reliability methods - **3.1** First-order second-moment **3.2** FORM and SORM **3.3** Monte Carlo and importance sampling

Chapter 4. System reliability - **4.1** Series and parallel systems **4.2** Fault trees and event trees

Chapter 5. Reliability-based design - **5.1** Partial factors and code calibration **5.2** Eurocode and LRFD background

Chapter 6. Risk assessment and decision - **6.1** Risk matrices **6.2** Quantitative risk **6.3** Bayesian updating with monitoring data **6.4** Value of information

Computer practice: Reliability analysis of a footing, slope and retaining wall in Python

Assessment and passing requirements

Assessment: Assignments 40%; final examination 60%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **OpenIntro Statistics (4th ed.)** - Diez, Cetinkaya-Rundel, Barr. *CC BY-SA; free PDF (pay-what-you-want, $0 minimum).* Open PDF/online: https://leanpub.com/os. Contents: Appendix A, A09.
- Baecher & Christian, Reliability and Statistics in Geotechnical Engineering (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Ang & Tang, Probability Concepts in Engineering (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - CEEN519 Risk Assessment in Geotechnical Engineering (3 cr).** Probabilistic tools for failure probability and risk in geotechnical design. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf
- **Colorado School of Mines - CEEN546 Statistical Methods for Reliability and Engineering Design (3 cr).** Quantifying uncertainty and risk in infrastructure performance, with structural and underground design reliability. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf

GE27 Construction Project Management and BIM

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: Project-based: 15 h lectures + 30 h studio/computer + about 90 h project work (= 135 h)

Used in: Civil Engineering (recommended)

Syllabus

Chapter 1. Project management fundamentals - **1.1** Project life cycle **1.2** Stakeholders **1.3** Contract forms: *FIDIC*

Chapter 2. Planning and scheduling - **2.1** Work breakdown structure **2.2** Critical path method **2.3** Resource levelling **2.4** Earned value

Chapter 3. Cost estimation and control - **3.1** Estimating methods **3.2** Cash flow **3.3** Change management

Chapter 4. Building information modelling - **4.1** BIM standards: *ISO 19650* **4.2** Model authoring and coordination **4.3** 4D and 5D BIM **4.4** Clash detection

Chapter 5. Digital construction - **5.1** Digital twins **5.2** Sensors and drones **5.3** Data-driven construction management

Chapter 6. Risk, safety and sustainability - **6.1** Construction risk management **6.2** Health and safety **6.3** Life-cycle assessment and green construction

Computer practice:

- BIM model of a small structure with clash detection
- Project schedule with CPM and earned-value analysis

Assessment and passing requirements

Assessment: BIM and scheduling project 50%; final examination 50%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Project Management (2nd ed.)** - A. Watt (BCcampus Open Education). *CC BY 4.0; free web and PDF.* Open PDF/online: https://opentextbc.ca/projectmanagement/. Contents: Appendix A, A38.
- Sacks, Eastman, Lee & Teicholz, BIM Handbook (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- No close graduate equivalent at the benchmark universities reviewed (Mines CEEN and MIT OCW lists); comparable content is taught in construction-management master's programmes.

GE28 Ground Improvement and Geosynthetics

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 45 h lectures/seminars + about 90 h self-study (= 135 h)

Used in: Civil Engineering (recommended)

Syllabus

Chapter 1. Problem soils - **1.1** Soft clays **1.2** Loose sands and liquefaction **1.3** Collapsible and expansive soils

Chapter 2. Densification - **2.1** Dynamic compaction **2.2** Vibro-compaction **2.3** Compaction grouting

Chapter 3. Drainage and consolidation - **3.1** Prefabricated vertical drains **3.2** Preloading and vacuum consolidation

Chapter 4. Reinforcement - **4.1** Stone columns **4.2** Rigid inclusions **4.3** Soil nailing

Chapter 5. Chemical and grouting methods - **5.1** Permeation and jet grouting **5.2** Deep soil mixing **5.3** Biocementation

Chapter 6. Geosynthetics - **6.1** Types and functions **6.2** Reinforced soil walls and slopes **6.3** Basal reinforcement **6.4** Liners in mine and waste facilities

Chapter 7. Design and quality control - **7.1** Design methods **7.2** Field verification tests

Assessment and passing requirements

Assessment: Design project 40%; final examination 60%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **An Introduction to Soil Mechanics** - A. Verruijt (Springer, 2018). *Open access PDF from Springer.* Open PDF/online: https://link.springer.com/book/10.1007/978-3-319-61185-3. Contents: Appendix A, A31.
- Han, Principles and Practice of Ground Improvement (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - CEEN510 Advanced Soil Mechanics (3 cr).** Seepage, consolidation, shear strength, failure criteria and constitutive models, solved numerically by FE and FD. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf

GE29 Earthquake Seismology and Seismic Hazard Analysis

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Civil Engineering (optional); Geophysics (recommended)

Syllabus

Chapter 1. Earthquake sources - **1.1** Faulting and elastic rebound **1.2** Moment tensors and focal mechanisms **1.3** Magnitude scales

Chapter 2. Seismic networks and location - **2.1** Seismometers and arrays **2.2** Earthquake location methods **2.3** Catalogues

Chapter 3. Seismotectonics - **3.1** Plate-boundary and intraplate seismicity **3.2** Active faults and paleoseismology

Chapter 4. Statistical seismology - **4.1** Gutenberg-Richter relation **4.2** Recurrence and Omori law **4.3** Declustering

Chapter 5. Ground motion - **5.1** Attenuation and ground-motion models **5.2** Site effects

Chapter 6. Seismic hazard analysis - **6.1** Deterministic and probabilistic seismic hazard: *PSHA* **6.2** Logic trees **6.3** Hazard maps and code spectra

Chapter 7. Induced seismicity - **7.1** Mining-, reservoir- and injection-induced seismicity **7.2** Monitoring and traffic-light systems

Computer practice:

- Earthquake location and focal mechanism with ObsPy
- PSHA for a site using OpenQuake

Assessment and passing requirements

Assessment: Laboratories 40%; final examination 60%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **New Manual of Seismological Observatory Practice (NMSOP-2)** - P. Bormann (ed.), GFZ Potsdam / IASPEI. *Open access (GFZ).* Open PDF/online: https://nmsop.gfz-potsdam.de. Contents: Appendix A, A39.
- **Earthquake-Resistant Design Concepts (FEMA P-749)** - FEMA / NIBS Building Seismic Safety Council. *Free PDF (U.S. Government).* Open PDF/online: https://www.fema.gov/sites/default/files/2020-10/earthquake-resistant-design-concepts_nehrp-seismic-provisions.pdf. Contents: Appendix A, A33.
- Stein & Wysession, An Introduction to Seismology, Earthquakes, and Earth Structure (Blackwell). *Commercial - no legal free PDF.* Library lookup: WorldCat
- McGuire, Seismic Hazard and Risk Analysis (EERI). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GPGN555 Earthquake Seismology (3 cr).** Earthquake location, mechanisms, tectonics, recurrence and seismic hazard. Source: https://catalog.mines.edu/coursesaz/geophyseng/

GE30 Advanced Rock Mechanics

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Mineral and Rock Engineering (recommended)

Syllabus

Chapter 1. Stress and strain in rock - **1.1** In-situ stress and its measurement: *overcoring, hydraulic fracturing* **1.2** Stress around openings: *Kirsch solution*

Chapter 2. Intact rock behaviour - **2.1** Elasticity and anisotropy **2.2** Brittle failure and post-peak behaviour **2.3** Time-dependent behaviour and creep

Chapter 3. Failure criteria - **3.1** Mohr-Coulomb **3.2** Hoek-Brown **3.3** Strength of discontinuities: *Barton-Bandis*

Chapter 4. Rock mass characterisation - **4.1** RMR, Q and GSI **4.2** Scale effects **4.3** Rock mass deformability

Chapter 5. Rock dynamics - **5.1** Wave propagation in rock **5.2** Rock burst and seismicity in mines

Chapter 6. Testing - **6.1** Uniaxial, triaxial and Brazilian tests **6.2** Shear tests on joints **6.3** Acoustic emission

Laboratory practice:

- UCS with strain gauges
- Triaxial test
- Direct shear of a rock joint
- Point-load test

Assessment and passing requirements

Assessment: Laboratory reports 30%; assignments 20%; final examination 50%.

To pass: Final mark at least 60/100; laboratory safety induction and all reports completed.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Practical Rock Engineering** - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).* Open PDF/online: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf. Contents: Appendix A, A34.
- Jaeger, Cook & Zimmerman, Fundamentals of Rock Mechanics (Blackwell). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Brady & Brown, Rock Mechanics for Underground Mining (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - MNGN508 Advanced Rock Mechanics (3 cr).** Analytical and numerical modelling of excavation stresses, failure criteria and rock-burst mitigation. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf
- **Colorado School of Mines - MNGN505 Rock Mechanics in Mining (3 cr).** Rock mechanics for mine layout: pillar design, caving, backfill, rock bursts and slope design. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf
- **Colorado School of Mines - CEEN513 Advanced Geomaterial Mechanics (4 cr).** Advanced soil and rock mechanics: rock-mass properties, failure theories, shear strength, stresses around underground openings. Source: https://catalog.mines.edu/coursesaz/civileng/civileng.pdf

GE31 Mine Planning and Optimisation

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: Project-based: 15 h lectures + 30 h studio/computer + about 90 h project work (= 135 h)

Used in: Mineral and Rock Engineering (recommended)

Syllabus

Chapter 1. Block models and economic values - **1.1** Block value calculation **1.2** Cut-off grades: *Lane's theory*

Chapter 2. Open-pit optimisation - **2.1** Lerchs-Grossmann and pseudoflow algorithms **2.2** Nested pits and pushbacks **2.3** Slope angle constraints

Chapter 3. Production scheduling - **3.1** Mixed-integer programming **3.2** Stockpiling **3.3** Blending constraints

Chapter 4. Underground mine optimisation - **4.1** Stope optimisation **4.2** Development and production scheduling

Chapter 5. Uncertainty in mine planning - **5.1** Geological uncertainty **5.2** Stochastic optimisation **5.3** Price uncertainty

Chapter 6. Mine economics - **6.1** Capital and operating costs **6.2** Cash-flow models **6.3** Strategic mine planning

Computer practice: Pit optimisation and scheduling with open tools (for example MineLib instances, Python-MIP) and commercial software

Assessment and passing requirements

Assessment: Planning project 50%; final examination 50%.

To pass: Final mark at least 60/100; project at least 50/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Convex Optimization** - S. Boyd, L. Vandenberghe (Cambridge University Press). *Free PDF from the authors' site.* Open PDF/online: https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf. Contents: Appendix A, A05.
- Hustrulid, Kuchta & Martin, Open Pit Mine Planning and Design (CRC). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - MNGN512 Surface Mine Design (3 cr).** Final pit limits, planning, road and dump layout and cost estimation. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf
- **Colorado School of Mines - MNGN536 Operations Research Techniques in the Mineral Industry (3 cr).** Statistics, simulation, Monte Carlo methods and linear programming for mining systems. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf

GE32 Numerical Methods in Geomechanics (FDM, BEM, DEM)

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Mineral and Rock Engineering (recommended)

Syllabus

Chapter 1. Overview of numerical methods - **1.1** Continuum versus discontinuum **1.2** Choice of method

Chapter 2. Finite difference (explicit Lagrangian) - **2.1** FLAC-type formulation **2.2** Large strain **2.3** Constitutive models

Chapter 3. Boundary element method - **3.1** Displacement discontinuity method **3.2** Tabular deposits

Chapter 4. Discrete element method - **4.1** Distinct element (UDEC/3DEC-type) for jointed rock **4.2** Particle-based DEM **4.3** Bonded-particle models

Chapter 5. Hybrid and fracture-mechanics methods - **5.1** FDEM **5.2** Fracture propagation **5.3** Lattice methods

Chapter 6. Coupled problems - **6.1** Hydro-mechanical coupling **6.2** Thermal effects

Chapter 7. Model building and validation - **7.1** Parameter selection **7.2** Back analysis **7.3** Reporting numerical models

Computer practice:

- FD model of a stope
- BEM model of a tabular orebody
- DEM model of a jointed slope
- Back analysis of monitoring data

Assessment and passing requirements

Assessment: Modelling assignments 60%; final examination 40%.

To pass: Final mark at least 60/100; each model accompanied by a verification check.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Practical Rock Engineering** - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).* Open PDF/online: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf. Contents: Appendix A, A34.
- **Solving PDEs in Python: The FEniCS Tutorial I** - H.P. Langtangen, A. Logg (Springer, Simula SpringerBriefs). *Open access (CC BY 4.0), free PDF from Springer.* Open PDF/online: https://link.springer.com/book/10.1007/978-3-319-52462-7. Contents: Appendix A, A13.
- Jing & Stephansson, Fundamentals of Discrete Element Methods for Rock Engineering (Elsevier). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEGN563 Applied Numerical Modelling for Geomechanics (3 cr).** FEM, FDM, DEM and BEM for design of rock and soil excavations and slopes. Source: https://catalog.mines.edu/coursesaz/geoleng/
- **Colorado School of Mines - MNGN525 Introduction to Numerical Techniques in Rock Mechanics (3 cr).** Boundary-element and finite-element methods, emphasising displacement discontinuities for tabular deposits. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf
- **Colorado School of Mines - MNGN526 Modeling and Measuring in Geomechanics (3 cr).** Field instrumentation, rock-strength testing and numerical modelling (BEM, FEM, FLAC). Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf

GE33 Advanced Mineral Processing and Plant Simulation

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Mineral and Rock Engineering (recommended)

Syllabus

Chapter 1. Process mineralogy and geometallurgy - **1.1** Liberation and textures **1.2** Geometallurgical domains **1.3** Ore variability testing

Chapter 2. Comminution - **2.1** Breakage mechanics **2.2** Crushing and grinding circuits **2.3** Bond and JK models **2.4** HPGR

Chapter 3. Classification and physical separation - **3.1** Screens and hydrocyclones **3.2** Gravity, magnetic and electrostatic separation **3.3** Ore sorting

Chapter 4. Flotation - **4.1** Surface chemistry and reagents **4.2** Kinetics **4.3** Circuit design

Chapter 5. Solid-liquid separation and tailings - **5.1** Thickening and filtration **5.2** Paste and filtered tailings

Chapter 6. Process modelling and simulation - **6.1** Mass balancing and data reconciliation **6.2** Unit models **6.3** Steady-state and dynamic plant simulation **6.4** Process control

Chapter 7. Hydrometallurgy overview - **7.1** Leaching **7.2** Solvent extraction **7.3** Electrowinning

Laboratory practice:

- Bond work-index test
- Batch flotation kinetics
- Hydrocyclone test

Computer practice: Mass balance of a concentrator and simulation of a grinding-flotation circuit

Assessment and passing requirements

Assessment: Laboratories 30%; simulation project 30%; final examination 40%.

To pass: Final mark at least 60/100; laboratory safety and reports completed.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- Wills & Finch, Wills' Mineral Processing Technology (Butterworth-Heinemann). *Commercial - no legal free PDF.* Library lookup: WorldCat
- King, Modeling and Simulation of Mineral Processing Systems (SME). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - MNGN532 Particulate Material Processing I: Comminution and Physical Separations (3 cr).** Crushing, grinding, screening, thickening and filtration design with plant simulation software. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf
- **Colorado School of Mines - MNGN533 Particulate Material Processing II: Applied Separations (3 cr).** Ore sorting, magnetic, gravity, electrostatic and flotation separations with plant simulation. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf

GE34 Rock Fragmentation and Blasting Engineering

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 15 h lectures + 2 weeks of field work + about 40 h report writing (= 135 h)

Used in: Mineral and Rock Engineering (recommended)

Syllabus

Chapter 1. Rock fragmentation theory - **1.1** Stress waves and gas pressure **1.2** Fracture mechanics of blasting

Chapter 2. Explosives - **2.1** Types and properties **2.2** Detonation physics **2.3** Initiation systems and electronic detonators

Chapter 3. Surface blast design - **3.1** Burden, spacing and stemming **3.2** Delay timing **3.3** Fragmentation models: *Kuz-Ram*

Chapter 4. Underground blast design - **4.1** Tunnel and drift rounds **4.2** Stope and raise blasting **4.3** Pre-splitting and smooth blasting

Chapter 5. Blast effects - **5.1** Ground vibration and airblast **5.2** Flyrock **5.3** Damage control

Chapter 6. Drilling - **6.1** Drilling methods and drillability **6.2** Measurement while drilling

Chapter 7. Safety and regulation - **7.1** Storage and transport **7.2** Blasting regulations

Field practice:

- Blast observation at a quarry or mine
- Fragmentation analysis from images
- Vibration monitoring

Assessment and passing requirements

Assessment: Blast design project 40%; field report 20%; final examination 40%.

To pass: Final mark at least 60/100; safety training completed.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Practical Rock Engineering** - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).* Open PDF/online: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf. Contents: Appendix A, A34.
- Persson, Holmberg & Lee, Rock Blasting and Explosives Engineering (CRC). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - MNGN507 Advanced Drilling and Blasting (3 cr).** Theory of rock penetration and fragmentation applied to drilling and blasting practice. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf

GE35 Underground Mine Design and Ground Control

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: Project-based: 15 h lectures + 30 h studio/computer + about 90 h project work (= 135 h)

Used in: Mineral and Rock Engineering (recommended)

Syllabus

Chapter 1. Selection of underground mining methods - **1.1** Geometry, ground conditions and grade **1.2** Method selection criteria

Chapter 2. Stoping methods - **2.1** Sublevel and long-hole stoping **2.2** Cut-and-fill **2.3** Room-and-pillar

Chapter 3. Caving methods - **3.1** Block and panel caving **3.2** Sublevel caving **3.3** Cave propagation

Chapter 4. Pillar and stope design - **4.1** Pillar strength **4.2** Stability graph method **4.3** Dilution

Chapter 5. Ground support and backfill - **5.1** Bolting and mesh **5.2** Cable bolts **5.3** Backfill types and design

Chapter 6. Mine seismicity and rock burst - **6.1** Seismic monitoring **6.2** Burst-prone ground management

Chapter 7. Underground infrastructure - **7.1** Shafts and declines **7.2** Haulage and hoisting **7.3** Automation

Computer practice:

- Stope design with the stability graph and numerical check
- Cave footprint layout

Assessment and passing requirements

Assessment: Design project 50%; final examination 50%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Practical Rock Engineering** - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).* Open PDF/online: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf. Contents: Appendix A, A34.
- Brady & Brown, Rock Mechanics for Underground Mining (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Hustrulid & Bullock (eds), Underground Mining Methods (SME). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - MNGN516 Underground Mine Design (3 cr).** Selection and design of underground mining methods, development and production planning, cost estimates. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf
- **Colorado School of Mines - MNGN518 Advanced Bulk Underground Mining Techniques (3 cr).** Sublevel, block and stoping systems: layout, scheduling, ventilation and costs. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf
- **Colorado School of Mines - MNGN505 Rock Mechanics in Mining (3 cr).** Rock mechanics for mine layout: pillar design, caving, backfill, rock bursts and slope design. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf

GE36 Mine Ventilation, Health and Safety

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Mineral and Rock Engineering (recommended)

Syllabus

Chapter 1. Underground atmosphere - **1.1** Mine gases **1.2** Dust **1.3** Heat sources

Chapter 2. Airflow fundamentals - **2.1** Atkinson's equation **2.2** Resistance and friction factors **2.3** Natural ventilation

Chapter 3. Ventilation networks - **3.1** Kirchhoff laws **3.2** Hardy Cross and numerical network solution

Chapter 4. Fans and controls - **4.1** Fan characteristics **4.2** Booster fans **4.3** Regulators and doors

Chapter 5. Ventilation planning - **5.1** Air quantity requirements **5.2** Diesel and battery-electric equipment **5.3** Ventilation on demand

Chapter 6. Heat and cooling - **6.1** Thermodynamics of mine air **6.2** Refrigeration and cooling systems

Chapter 7. Mine safety - **7.1** Fires and explosions **7.2** Emergency response and refuge chambers **7.3** Health and safety management systems

Computer practice: Ventilation network model of a mine with open or commercial software

Assessment and passing requirements

Assessment: Network project 40%; final examination 60%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- McPherson, Subsurface Ventilation and Environmental Engineering (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - MNGN524 Advanced Mine Ventilation (3 cr).** Ventilation design by mining method, numerical modelling, air cooling, fires and dust control. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf

GE37 Tailings and Mine Waste Engineering

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 45 h lectures/seminars + about 90 h self-study (= 135 h)

Used in: Mineral and Rock Engineering (recommended)

Syllabus

Chapter 1. Tailings properties - **1.1** Geotechnical and geochemical characterisation **1.2** Rheology

Chapter 2. Tailings storage facilities - **2.1** Upstream, downstream and centreline dams **2.2** Filtered and paste tailings

Chapter 3. Design and stability - **3.1** Seepage and liquefaction **3.2** Seismic design **3.3** Dam-break and run-out

Chapter 4. Water and geochemistry - **4.1** Water balance **4.2** Acid and metalliferous drainage **4.3** Water treatment

Chapter 5. Monitoring and governance - **5.1** Instrumentation **5.2** Global Industry Standard on Tailings Management **5.3** Engineer of record

Chapter 6. Closure and reprocessing - **6.1** Closure design **6.2** Reuse of tailings and waste rock

Computer practice:

- Seepage and stability model of a tailings dam
- Water balance

Assessment and passing requirements

Assessment: Design project 50%; final examination 50%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Global Industry Standard on Tailings Management** - ICMM, UNEP and PRI (Global Tailings Review, 2020). *Free PDF.* Open PDF/online: https://globaltailingsreview.org/global-industry-standard/. Contents: Appendix A, A40.
- Vick, Planning, Design, and Analysis of Tailings Dams (BiTech). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - MNGN582 Fundamentals of Tailings Engineering II (catalogue title).** Graduate tailings engineering course listed in the Mining Engineering catalogue. Source: https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf

GE38 Geophysical Inverse Theory

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Geophysics (recommended)

Syllabus

Chapter 1. Inverse problems - **1.1** Forward and inverse problems **1.2** Existence, uniqueness and stability **1.3** Linear and nonlinear problems

Chapter 2. Least squares - **2.1** Discrete linear inverse problems **2.2** Generalised inverse **2.3** Resolution and covariance

Chapter 3. Regularisation - **3.1** Tikhonov regularisation **3.2** Smoothness and sparsity: *L1, total variation* **3.3** Choice of trade-off: *L-curve, discrepancy, GCV*

Chapter 4. Nonlinear inversion - **4.1** Gauss-Newton and Levenberg-Marquardt **4.2** Adjoint-state gradients **4.3** Line search

Chapter 5. Probabilistic inversion - **5.1** Bayesian formulation **5.2** Markov chain Monte Carlo **5.3** Trans-dimensional inversion

Chapter 6. Joint and constrained inversion - **6.1** Petrophysical and structural coupling **6.2** Incorporating geological constraints

Chapter 7. Machine-learning inversion - **7.1** Neural-network surrogates **7.2** Learned regularisation

Computer practice:

- Weekly Python exercises: line-fit and tomography
- regularised 1D DC resistivity inversion
- MCMC for a two-parameter problem
- SimPEG 3D gravity inversion

Assessment and passing requirements

Assessment: Exercises 50%; final examination 50%.

To pass: Final mark at least 60/100; all exercises submitted.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Inverse Problem Theory and Methods for Model Parameter Estimation** - A. Tarantola (SIAM, 2005). *Free PDF made available by the author (listed by e-booksdirectory).* Open PDF/online: https://www.ipgp.fr/~tarantola/Files/Professional/Books/InverseProblemTheory.pdf. Contents: Appendix A, A41.
- **SimPEG documentation and tutorials (Simulation and Parameter Estimation in Geophysics)** - SimPEG developers (open source). *Free web resource (MIT licence).* Open PDF/online: https://simpeg.xyz. Contents: Appendix A, A19.
- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* Open PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Contents: Appendix A, A18.
- Aster, Borchers & Thurber, Parameter Estimation and Inverse Problems (Elsevier). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Menke, Geophysical Data Analysis: Discrete Inverse Theory (Academic Press). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GPGN509 Inversion (3 cr).** Probabilistic and deterministic inverse problem theory, optimisation, error and resolution analysis, with programming homework. Source: https://catalog.mines.edu/coursesaz/geophyseng/
- **Colorado School of Mines - GPGN510 Machine Learning Inversion in Applied Geoscience (3 cr).** Building geoscience models through Tikhonov regularisation, machine learning and generative AI, including uncertainty quantification. Source: https://catalog.mines.edu/coursesaz/geophyseng/
- **TU Bergakademie Freiberg (MSc Applied Geoscience) - Inverse Problems in Geophysics (6 LP).** Elective module in the computational and mathematical geoscience area. Source: https://tu-freiberg.de/sites/default/files/2025-12/english-taught-modules_applied_geoscience.pdf
- **China University of Petroleum (master's training plans) - Fundamentals of Geophysical Inversion.** Degree/elective course in geophysics master's plans. Source: https://cie.upc.edu.cn/pdf/Master/dizhiziyuanyudizhigongcheng.pdf

GE39 Advanced Seismology and Wave Propagation

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Geophysics (recommended)

Syllabus

Chapter 1. Elasticity and wave equations - **1.1** Stress, strain and Hooke's law **1.2** Elastic wave equation **1.3** P and S waves

Chapter 2. Body waves in layered media - **2.1** Reflection and transmission **2.2** Snell's law **2.3** Head waves

Chapter 3. Ray theory - **3.1** Travel times **3.2** Ray tracing **3.3** Geometrical spreading

Chapter 4. Surface waves and normal modes - **4.1** Rayleigh and Love waves **4.2** Dispersion **4.3** Free oscillations

Chapter 5. Attenuation and anisotropy - **5.1** Q **5.2** Anisotropic media **5.3** Shear-wave splitting

Chapter 6. Seismic sources - **6.1** Point sources and moment tensors **6.2** Radiation patterns

Chapter 7. Seismic tomography - **7.1** Travel-time tomography **7.2** Ambient-noise and full-waveform tomography

Computer practice:

- Ray tracing
- Surface-wave dispersion analysis
- Ambient-noise cross-correlation (ObsPy)

Assessment and passing requirements

Assessment: Problem sets 40%; final examination 60%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **New Manual of Seismological Observatory Practice (NMSOP-2)** - P. Bormann (ed.), GFZ Potsdam / IASPEI. *Open access (GFZ).* Open PDF/online: https://nmsop.gfz-potsdam.de. Contents: Appendix A, A39.
- Shearer, Introduction to Seismology (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Aki & Richards, Quantitative Seismology (University Science Books). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Massachusetts Institute of Technology (MIT OpenCourseWare) - 12.510 Introduction to Seismology (graduate, Spring 2010).** Elastic wave propagation, ray theory, travel times, finite-frequency effects, surface-wave dispersion, seismic tomography. Source: https://ocw.mit.edu/courses/12-510-introduction-to-seismology-spring-2010/pages/syllabus/
- **Colorado School of Mines - GPGN552/553 Introduction to Seismology I and II (3+3 cr).** Elasticity, seismic sources, plane-wave reflection and transmission in stratified media; spherical waves, point sources in layered media, surface waves, anisotropy and attenuation. Source: https://catalog.mines.edu/coursesaz/geophyseng/

GE40 Advanced Gravity and Magnetic Methods

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Geophysics (recommended)

Syllabus

Chapter 1. Potential-field theory - **1.1** Newtonian potential **1.2** Laplace and Poisson equations **1.3** Green's functions

Chapter 2. Gravity data - **2.1** Corrections and anomalies **2.2** Gravity gradiometry **2.3** Satellite gravity: *GRACE, GOCE*

Chapter 3. Magnetic data - **3.1** Geomagnetic field **3.2** Remanent and induced magnetisation **3.3** Aeromagnetic surveys

Chapter 4. Processing - **4.1** Fourier-domain filtering **4.2** Continuation and derivatives **4.3** Reduction to pole **4.4** Edge detection

Chapter 5. Forward modelling - **5.1** 2D and 3D bodies **5.2** Voxel models

Chapter 6. Inversion - **6.1** Depth weighting **6.2** 3D inversion with constraints **6.3** Magnetisation-vector inversion

Chapter 7. Interpretation - **7.1** Euler and spectral depth estimation **7.2** Integrated interpretation with geology

Computer practice:

- Processing of a regional gravity and aeromagnetic grid
- 3D inversion

Assessment and passing requirements

Assessment: Exercises 50%; final examination 50%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* Open PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Contents: Appendix A, A18.
- **SimPEG documentation and tutorials (Simulation and Parameter Estimation in Geophysics)** - SimPEG developers (open source). *Free web resource (MIT licence).* Open PDF/online: https://simpeg.xyz. Contents: Appendix A, A19.
- Blakely, Potential Theory in Gravity and Magnetic Applications (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GPGN511 Advanced Gravity and Magnetic Methods (3 cr).** Processing, 3D inversion and integrated interpretation of gravity and magnetic data to extract geologic information. Source: https://catalog.mines.edu/coursesaz/geophyseng/

GE41 Advanced Formation Evaluation and Borehole Geophysics

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Geophysics (optional); Petroleum Exploration and Engineering (recommended)

Syllabus

Chapter 1. Logging tools and measurements - **1.1** Wireline and LWD **1.2** Resistivity, nuclear, acoustic and NMR tools **1.3** Image logs

Chapter 2. Porosity and lithology - **2.1** Density-neutron **2.2** Sonic porosity **2.3** Cross-plots **2.4** Mineral inversion

Chapter 3. Saturation - **3.1** Archie and shaly-sand models **3.2** Formation water resistivity

Chapter 4. Permeability and productivity - **4.1** Log-based permeability **4.2** NMR permeability **4.3** Core-log integration

Chapter 5. Borehole imaging and dip analysis - **5.1** Fractures and bedding **5.2** Stress indicators: *breakouts*

Chapter 6. Log-seismic tie - **6.1** Synthetic seismograms **6.2** Check-shot and VSP

Chapter 7. Applications beyond petroleum - **7.1** Mineral, groundwater and geothermal logging

Computer practice:

- Petrophysical interpretation of a multi-well data set (Python, open log-analysis tools)
- Synthetic seismogram

Assessment and passing requirements

Assessment: Interpretation project 50%; final examination 50%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **PetroWiki** - Society of Petroleum Engineers (based on the SPE Petroleum Engineering Handbook). *Free web resource.* Open PDF/online: https://petrowiki.spe.org. Contents: Appendix A, A42.
- **Seismic Data Analysis: Processing, Inversion, and Interpretation of Seismic Data** - Oz Yilmaz (SEG). *Open access on the SEG Wiki (web, not PDF).* Open PDF/online: https://wiki.seg.org/wiki/Seismic_Data_Analysis. Contents: Appendix A, A43.
- Ellis & Singer, Well Logging for Earth Scientists (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GPGN519 Advanced Formation Evaluation (3 cr).** Well-logging tools, conversion of log measurements to rock and fluid properties, and ties of logs to seismic data. Source: https://catalog.mines.edu/coursesaz/geophyseng/
- **Colorado School of Mines - PEGN519 Advanced Formation Evaluation (3 cr).** Wireline logging methods and log interpretation for reservoir properties and volumetrics. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf

GE42 Seismic Data Processing and Imaging

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Geophysics (recommended); Petroleum Exploration and Engineering (optional)

Syllabus

Chapter 1. Seismic acquisition - **1.1** Land and marine geometries **1.2** Sources and receivers **1.3** Noise

Chapter 2. Signal processing - **2.1** Fourier analysis **2.2** Convolution and deconvolution **2.3** Filtering

Chapter 3. Pre-processing - **3.1** Geometry and trace editing **3.2** Static corrections **3.3** Noise attenuation

Chapter 4. Velocity analysis and stacking - **4.1** NMO and DMO **4.2** Velocity spectra **4.3** Stacking

Chapter 5. Migration - **5.1** Kirchhoff, phase-shift and finite-difference migration **5.2** Pre-stack time and depth migration **5.3** Reverse-time migration

Chapter 6. Advanced topics - **6.1** Multiple attenuation **6.2** AVO-preserving processing **6.3** Full-waveform inversion

Computer practice: Processing of a 2D line from raw shot gathers to migrated section (Seismic Unix or Madagascar)

Assessment and passing requirements

Assessment: Processing project 50%; final examination 50%.

To pass: Final mark at least 60/100; processed section and processing report submitted.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Seismic Data Analysis: Processing, Inversion, and Interpretation of Seismic Data** - Oz Yilmaz (SEG). *Open access on the SEG Wiki (web, not PDF).* Open PDF/online: https://wiki.seg.org/wiki/Seismic_Data_Analysis. Contents: Appendix A, A43.

Similar subjects at other universities

- **Colorado School of Mines - GPGN561 Seismic Data Processing I (4 cr).** Processing workflow for 2D and 3D reflection seismic data from velocity analysis and statics to imaging, with a Seismic Unix laboratory. Source: https://catalog.mines.edu/coursesaz/geophyseng/

GE43 Electrical and Electromagnetic Methods

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Geophysics (recommended)

Syllabus

Chapter 1. Electromagnetic theory - **1.1** Maxwell's equations **1.2** Diffusion and propagation regimes **1.3** Skin depth

Chapter 2. DC resistivity and IP - **2.1** Arrays **2.2** 2D/3D ERT **2.3** Spectral IP

Chapter 3. Controlled-source EM - **3.1** Frequency- and time-domain systems **3.2** Airborne EM **3.3** Marine CSEM

Chapter 4. Magnetotellurics - **4.1** Impedance tensor **4.2** Dimensionality **4.3** Static shift

Chapter 5. Ground-penetrating radar - **5.1** Propagation and attenuation **5.2** Data processing

Chapter 6. Forward modelling and inversion - **6.1** 1D layered earth **6.2** 2D/3D numerical modelling **6.3** Joint inversion

Chapter 7. Applications - **7.1** Minerals, groundwater, geothermal and engineering

Computer practice:

- ERT and TEM inversion
- MT data analysis with open tools

Field practice: Two-day ERT and TEM survey

Assessment and passing requirements

Assessment: Computer and field exercises 50%; final examination 50%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* Open PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Contents: Appendix A, A18.
- **SimPEG documentation and tutorials (Simulation and Parameter Estimation in Geophysics)** - SimPEG developers (open source). *Free web resource (MIT licence).* Open PDF/online: https://simpeg.xyz. Contents: Appendix A, A19.
- Zhdanov, Geophysical Electromagnetic Theory and Methods (Elsevier). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GPGN520 Electrical and Electromagnetic Exploration (3 cr).** Electromagnetic theory, instrumentation, survey planning, data processing and geologic interpretation of electrical methods. Source: https://catalog.mines.edu/coursesaz/geophyseng/

GE44 Computational Geophysics (numerical simulation of geophysical fields)

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Geophysics (recommended)

Syllabus

Chapter 1. Numerical solution of geophysical PDEs - **1.1** Finite differences for the wave equation **1.2** Staggered grids

Chapter 2. Finite-element and spectral-element methods - **2.1** Weak form **2.2** Spectral elements for seismic waves

Chapter 3. Potential-field and EM modelling - **3.1** Integral equations **3.2** FE/FV for EM diffusion

Chapter 4. Absorbing boundaries and sources - **4.1** PML **4.2** Source implementation

Chapter 5. Performance and parallel computing - **5.1** Vectorisation **5.2** MPI and GPU basics **5.3** Reproducible scientific software: *Git, testing*

Chapter 6. Imaging and inversion engines - **6.1** Adjoint methods **6.2** Reverse-time migration implementation

Computer practice:

- Own 2D acoustic and elastic FD wave codes
- FE solution of a DC resistivity problem
- Parallel run on a cluster

Assessment and passing requirements

Assessment: Programming projects 70%; oral examination 30%.

To pass: Final mark at least 60/100; codes pass the instructor's test cases.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Finite Difference Computing with PDEs: A Modern Software Approach** - H.P. Langtangen, S. Linge (Springer, Texts in Computational Science and Engineering 16). *Open access (CC BY 4.0), free PDF from Springer.* Open PDF/online: https://link.springer.com/book/10.1007/978-3-319-55456-3. Contents: Appendix A, A12.
- **Solving PDEs in Python: The FEniCS Tutorial I** - H.P. Langtangen, A. Logg (Springer, Simula SpringerBriefs). *Open access (CC BY 4.0), free PDF from Springer.* Open PDF/online: https://link.springer.com/book/10.1007/978-3-319-52462-7. Contents: Appendix A, A13.
- Igel, Computational Seismology: A Practical Introduction (Oxford). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GPGN536 Advanced Geophysical Computing I (3 cr).** 2D and 3D numerical solutions of geophysical PDEs and data-processing problems in high- and low-level languages. Source: https://catalog.mines.edu/coursesaz/geophyseng/
- **TU Bergakademie Freiberg (MSc Applied Geoscience) - Numerical Simulation Methods in Geophysics (6 LP).** Elective module in the computational and mathematical geoscience area. Source: https://tu-freiberg.de/sites/default/files/2025-12/english-taught-modules_applied_geoscience.pdf
- **China University of Petroleum (master's training plans) - Seismic Wave Simulation and Imaging.** Elective course in geophysics master's plans. Source: https://cie.upc.edu.cn/pdf/Master/dizhiziyuanyudizhigongcheng.pdf

GE45 Rock Physics and Petrophysics

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Geophysics (recommended); Petroleum Exploration and Engineering (recommended)

Syllabus

Chapter 1. Rock and pore-fluid properties - **1.1** Mineral moduli **1.2** Fluid properties: *Batzle-Wang* **1.3** Porosity and pore geometry

Chapter 2. Elastic bounds and effective media - **2.1** Voigt-Reuss-Hill **2.2** Hashin-Shtrikman **2.3** Inclusion models

Chapter 3. Fluid substitution - **3.1** Gassmann equation **3.2** Patchy saturation **3.3** Dispersion and attenuation

Chapter 4. Rock-physics templates - **4.1** Sand, shale and carbonate models **4.2** Velocity-porosity trends

Chapter 5. Rock mechanics links - **5.1** Static and dynamic moduli **5.2** Stress sensitivity

Chapter 6. Digital rock physics - **6.1** Micro-CT imaging **6.2** Pore-scale simulation

Chapter 7. Seismic applications - **7.1** AVO **7.2** Quantitative interpretation **7.3** 4D seismic

Laboratory practice:

- Ultrasonic velocity measurements on core
- Porosity and permeability tests

Computer practice: Fluid-substitution and rock-physics template exercises

Assessment and passing requirements

Assessment: Laboratory and computer exercises 50%; final examination 50%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Seismic Data Analysis: Processing, Inversion, and Interpretation of Seismic Data** - Oz Yilmaz (SEG). *Open access on the SEG Wiki (web, not PDF).* Open PDF/online: https://wiki.seg.org/wiki/Seismic_Data_Analysis. Contents: Appendix A, A43.
- **PetroWiki** - Society of Petroleum Engineers (based on the SPE Petroleum Engineering Handbook). *Free web resource.* Open PDF/online: https://petrowiki.spe.org. Contents: Appendix A, A42.
- Mavko, Mukerji & Dvorkin, The Rock Physics Handbook (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GPGN547 Physics, Mechanics and Petrophysics of Rocks (3 cr).** Rock physics, rock mechanics and petrophysics: elastic moduli, elastic bounds, seismic and log expression of formations. Source: https://catalog.mines.edu/coursesaz/geophyseng/
- **China University of Petroleum (master's training plans) - Digital Rock Physics.** Elective course in geological resources and engineering plans. Source: https://cie.upc.edu.cn/pdf/Master/dizhiziyuanyudizhigongcheng.pdf

GE46 Seismic Interpretation and Reservoir Characterisation

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: Project-based: 15 h lectures + 30 h studio/computer + about 90 h project work (= 135 h)

Used in: Geophysics (recommended); Petroleum Exploration and Engineering (recommended)

Syllabus

Chapter 1. Seismic stratigraphy - **1.1** Reflection terminations **1.2** Seismic facies **1.3** Sequence stratigraphy

Chapter 2. Structural interpretation - **2.1** Faults and folds **2.2** Salt and inversion structures **2.3** Depth conversion

Chapter 3. Seismic attributes - **3.1** Amplitude, frequency, coherence and curvature **3.2** Spectral decomposition

Chapter 4. Seismic inversion - **4.1** Post-stack and pre-stack inversion **4.2** AVO analysis

Chapter 5. Reservoir characterisation - **5.1** Facies classification **5.2** Property prediction **5.3** Integration with wells

Chapter 6. Prospect evaluation - **6.1** Mapping of traps **6.2** Volumetrics **6.3** Risk

Computer practice: 3D seismic interpretation project in OpendTect or Petrel-type software, with attribute and inversion analysis

Assessment and passing requirements

Assessment: Interpretation project 60%; final examination 40%.

To pass: Final mark at least 60/100; interpretation maps and report submitted.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Seismic Data Analysis: Processing, Inversion, and Interpretation of Seismic Data** - Oz Yilmaz (SEG). *Open access on the SEG Wiki (web, not PDF).* Open PDF/online: https://wiki.seg.org/wiki/Seismic_Data_Analysis. Contents: Appendix A, A43.
- Brown, Interpretation of Three-Dimensional Seismic Data (AAPG/SEG). *Commercial - no legal free PDF.* Library lookup: WorldCat

- Chopra & Marfurt, Seismic Attributes for Prospect Identification and Reservoir Characterization (SEG). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GPGN558 Seismic Data Interpretation and Quantitative Analysis (3 cr).** Links seismic data to rock properties, wavelets, tuning, attributes and inversion through team exercises in industry software. Source: https://catalog.mines.edu/coursesaz/geophyseng/
- **Colorado School of Mines - GEOL502 Structural Methods for Seismic Interpretation (3 cr).** Structural techniques for interpreting 2D/3D seismic data in complex areas, with a 4-day field trip. Source: https://catalog.mines.edu/coursesaz/geology
- **China University of Petroleum (master's training plans) - Reservoir Characterization.** Elective course in geological resources and engineering plans. Source: https://cie.upc.edu.cn/pdf/Master/dizhiziyuanyudizhigongcheng.pdf

GE47 Engineering, Environmental and Hydrogeophysics

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 15 h lectures + 2 weeks of field work + about 40 h report writing (= 135 h)

Used in: Geophysics (recommended)

Syllabus

Chapter 1. Near-surface geophysics - **1.1** Objectives and constraints **1.2** Resolution and depth of investigation

Chapter 2. Seismic methods - **2.1** Refraction tomography **2.2** MASW and passive surface waves **2.3** Shallow reflection

Chapter 3. Electrical and EM methods - **3.1** ERT for groundwater and contamination **3.2** EM induction **3.3** GPR

Chapter 4. Hydrogeophysics - **4.1** Petrophysical links to hydrology **4.2** Monitoring infiltration and solute transport **4.3** NMR

Chapter 5. Geotechnical and environmental applications - **5.1** Cavities and karst **5.2** Landfills and mine sites **5.3** Infrastructure inspection

Chapter 6. Archaeological and UXO surveys - **6.1** Magnetometry **6.2** GPR

Field practice: Five-day near-surface survey: refraction, MASW, ERT, GPR and magnetics at a test site

Assessment and passing requirements

Assessment: Field report 50%; final examination 50%.

To pass: Field participation and final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* Open PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Contents: Appendix A, A18.
- **Groundwater** - R. Allen Freeze, John A. Cherry (1979). *Free PDF from The Groundwater Project (preserved book).* Open PDF/online: https://gw-project.org/books/groundwater/. Contents: Appendix A, A44.
- Reynolds, An Introduction to Applied and Environmental Geophysics (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Everett, Near-Surface Applied Geophysics (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GPGN574 Advanced Hydrogeophysics (3 cr).** Electrical, seismic, logging and NMR methods for groundwater flow and solute transport, with forward and inverse modelling. Source: https://catalog.mines.edu/coursesaz/geophyseng/

GE48 Geophysical Field School and Instrumentation

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 15 h lectures + 2 weeks of field work + about 40 h report writing (= 135 h)

Used in: Geophysics (recommended)

Syllabus

Chapter 1. Instrumentation - **1.1** Sensors, data loggers and telemetry **1.2** Calibration **1.3** Noise and drift

Chapter 2. Field data management - **2.1** Data formats **2.2** Metadata **2.3** Field QA/QC

Chapter 3. Integrated interpretation - **3.1** Joint interpretation of methods **3.2** Final report and presentation

Field practice:

- Two-week geophysical field school: survey planning and permits
- Gravity and magnetic survey with differential GNSS
- Seismic refraction or passive seismic array
- ERT/IP and TEM
- Borehole logging if available
- Daily processing and quality control

Assessment and passing requirements

Assessment: Field notebook and QC 30%; report 50%; presentation 20%.

To pass: Participation in at least 90% of field days and final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Geophysics for Practicing Geoscientists (GPG)** - UBC Geophysical Inversion Facility. *Open web resource (CC BY).* Open PDF/online: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm. Contents: Appendix A, A18.
- Milsom & Eriksen, Field Geophysics (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GPGN590 Instrumental Design in Applied Geosciences (3 cr).** Teams design and build sensing prototypes (magnetometers, telemetered monitoring units) for applied geoscience and engineering. Source: https://catalog.mines.edu/coursesaz/geophyseng/

GE49 Petroleum Systems Analysis and Basin Modelling

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Petroleum Exploration and Engineering (recommended)

Syllabus

Chapter 1. Petroleum system elements - **1.1** Source, reservoir, seal, overburden **1.2** Trap formation **1.3** Timing

Chapter 2. Source rocks and geochemistry - **2.1** Kerogen types **2.2** Rock-Eval and TOC **2.3** Biomarkers **2.4** Oil-source correlation

Chapter 3. Basin analysis - **3.1** Subsidence and backstripping **3.2** Thermal history **3.3** Burial history

Chapter 4. Basin and petroleum system modelling - **4.1** 1D, 2D and 3D models **4.2** Generation, expulsion and migration **4.3** Calibration with vitrinite reflectance and temperature

Chapter 5. Play and prospect analysis - **5.1** Play fairway mapping **5.2** Common risk segment maps

Chapter 6. Risk and volumes - **6.1** Chance of success **6.2** Monte Carlo volumetrics

Chapter 7. Carbon storage and energy transition - **7.1** Storage plays **7.2** Hydrogen and geothermal in basins

Computer practice:

- 1D/2D basin model of a well and section
- Play risk mapping

Assessment and passing requirements

Assessment: Modelling project 50%; final examination 50%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **PetroWiki** - Society of Petroleum Engineers (based on the SPE Petroleum Engineering Handbook). *Free web resource.* Open PDF/online: https://petrowiki.spe.org. Contents: Appendix A, A42.
- Hantschel & Kauerauf, Fundamentals of Basin and Petroleum Systems Modeling (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Allen & Allen, Basin Analysis (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEOL575 Petroleum Systems Analysis (3 cr).** Petroleum-system analysis using petroleum geochemistry and basin modelling. Source: https://catalog.mines.edu/coursesaz/geology
- **Colorado School of Mines - GEOL550 Integrated Basin Modeling (3 cr).** Computer-based structural restoration, thermal and hydrocarbon-generation and stratigraphic modelling with seismic and well data. Source: https://catalog.mines.edu/coursesaz/geology
- **Colorado School of Mines - GEOL565 Risks and Volumes Assessment for Prospects and Plays (3 cr).** Translating geological knowledge into risk and petroleum volume estimates by deterministic and Monte Carlo methods. Source: https://catalog.mines.edu/coursesaz/geology

GE50 Advanced Reservoir Engineering

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 45 h lectures/seminars + about 90 h self-study (= 135 h)

Used in: Petroleum Exploration and Engineering (recommended)

Syllabus

Chapter 1. Reservoir fluids - **1.1** Phase behaviour **1.2** PVT experiments **1.3** Equations of state

Chapter 2. Rock-fluid properties - **2.1** Wettability **2.2** Capillary pressure **2.3** Relative permeability

Chapter 3. Material balance - **3.1** General material balance **3.2** Drive mechanisms **3.3** Aquifer influx

Chapter 4. Fluid flow in porous media - **4.1** Diffusivity equation **4.2** Radial flow **4.3** Well productivity

Chapter 5. Displacement - **5.1** Fractional flow **5.2** Buckley-Leverett **5.3** Sweep efficiency

Chapter 6. Reservoir performance forecasting - **6.1** Decline curves **6.2** Rate-transient analysis **6.3** Reserves classification: *SPE-PRMS*

Computer practice:

- Material balance and decline analysis of a field data set
- PVT matching

Assessment and passing requirements

Assessment: Assignments 40%; final examination 60%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **PetroWiki** - Society of Petroleum Engineers (based on the SPE Petroleum Engineering Handbook). *Free web resource.* Open PDF/online: https://petrowiki.spe.org. Contents: Appendix A, A42.
- **An Introduction to Reservoir Simulation Using MATLAB/GNU Octave (MRST user guide)** - K.-A. Lie (SINTEF; Cambridge University Press, 2019). *Open access (CC BY-NC-ND 4.0) on Cambridge Core.* Open PDF/online: https://www.cambridge.org/core/books/an-introduction-to-reservoir-simulation-using-matlabgnu-octave/introduction/C66C3A56440AA6B95D7F814CAD3CD971. Contents: Appendix A, A14.
- Dake, Fundamentals of Reservoir Engineering (Elsevier). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Ahmed, Reservoir Engineering Handbook (Gulf). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - PEGN515 Reservoir Engineering Principles (3 cr).** Hydrocarbons in place, material balance, well productivity, performance prediction, simulation fundamentals. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf
- **Colorado School of Mines - PEGN511 Advanced Thermodynamics and Petroleum Fluids Phase Behavior (3 cr).** Equations of state, PVT, asphaltenes and hydrates. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf

GE51 Reservoir Simulation

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Petroleum Exploration and Engineering (recommended)

Syllabus

Chapter 1. Mathematical models - **1.1** Mass conservation and Darcy's law **1.2** Black-oil and compositional formulations

Chapter 2. Discretisation - **2.1** Finite-volume methods **2.2** Grids: *Cartesian, corner-point, unstructured* **2.3** Well models: *Peaceman*

Chapter 3. Solution methods - **3.1** IMPES and fully implicit **3.2** Newton-Raphson **3.3** Linear solvers and preconditioners

Chapter 4. Multiphase flow - **4.1** Two- and three-phase flow **4.2** Upwinding **4.3** Numerical dispersion

Chapter 5. Upscaling and model building - **5.1** Static to dynamic model **5.2** Upscaling of permeability

Chapter 6. History matching and forecasting - **6.1** Manual and assisted history matching **6.2** Ensemble methods **6.3** Uncertainty

Chapter 7. Special processes - **7.1** Dual porosity **7.2** Thermal and CO2 storage simulation

Computer practice:

- Own 1D/2D two-phase simulator in Python
- Simulation studies in MRST and OPM Flow (open source)

Assessment and passing requirements

Assessment: Simulation projects 60%; final examination 40%.

To pass: Final mark at least 60/100; simulator verified against Buckley-Leverett solution.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **An Introduction to Reservoir Simulation Using MATLAB/GNU Octave (MRST user guide)** - K.-A. Lie (SINTEF; Cambridge University Press, 2019). *Open access (CC BY-NC-ND 4.0) on Cambridge Core.* Open PDF/online: https://www.cambridge.org/core/books/an-introduction-to-reservoir-simulation-using-matlabgnu-octave/introduction/C66C3A56440AA6B95D7F814CAD3CD971. Contents: Appendix A, A14.
- Aziz & Settari, Petroleum Reservoir Simulation (Elsevier). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Ertekin, Abou-Kassem & King, Basic Applied Reservoir Simulation (SPE). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - PEGN513 Reservoir Simulation I (3 cr).** Flow equations, finite-difference and control-volume methods, solvers and multiphase models. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf
- **Colorado School of Mines - PEGN541 Applied Reservoir Simulation (3 cr).** Using flow simulators to meet reservoir-management goals in an open-ended design problem. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf
- **Colorado School of Mines - PEGN550 Modern Reservoir Simulators (3 cr).** Commercial simulators with pre- and post-processors; black-oil and compositional models. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf

GE52 Advanced Drilling Engineering and Well Design

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Petroleum Exploration and Engineering (recommended)

Syllabus

Chapter 1. Well planning - **1.1** Pore and fracture pressure prediction **1.2** Casing setting depths **1.3** Well design basis

Chapter 2. Drilling hydraulics - **2.1** Rheology **2.2** Pressure losses **2.3** Hole cleaning **2.4** ECD management

Chapter 3. Directional and horizontal drilling - **3.1** Trajectory design **3.2** Survey calculations **3.3** Torque and drag

Chapter 4. Tubular design - **4.1** Casing and tubing design: *burst, collapse, tension* **4.2** Drill-string design

Chapter 5. Well control - **5.1** Kick detection **5.2** Kill methods **5.3** Managed pressure drilling

Chapter 6. Drilling optimisation and automation - **6.1** Bit selection **6.2** Rate of penetration models **6.3** Real-time drilling data

Chapter 7. Geothermal and scientific drilling - **7.1** High-temperature drilling **7.2** Coring

Computer practice:

- Well plan with casing design, hydraulics and torque-drag calculations
- Well-control simulator session

Assessment and passing requirements

Assessment: Well-design project 50%; final examination 50%.

To pass: Final mark at least 60/100; well-control simulator exercise passed.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **PetroWiki** - Society of Petroleum Engineers (based on the SPE Petroleum Engineering Handbook). *Free web resource.* Open PDF/online: https://petrowiki.spe.org. Contents: Appendix A, A42.
- **Handbook of Best Practices for Geothermal Drilling** - J. Finger, D. Blankenship (Sandia National Laboratories). *Free PDF (U.S. DOE).* Open PDF/online: https://www1.eere.energy.gov/geothermal/pdfs/drillinghandbook.pdf. Contents: Appendix A, A45.
- Bourgoyne, Millheim, Chenevert & Young, Applied Drilling Engineering (SPE). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Mitchell & Miska, Fundamentals of Drilling Engineering (SPE). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - PEGN517 Drilling Engineering Principles (3 cr).** Hydraulics, drill string and bit selection, pore pressure, casing design, cementing and directional drilling. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf
- **Colorado School of Mines - PEGN594 Advanced Directional Drilling (3 cr).** Directional and horizontal well planning, survey and trajectory calculations, collision diagrams. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf
- **Colorado School of Mines - PEGN596 Advanced Well Control (3 cr).** Pressure control, blowout equipment, kick control and abnormal-pressure detection on a full-scale simulator. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf

GE53 Reservoir Geomechanics

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Petroleum Exploration and Engineering (recommended)

Syllabus

Chapter 1. Stress in the crust - **1.1** Stress magnitudes and orientation **1.2** Pore pressure

Chapter 2. Rock mechanical properties - **2.1** Laboratory tests **2.2** Log-derived properties

Chapter 3. Wellbore stability - **3.1** Breakouts and tensile fractures **3.2** Mud-weight window

Chapter 4. Faults and fractures - **4.1** Critically stressed faults **4.2** Fault reactivation

Chapter 5. Depletion and injection effects - **5.1** Compaction and subsidence **5.2** Induced seismicity **5.3** Caprock integrity in CO2 storage

Chapter 6. Hydraulic fracturing - **6.1** Fracture mechanics **6.2** Fracture design **6.3** Microseismic monitoring

Chapter 7. Geomechanical modelling - **7.1** 1D mechanical earth models **7.2** 3D coupled models

Computer practice:

- 1D mechanical earth model and mud-weight window for a well
- Fault reactivation analysis

Assessment and passing requirements

Assessment: Project 50%; final examination 50%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **PetroWiki** - Society of Petroleum Engineers (based on the SPE Petroleum Engineering Handbook). *Free web resource.* Open PDF/online: https://petrowiki.spe.org. Contents: Appendix A, A42.
- **Practical Rock Engineering** - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).* Open PDF/online: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf. Contents: Appendix A, A34.
- Zoback, Reservoir Geomechanics (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - PEGN590 Reservoir Geomechanics (3 cr).** Stress, pore pressure, fracture gradient, rock failure, wellbore stability and hydraulic fracturing. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf

GE54 Enhanced Oil Recovery and CO2 Storage

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 45 h lectures/seminars + about 90 h self-study (= 135 h)

Used in: Petroleum Exploration and Engineering (recommended)

Syllabus

Chapter 1. Recovery mechanisms - **1.1** Displacement and sweep efficiency **1.2** Mobility ratio **1.3** Capillary number

Chapter 2. Waterflooding - **2.1** Design and surveillance **2.2** Smart water

Chapter 3. Gas injection - **3.1** Miscibility: *MMP* **3.2** Gas and WAG injection

Chapter 4. Chemical EOR - **4.1** Polymer, surfactant and alkaline flooding

Chapter 5. Thermal EOR - **5.1** Steam injection **5.2** In-situ combustion

Chapter 6. CO2-EOR and CCUS - **6.1** CO2 storage mechanisms **6.2** Monitoring, measurement and verification

Chapter 7. Screening and economics - **7.1** EOR screening criteria **7.2** Pilot design

Computer practice:

- Simulation of waterflood and polymer flood
- EOR screening exercise

Assessment and passing requirements

Assessment: Project 40%; final examination 60%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **PetroWiki** - Society of Petroleum Engineers (based on the SPE Petroleum Engineering Handbook). *Free web resource.* Open PDF/online: https://petrowiki.spe.org. Contents: Appendix A, A42.
- **An Introduction to Reservoir Simulation Using MATLAB/GNU Octave (MRST user guide)** - K.-A. Lie (SINTEF; Cambridge University Press, 2019). *Open access (CC BY-NC-ND 4.0) on Cambridge Core.* Open PDF/online: https://www.cambridge.org/core/books/an-introduction-to-reservoir-simulation-using-matlabgnu-octave/introduction/C66C3A56440AA6B95D7F814CAD3CD971. Contents: Appendix A, A14.

- Lake, Johns, Rossen & Pope, Fundamentals of Enhanced Oil Recovery (SPE). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - PEGN506 Enhanced Oil Recovery Methods (3 cr).** EOR mechanisms and design procedures reviewed with field case histories. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf

GE55 Advanced Production Engineering

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 45 h lectures/seminars + about 90 h self-study (= 135 h)

Used in: Petroleum Exploration and Engineering (recommended)

Syllabus

Chapter 1. Inflow performance - **1.1** IPR models **1.2** Skin **1.3** Horizontal and multi-fractured wells

Chapter 2. Multiphase flow in wells and pipelines - **2.1** Flow regimes **2.2** Correlations and mechanistic models

Chapter 3. Nodal analysis - **3.1** System design **3.2** Production optimisation

Chapter 4. Artificial lift - **4.1** Gas lift **4.2** ESP **4.3** Rod pump **4.4** Selection

Chapter 5. Well stimulation and completion - **5.1** Matrix acidising **5.2** Sand control **5.3** Completion design

Chapter 6. Flow assurance and facilities - **6.1** Hydrates, wax and scale **6.2** Separation and processing

Computer practice: Nodal analysis and artificial-lift design with open or commercial software

Assessment and passing requirements

Assessment: Design project 40%; final examination 60%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **PetroWiki** - Society of Petroleum Engineers (based on the SPE Petroleum Engineering Handbook). *Free web resource.* Open PDF/online: https://petrowiki.spe.org. Contents: Appendix A, A42.
- Economides, Hill, Ehlig-Economides & Zhu, Petroleum Production Systems (Prentice Hall). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - PEGN518 Advanced Production Engineering (3 cr).** Nodal analysis, artificial-lift design, flow assurance and multiphase simulators. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf

GE56 Well Test and Rate-Transient Analysis

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Petroleum Exploration and Engineering (recommended)

Syllabus

Chapter 1. Pressure transient theory - **1.1** Diffusivity equation **1.2** Superposition

Chapter 2. Test types - **2.1** Drawdown, build-up, injection and interference tests **2.2** DST

Chapter 3. Diagnostic analysis - **3.1** Log-log derivative **3.2** Flow regimes

Chapter 4. Reservoir and boundary models - **4.1** Wellbore storage and skin **4.2** Fractured wells **4.3** Boundaries **4.4** Dual porosity

Chapter 5. Gas wells and multiphase - **5.1** Pseudo-pressure **5.2** Deliverability

Chapter 6. Rate-transient analysis - **6.1** Production data analysis **6.2** Unconventional wells

Computer practice:

- Interpretation of field build-up and interference tests
- Rate-transient analysis

Assessment and passing requirements

Assessment: Interpretation project 50%; final examination 50%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **PetroWiki** - Society of Petroleum Engineers (based on the SPE Petroleum Engineering Handbook). *Free web resource.* Open PDF/online: https://petrowiki.spe.org. Contents: Appendix A, A42.
- Horne, Modern Well Test Analysis (Petroway). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Bourdet, Well Test Analysis: The Use of Advanced Interpretation Models (Elsevier). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - PEGN515 Reservoir Engineering Principles (3 cr).** Hydrocarbons in place, material balance, well productivity, performance prediction, simulation fundamentals. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf

GE57 Petroleum Economics, Risk and Decision Analysis

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: Project-based: 15 h lectures + 30 h studio/computer + about 90 h project work (= 135 h)

Used in: Petroleum Exploration and Engineering (recommended)

Syllabus

Chapter 1. Petroleum fiscal systems - **1.1** Royalty-tax and production-sharing contracts

Chapter 2. Cash-flow analysis - **2.1** Capital and operating costs **2.2** NPV, IRR and payback

Chapter 3. Decision analysis - **3.1** Decision trees **3.2** Expected value **3.3** Value of information

Chapter 4. Risk and uncertainty - **4.1** Monte Carlo simulation **4.2** Portfolio analysis

Chapter 5. Reserves and resources - **5.1** SPE-PRMS classification **5.2** Reporting

Chapter 6. Energy transition - **6.1** Carbon pricing **6.2** Economics of CCS and hydrogen

Computer practice: Field development economic model with risk analysis in a spreadsheet or Python

Assessment and passing requirements

Assessment: Economic evaluation project 50%; final examination 50%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **PetroWiki** - Society of Petroleum Engineers (based on the SPE Petroleum Engineering Handbook). *Free web resource.* Open PDF/online: https://petrowiki.spe.org. Contents: Appendix A, A42.
- Mian, Project Economics and Decision Analysis (PennWell). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - PEGN523 Advanced Economic Analysis of Oil and Gas Projects (3 cr).** Present value, taxes, profitability measures, decision trees and Monte Carlo methods. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf
- **Colorado School of Mines - GEOL565 Risks and Volumes Assessment for Prospects and Plays (3 cr).** Translating geological knowledge into risk and petroleum volume estimates by deterministic and Monte Carlo methods. Source: https://catalog.mines.edu/coursesaz/geology

GE58 Unconventional Oil and Gas Resources

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 45 h lectures/seminars + about 90 h self-study (= 135 h)

Used in: Petroleum Exploration and Engineering (optional)

Syllabus

Chapter 1. Unconventional resources - **1.1** Shale gas and oil **1.2** Tight gas **1.3** Coalbed methane **1.4** Gas hydrates

Chapter 2. Characterisation - **2.1** Organic richness and maturity **2.2** Mineralogy and brittleness **2.3** Nanopore storage

Chapter 3. Geomechanics and completions - **3.1** Horizontal wells and multi-stage fracturing **3.2** Stimulated reservoir volume

Chapter 4. Production and forecasting - **4.1** Flow regimes **4.2** Rate-transient and decline analysis

Chapter 5. Environmental issues - **5.1** Water management **5.2** Induced seismicity **5.3** Methane emissions

Assessment and passing requirements

Assessment: Assignment 40%; final examination 60%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **PetroWiki** - Society of Petroleum Engineers (based on the SPE Petroleum Engineering Handbook). *Free web resource.* Open PDF/online: https://petrowiki.spe.org. Contents: Appendix A, A42.
- Zoback & Kohli, Unconventional Reservoir Geomechanics (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEOL552 Unconventional Petroleum Systems (3 cr).** Main unconventional oil and gas systems, technology, environmental issues and resource estimates, with core study and field trip. Source: https://catalog.mines.edu/coursesaz/geology

- **Colorado School of Mines - PEGN591 Shale Reservoir Engineering (3 cr).** Shale production, fractured horizontal-well flow, dual porosity and rate-transient analysis. Source: https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf

GE59 Satellite Geodesy and GNSS Positioning

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Geographic Information and Space (recommended)

Syllabus

Chapter 1. Reference systems - **1.1** Celestial and terrestrial frames: *ICRF, ITRF* **1.2** Time systems **1.3** Datums and transformations

Chapter 2. GNSS signals and systems - **2.1** GPS, GLONASS, Galileo, BeiDou **2.2** Signal structure **2.3** Satellite orbits and ephemerides

Chapter 3. Observation equations - **3.1** Pseudorange and carrier phase **3.2** Error sources: *clock, ionosphere, troposphere, multipath*

Chapter 4. Positioning methods - **4.1** Single point **4.2** Differential and RTK **4.3** Precise point positioning

Chapter 5. Ambiguity resolution and estimation - **5.1** Least squares and Kalman filtering **5.2** LAMBDA method

Chapter 6. Geodetic applications - **6.1** Crustal deformation and plate motion **6.2** Monitoring of structures and slopes **6.3** GNSS meteorology

Chapter 7. Gravity field and heights - **7.1** Geoid **7.2** Satellite gravimetry **7.3** Height systems

Computer practice:

- Processing of GNSS observations with RTKLIB
- PPP solution
- Time-series analysis of a CORS station

Field practice: Static and RTK survey of a control network

Assessment and passing requirements

Assessment: Laboratory and field reports 40%; final examination 60%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **GNSS Data Processing, Vol. I: Fundamentals and Algorithms (ESA TM-23/1)** - J. Sanz Subirana, J.M. Juan Zornoza, M. Hernandez-Pajares (ESA). *Free PDF from ESA Navipedia.* Open PDF/online: https://gssc.esa.int/navipedia/GNSS_Book/ESA_GNSS-Book_TM-23_Vol_I.pdf. Contents: Appendix A, A46.
- Hofmann-Wellenhof, Lichtenegger & Wasle, GNSS - Global Navigation Satellite Systems (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Teunissen & Montenbruck (eds), Springer Handbook of Global Navigation Satellite Systems (Springer). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Massachusetts Institute of Technology (MIT OpenCourseWare) - 12.540 Principles of the Global Positioning System (graduate, Spring 2012).** GPS signals and measurement models, coordinate and time systems, orbits, estimation and error sources, precise geodetic applications. Source: https://ocw.mit.edu/courses/12-540-principles-of-the-global-positioning-system-spring-2012/pages/syllabus/

GE60 Orbital Mechanics and Space Mission Analysis

Credits: 3 | **Type:** Elective | **Offered:** Semester 1

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Geographic Information and Space (recommended)

Syllabus

Chapter 1. Two-body problem - **1.1** Kepler's laws **1.2** Orbital elements **1.3** Conic sections

Chapter 2. Orbit propagation - **2.1** Kepler's equation **2.2** Numerical propagation **2.3** Coordinate frames

Chapter 3. Perturbations - **3.1** Earth oblateness: *J2* **3.2** Atmospheric drag **3.3** Solar radiation pressure **3.4** Third-body effects

Chapter 4. Earth-observation orbits - **4.1** Sun-synchronous and repeat-ground-track orbits **4.2** Constellations and coverage

Chapter 5. Orbital manoeuvres - **5.1** Hohmann and bi-elliptic transfers **5.2** Plane changes **5.3** Rendezvous basics

Chapter 6. Interplanetary trajectories - **6.1** Patched conics **6.2** Launch windows

Chapter 7. Mission analysis - **7.1** Link budgets and access times **7.2** Space debris and collision avoidance

Computer practice: Orbit propagation and coverage analysis in Python (poliastro or Orekit) or GMAT

Assessment and passing requirements

Assessment: Assignments 40%; final examination 60%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Basics of Space Flight** - NASA Jet Propulsion Laboratory. *Free (U.S. Government work); web tutorial.* Open PDF/online: https://science.nasa.gov/learn/basics-of-space-flight/participants-guide. Contents: Appendix A, A47.
- Curtis, Orbital Mechanics for Engineering Students (Elsevier). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Vallado, Fundamentals of Astrodynamics and Applications (Microcosm). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Massachusetts Institute of Technology (MIT OpenCourseWare) - 16.346 Astrodynamics (graduate, Fall 2008).** Two-body problem, orbit determination, Lambert's problem, perturbations and space-mission trajectories. Source: https://ocw.mit.edu/courses/16-346-astrodynamics-fall-2008/pages/syllabus
- **Colorado School of Mines - SPRS505 Space Operations (3 cr).** Orbital mechanics, space environments and mission design applied to a preliminary mission design. Source: https://catalog.mines.edu/coursesaz/spaceresources/spaceresources.pdf

GE61 Spatial Databases and WebGIS Development

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: Project-based: 15 h lectures + 30 h studio/computer + about 90 h project work (= 135 h)

Used in: Geographic Information and Space (recommended)

Syllabus

Chapter 1. Spatial database concepts - **1.1** Spatial data types and indexing: *R-tree* **1.2** Spatial SQL

Chapter 2. PostGIS - **2.1** Geometry and raster functions **2.2** Topology **2.3** Performance tuning

Chapter 3. Spatial data infrastructures - **3.1** OGC standards: *WMS, WFS, WCS, OGC API* **3.2** Metadata catalogues

Chapter 4. WebGIS architecture - **4.1** GeoServer **4.2** Tile servers and vector tiles **4.3** Cloud-native formats: *COG, GeoParquet*

Chapter 5. Web mapping programming - **5.1** JavaScript libraries: *Leaflet, OpenLayers, MapLibre* **5.2** REST APIs

Chapter 6. Big geodata - **6.1** Distributed processing **6.2** Spatio-temporal databases

Computer practice: Build a PostGIS database and a web map application for a geoscience data set

Assessment and passing requirements

Assessment: Application project 60%; final test 40%.

To pass: Final mark at least 60/100; working application deployed and documented.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Database Design (2nd ed.)** - Adrienne Watt, Nelson Eng (BCcampus). *CC BY; free PDF.* Open PDF/online: https://yc-prod-openlibrary-repo.ecampusontario.ca/jspui/bitstream/123456789/247/16/Database-Design-2nd-Edition-1540829576.pdf. Contents: Appendix A, A48.
- **Spatial Data Science: With Applications in R** - E. Pebesma, R. Bivand (CRC Press). *Free online edition from the authors.* Open PDF/online: https://r-spatial.org/book/. Contents: Appendix A, A11.
- Obe & Hsu, PostGIS in Action (Manning). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEGN579 Python Scripting for GIS (3 cr).** Python with ArcGIS to automate geoprocessing and build standalone GIS scripts and tools. Source: https://catalog.mines.edu/coursesaz/geoleng/

GE62 Photogrammetry, LiDAR and UAV Mapping

Credits: 3 | **Type:** Elective | **Offered:** Semester 2

Workload: 15 h lectures + 2 weeks of field work + about 40 h report writing (= 135 h)

Used in: Geographic Information and Space (recommended)

Syllabus

Chapter 1. Photogrammetry principles - **1.1** Camera models and calibration **1.2** Collinearity **1.3** Stereo geometry

Chapter 2. Aerial and UAV photogrammetry - **2.1** Flight planning **2.2** Ground control **2.3** Structure from motion and dense matching

Chapter 3. Products - **3.1** Orthophotos **3.2** DSM and DTM **3.3** Accuracy assessment

Chapter 4. LiDAR - **4.1** Airborne, terrestrial and mobile laser scanning **4.2** Point-cloud processing **4.3** Classification and ground filtering

Chapter 5. Applications - **5.1** Topographic mapping **5.2** Mine and quarry volume surveys **5.3** Geological outcrop models **5.4** Change detection

Chapter 6. Standards and quality - **6.1** Accuracy standards **6.2** Data deliverables

Field practice:

- UAV photogrammetry survey with ground control
- Terrestrial laser scan of an outcrop or structure

Computer practice: SfM processing and point-cloud analysis (open software such as CloudCompare, OpenDroneMap)

Assessment and passing requirements

Assessment: Field and processing project 60%; final test 40%.

To pass: Final mark at least 60/100; drone safety and regulation training completed.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Fundamentals of Remote Sensing (tutorial)** - Canada Centre for Remote Sensing / Natural Resources Canada. *Free web tutorial.* Open PDF/online: https://natural-resources.canada.ca/science-data/science-research/geomatics/remote-sensing/tutorial-fundamentals-remote-sensing. Contents: Appendix A, A25.
- Wolf, Dewitt & Wilkinson, Elements of Photogrammetry with Applications in GIS (McGraw-Hill). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Vosselman & Maas (eds), Airborne and Terrestrial Laser Scanning (Whittles). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEGN542 Advanced Digital Terrain Analysis (3 cr).** GIS and remote sensing to model the Earth's surface and assess hazards such as erosion and landslides. Source: https://catalog.mines.edu/coursesaz/geoleng/

GE63 Spacecraft Systems Engineering and Small Satellites

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: Project-based: 15 h lectures + 30 h studio/computer + about 90 h project work (= 135 h)

Used in: Geographic Information and Space (recommended)

Syllabus

Chapter 1. Space systems engineering - **1.1** Life cycle and reviews **1.2** Requirements **1.3** Trade studies

Chapter 2. Spacecraft subsystems - **2.1** Structures **2.2** Power **2.3** Thermal control **2.4** Attitude determination and control **2.5** Communications **2.6** On-board data handling **2.7** Propulsion

Chapter 3. Small satellites - **3.1** CubeSat standards **3.2** Commercial components **3.3** Launch and deployment

Chapter 4. Payloads for Earth observation - **4.1** Optical, radar and other sensors **4.2** Payload sizing

Chapter 5. Ground segment and operations - **5.1** Ground stations **5.2** Mission operations **5.3** Data downlink and processing

Chapter 6. Assembly, integration and test - **6.1** Environmental testing **6.2** Verification

Chapter 7. Space law and policy - **7.1** Outer Space Treaty **7.2** Licensing **7.3** Space debris mitigation

Project work: Team preliminary design of a small Earth-observation or space-resources mission with a design review

Assessment and passing requirements

Assessment: Team design project 60%; individual examination 40%.

To pass: Final mark at least 60/100; individual contribution satisfactory in peer assessment.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **NASA Systems Engineering Handbook (SP-2016-6105 Rev 2)** - NASA. *Free PDF (U.S. Government).* Open PDF/online: https://www.nasa.gov/wp-content/uploads/2018/09/nasa_systems_engineering_handbook_0.pdf. Contents: Appendix A, A49.
- **State-of-the-Art of Small Spacecraft Technology** - NASA Small Spacecraft Systems Virtual Institute. *Free (U.S. Government), web and PDF.* Open PDF/online: https://www.nasa.gov/smallsat-institute/sst-soa/. Contents: Appendix A, A50.
- **CubeSat 101: Basic Concepts and Processes for First-Time CubeSat Developers** - NASA CubeSat Launch Initiative (2017). *Free PDF (U.S. Government work).* Open PDF/online: https://www.nasa.gov/wp-content/uploads/2017/03/nasa_csli_cubesat_101_508.pdf. Contents: Appendix A, A51.
- Wertz, Everett & Puschell (eds), Space Mission Engineering: The New SMAD (Microcosm). *Commercial - no legal free PDF.* Library lookup: WorldCat
- Fortescue, Swinerd & Stark (eds), Spacecraft Systems Engineering (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Massachusetts Institute of Technology (MIT OpenCourseWare) - 16.851 Satellite Engineering (graduate, Fall 2003).** Satellite subsystems (power, thermal, attitude, communications, propulsion, structures) and spacecraft design projects. Source: https://ocw.mit.edu/courses/16-851-satellite-engineering-fall-2003/pages/syllabus
- **Colorado School of Mines - SPRS502 Space Systems Engineering (3 cr).** Systems-engineering process for large space systems from requirements through design, analysis and verification. Source: https://catalog.mines.edu/coursesaz/spaceresources/spaceresources.pdf
- **Colorado School of Mines - SPRS506 International Space Law and Policy (3 cr).** Fundamentals of international space law and analysis of space-law and policy issues. Source: https://catalog.mines.edu/coursesaz/spaceresources/spaceresources.pdf

GE64 Digital Terrain Analysis and Geohazard Modelling

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 30 h lectures + 30 h laboratory/computer + about 75 h self-study (= 135 h)

Used in: Geographic Information and Space (recommended)

Syllabus

Chapter 1. Digital elevation models - **1.1** Data sources: *SRTM, Copernicus DEM, LiDAR* **1.2** Errors and resolution

Chapter 2. Geomorphometry - **2.1** Slope, aspect and curvature **2.2** Terrain indices **2.3** Landform classification

Chapter 3. Hydrological modelling - **3.1** Flow direction and accumulation **3.2** Catchments **3.3** Flood inundation modelling

Chapter 4. Geohazard susceptibility - **4.1** Landslide, flood and erosion susceptibility **4.2** Statistical and machine-learning models

Chapter 5. Process-based hazard models - **5.1** Physically based slope stability: *infinite slope, TRIGRS-type* **5.2** Run-out and debris-flow models

Chapter 6. Risk and early warning - **6.1** Exposure and vulnerability **6.2** Risk maps **6.3** Monitoring and warning systems

Computer practice:

- Terrain analysis and susceptibility mapping in QGIS/WhiteboxTools
- Flood or run-out simulation

Assessment and passing requirements

Assessment: Project 60%; final test 40%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Essentials of Geographic Information Systems** - Jonathan Campbell, Michael Shin. *CC BY-NC-SA; free (Open Textbook Library).* Open PDF/online: https://open.umn.edu/opentextbooks/textbooks/67. Contents: Appendix A, A21.
- **The Landslide Handbook - A Guide to Understanding Landslides (USGS Circular 1325)** - L.M. Highland, P. Bobrowsky (2008). *Public domain; free PDF.* Open PDF/online: https://pubs.usgs.gov/publication/cir1325. Contents: Appendix A, A37.
- Wilson & Gallant (eds), Terrain Analysis: Principles and Applications (Wiley). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - GEGN542 Advanced Digital Terrain Analysis (3 cr).** GIS and remote sensing to model the Earth's surface and assess hazards such as erosion and landslides. Source: https://catalog.mines.edu/coursesaz/geoleng/

GE65 Planetary Geology and Space Resources

Credits: 3 | **Type:** Elective | **Offered:** Semester 3

Workload: 45 h lectures/seminars + about 90 h self-study (= 135 h)

Used in: Geographic Information and Space (recommended)

Syllabus

Chapter 1. Solar system formation and planetary interiors - **1.1** Accretion and differentiation **1.2** Interior structure

Chapter 2. Planetary surface processes - **2.1** Impact cratering **2.2** Volcanism **2.3** Tectonics **2.4** Aeolian, fluvial and glacial processes

Chapter 3. Planetary remote sensing - **3.1** Imaging and spectroscopy missions **3.2** Planetary GIS and mapping

Chapter 4. The Moon and Mars - **4.1** Geology and stratigraphy **4.2** Regolith **4.3** Water ice

Chapter 5. Space resources - **5.1** Resource types: *water, regolith, metals* **5.2** Prospecting **5.3** In-situ resource utilisation

Chapter 6. Planetary exploration and law - **6.1** Missions **6.2** Planetary protection **6.3** Space resources law

Computer practice: Geological mapping of a lunar or Martian region from orbital data in GIS

Assessment and passing requirements

Assessment: Mapping project 40%; final examination 60%.

To pass: Final mark at least 60/100.

General rules (Part 2.5) also apply: at least 80% attendance and one resit in the following semester.

Recommended textbooks

- **Lunar Sourcebook: A User's Guide to the Moon** - G. Heiken, D. Vaniman, B. French (eds) (Cambridge, 1991; LPI). *Free PDF from the Lunar and Planetary Institute.* Open PDF/online: https://www.lpi.usra.edu/publications/books/lunar_sourcebook/. Contents: Appendix A, A52.
- Melosh, Planetary Surface Processes (Cambridge). *Commercial - no legal free PDF.* Library lookup: WorldCat

Similar subjects at other universities

- **Colorado School of Mines - SPRS507 Advanced Planetary Geology (3 cr).** Differentiation, impact cratering, volcanism and other processes that shape planetary bodies and form space resources. Source:

https://catalog.mines.edu/coursesaz/spaceresources/spaceresources.pdf

- **Colorado School of Mines - GEGN588 Advanced Planetary GIS (3 cr).** Planetary GIS data sets, mapping, geospatial analysis and digital terrain modelling (online, project based). Source: https://catalog.mines.edu/coursesaz/geoleng/
- **Colorado School of Mines - SPRS501 Space Resources Fundamentals (3 cr).** Known solar-system resources, extraction systems, feasibility studies, legal issues and resource-enabled exploration. Source: https://catalog.mines.edu/coursesaz/spaceresources/spaceresources.pdf

Appendix A. Open-access textbooks: links and tables of contents

Only legally free editions are listed (open licence, author- or publisher-hosted free edition, or government publication). Where a table of contents is not given, it could not be checked for this report; open the link for the current contents.

A01 Academic Phrasebank - J. Morley (University of Manchester). *Free web resource.*

Link: https://www.phrasebank.manchester.ac.uk/ | Used for: GC01, GM07, GM08

Table of contents: not checked for this report - see the contents page at the link.

A02 Technical Writing Essentials - S. Last (University of Victoria; BCcampus). *CC BY 4.0; free web and PDF.*

Link: https://pressbooks.bccampus.ca/technicalwriting/ | Used for: GC01, GC03, GM02, GM08

Table of contents: not checked for this report - see the contents page at the link.

A03 Python Programming and Numerical Methods: A Guide for Engineers and Scientists - Q. Kong, T. Siauw, A. Bayen (UC Berkeley; Elsevier). *Free online edition from the authors (print edition commercial).*

Link: https://bayen.berkeley.edu/python-programming-and-numerical-methods | Used for: GC02

Table of contents: **1** Python Basics **2** Variables and Basic Data Structures **3** Functions **4** Branching Statements **5** Iteration **6** Recursion **7** Object Oriented Programming **8** Complexity **9** Representation of Numbers **10** Errors, Good Programming Practices, and Debugging **11** Reading and Writing Data **12** Visualization and Plotting **13** Parallel Your Python **14** Linear Algebra and Systems of Linear Equations **15** Eigenvalues and Eigenvectors **16** Least Squares Regression **17** Interpolation **18** Series **19** Root Finding **20** Numerical Differentiation **21** Numerical Integration **22** Ordinary Differential Equations: Initial-Value Problems **23** Boundary-Value Problems for ODEs **24** Fourier Transform **25** Introduction to Machine Learning

A04 Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares - S. Boyd, L. Vandenberghe (Cambridge University Press). *Free PDF from the authors' site.*

Link: https://web.stanford.edu/~boyd/vmls/vmls.pdf | Used for: GC02

Table of contents: **1** Vectors **2** Linear functions **3** Norm and distance **4** Clustering **5** Linear independence **6** Matrices **7** Matrix examples **8** Linear equations **9** Linear dynamical systems **10** Matrix multiplication **11** Matrix inverses **12** Least squares **13** Least squares data fitting **14** Least squares classification **15** Multi-objective least squares **16** Constrained least squares **17** Constrained least squares applications **18** Nonlinear least squares **19** Constrained nonlinear least squares

A05 Convex Optimization - S. Boyd, L. Vandenberghe (Cambridge University Press). *Free PDF from the authors' site.*

Link: https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf | Used for: GC02, GE31

Table of contents: **1** Introduction **2** Convex sets **3** Convex functions **4** Convex optimization problems **5** Duality **6** Approximation and fitting **7** Statistical estimation **8** Geometric problems **9** Unconstrained minimization **10** Equality constrained minimization **11** Interior-point methods

A06 Notes on Diffy Qs: Differential Equations for Engineers - Jiri Lebl. *CC BY-SA / CC BY-NC-SA; free PDF.*

Link: https://jirka.org/diffyqs/diffyqs.pdf | Used for: GC02

Table of contents: **1** Introduction **2** First-order equations **3** Higher-order linear ODEs **4** Systems of ODEs **5** Fourier series and PDEs **6** More on eigenvalue problems **7** The Laplace transform **8** Power-series methods **9** Nonlinear systems **10** Appendix A: Linear algebra

A07 On Being a Scientist: A Guide to Responsible Conduct in Research (3rd ed.) - National Academy of Sciences, NAE, IOM. *Free PDF from National Academies Press.*

Link: https://nap.nationalacademies.org/catalog/12192 | Used for: GC03, GM01, GM02

Table of contents: **1** Introduction to the Responsible Conduct of Research **2** Advising and Mentoring **3** The Treatment of Data **4** Mistakes and Negligence **5** Research Misconduct **6** Responding to Suspected Violations of Professional Standards **7** Human Participants and Animal Subjects in Research **8** Laboratory Safety in Research **9** Sharing of Research Results **10** Authorship and the Allocation of Credit **11** Intellectual Property **12** Competing Interests, Commitments, and Values **13** The Researcher in Society

A08 An Introduction to Statistical Learning (2nd ed.; R and Python editions) - James, Witten, Hastie, Tibshirani (Springer). *Free PDF from the authors' site.*

Link: https://www.statlearning.com | Used for: GC04, GE13, GE15

Table of contents: **1** Introduction **2** Statistical Learning **3** Linear Regression **4** Classification **5** Resampling Methods **6** Linear Model Selection and Regularization **7** Moving Beyond Linearity **8** Tree-Based Methods **9** Support Vector Machines **10** Deep Learning **11** Survival Analysis and Censored Data **12** Unsupervised Learning **13** Multiple Testing

A09 OpenIntro Statistics (4th ed.) - Diez, Cetinkaya-Rundel, Barr. *CC BY-SA; free PDF (pay-what-you-want, $0 minimum).*

Link: https://leanpub.com/os | Used for: GC04, GE02, GE26

Table of contents: not checked for this report - see the contents page at the link.

A10 Geostatistics Lessons - Centre for Computational Geostatistics, University of Alberta (C.V. Deutsch et al.). *Free peer-reviewed online lessons.*

Link: https://geostatisticslessons.com | Used for: GC04, GE10

Table of contents: not checked for this report - see the contents page at the link.

A11 Spatial Data Science: With Applications in R - E. Pebesma, R. Bivand (CRC Press). *Free online edition from the authors.*

Link: https://r-spatial.org/book/ | Used for: GC04, GE05, GE13, GE61

Table of contents: **1** Getting Started **2** Coordinates **3** Geometries **4** Spherical Geometries **5** Attributes and Support **6** Data Cubes **7** Introduction to sf and stars **8** Plotting spatial data **9** Large data and cloud native **10** Statistical modelling of spatial data **11** Point Pattern Analysis **12** Spatial Interpolation **13** Multivariate and Spatiotemporal Geostatistics **14** Proximity and Areal Data **15** Measures of Spatial Autocorrelation **16** Spatial Regression **17** Spatial Econometrics Models

A12 Finite Difference Computing with PDEs: A Modern Software Approach - H.P. Langtangen, S. Linge (Springer, Texts in Computational Science and Engineering 16). *Open access (CC BY 4.0), free PDF from Springer.*

Link: https://link.springer.com/book/10.1007/978-3-319-55456-3 | Used for: GC05, GE18, GE44

Table of contents: **1** Vibration ODEs **2** Wave Equations **3** Diffusion Equations **4** Advection-Dominated Equations **5** Nonlinear Problems **6** Appendix: Useful Formulas **7** Appendix: Truncation Error Analysis **8** Appendix: Software Engineering; Wave Equation Model

A13 Solving PDEs in Python: The FEniCS Tutorial I - H.P. Langtangen, A. Logg (Springer, Simula SpringerBriefs). *Open access (CC BY 4.0), free PDF from Springer.*

Link: https://link.springer.com/book/10.1007/978-3-319-52462-7 | Used for: GC05, GE18, GE32, GE44

Table of contents: **1** Preliminaries **2** Fundamentals: Solving the Poisson Equation **3** A Gallery of Finite Element Solvers (heat, nonlinear Poisson, elasticity, Navier-Stokes, reaction systems) **4** Subdomains and Boundary Conditions **5** Extensions: Improving the Poisson Solver

A14 An Introduction to Reservoir Simulation Using MATLAB/GNU Octave (MRST user guide) - K.-A. Lie (SINTEF; Cambridge University Press, 2019). *Open access (CC BY-NC-ND 4.0) on Cambridge Core.*

Link: https://www.cambridge.org/core/books/an-introduction-to-reservoir-simulation-using-matlabgnu-octave/introduction/C66C3A56440AA6B95D7F814CAD3CD971 | Used for: GC05, GE50, GE51, GE54

Table of contents: **1** Introduction **2** Part I: Geological Models and Grids **3** Part II: Single-Phase Flow **4** Part III: Multiphase Flow **5** Part IV: Reservoir Engineering Workflows **6** Appendix: The MATLAB Reservoir Simulation Toolbox

A15 Documentation for the MODFLOW 6 Groundwater Flow Model (USGS Techniques and Methods 6-A55) - C.D. Langevin et al., USGS 2017. *Free PDF (U.S. Government).*

Link: https://pubs.usgs.gov/publication/tm6A55 | Used for: GC05

Table of contents: not checked for this report - see the contents page at the link.

A16 Mineral Deposit Models (USGS Bulletin 1693) - D.P. Cox, D.A. Singer (eds), U.S. Geological Survey, 1986. *Public domain; free PDF.*

Link: https://pubs.usgs.gov/bul/b1693/ | Used for: GE01, GE06, GE11, GE12

Table of contents: **1** Part 1: 85 descriptive deposit models (geological environment, deposit description, exploration guides) **2** Part 2: 60 grade-tonnage models from over 3,900 deposits

A17 Analytical Chemistry 2.1 - David Harvey (DePauw University; LibreTexts). *Free; full-book PDF download.*

Link: https://chem.libretexts.org/Bookshelves/Analytical_Chemistry/Analytical_Chemistry_2.1_by_David_Harvey | Used for: GE02, GE07

Table of contents: **1** Introduction to Analytical Chemistry **2** Basic Tools of Analytical Chemistry **3** The Vocabulary of Analytical Chemistry **4** Evaluating Analytical Data **5** Standardizing Analytical Methods **6** Equilibrium Chemistry **7** Obtaining and Preparing Samples for Analysis **8** Gravimetric Methods **9** Titrimetric Methods **10** Spectroscopic Methods **11** Electrochemical Methods **12** Chromatographic and Electrophoretic Methods **13** Kinetic Methods **14** Developing a Standard Method **15** Quality Assurance

A18 Geophysics for Practicing Geoscientists (GPG) - UBC Geophysical Inversion Facility. *Open web resource (CC BY).*

Link: https://www.eoas.ubc.ca/courses/eosc350/content/gpg-outline.htm | Used for: GE03, GE21, GE38, GE40, GE43, GE47, GE48

Table of contents: **1** Welcome and getting started **2** Foundations: framework and physical properties **3** Geophysical survey methods: magnetics, gravity, DC resistivity and IP, seismic refraction, GPR, terrain conductivity (EM) **4** Inversion concepts **5** Exercises **6** Case histories **7** Software and manuals **8** Resources (glossary, references)

A19 SimPEG documentation and tutorials (Simulation and Parameter Estimation in Geophysics) - SimPEG developers (open source). *Free web resource (MIT licence).*

Link: https://simpeg.xyz | Used for: GE03, GE38, GE40, GE43

Table of contents: not checked for this report - see the contents page at the link.

A20 Geological Structures: a Practical Introduction - John W.F. Waldron, Morgan Snyder (Univ. of Alberta). *CC BY-NC; free web and PDF.*

Link: https://pressbooks.openeducationalberta.ca/introductorystructuralgeology/ | Used for: GE04

Table of contents: **1** Part 1: Topics (maps, orientation, stereographic projection, folds, faults, stress and strain, cross-sections) **2** Part 2: Labs (practical exercises for each topic)

A21 Essentials of Geographic Information Systems - Jonathan Campbell, Michael Shin. *CC BY-NC-SA; free (Open Textbook Library).*

Link: https://open.umn.edu/opentextbooks/textbooks/67 | Used for: GE05, GE64

Table of contents: **1** Introduction **2** Map Anatomy **3** Data, Information, and Where to Find Them **4** Data Models for GIS **5** Geospatial Data Management **6** Data Characteristics and Visualization **7** Geospatial Analysis I: Vector Operations **8** Geospatial Analysis II: Raster Data **9** Cartographic Principles **10** GIS Project Management

A22 Geographic Data Science with Python - S. Rey, D. Arribas-Bel, L.J. Wolf (CRC Press). *Free online edition (CC BY-NC-ND).*

Link: https://geographicdata.science/book/ | Used for: GE05

Table of contents: not checked for this report - see the contents page at the link.

A23 Mineralogy - Dexter Perkins et al. (Univ. of North Dakota; LibreTexts). *CC BY-NC-SA; free web and PDF export.*

Link: https://geo.libretexts.org/Bookshelves/Geology/Mineralogy_(Perkins_et_al.) | Used for: GE07

Table of contents: **1** Introduction **2** Mineral Chemistry **3** Mineral Properties **4** Crystals and Crystallization **5** Optical Mineralogy **6** Igneous Rocks and Silicate Minerals **7** Sedimentary Minerals and Sedimentary Rocks **8** Metamorphic Minerals and Metamorphic Rocks **9** Ore Deposits and Economic Minerals **10** Crystal Morphology and Symmetry **11** Crystallography **12** X-ray Diffraction and Mineral Analysis **13** Crystal Structures **14** Mineral Descriptions

A24 GemPy documentation and tutorials (implicit 3D geological modelling) - GemPy developers (open source). *Free web resource.*

Link: https://docs.gempy.org | Used for: GE08

Table of contents: not checked for this report - see the contents page at the link.

A25 Fundamentals of Remote Sensing (tutorial) - Canada Centre for Remote Sensing / Natural Resources Canada. *Free web tutorial.*

Link: https://natural-resources.canada.ca/science-data/science-research/geomatics/remote-sensing/tutorial-fundamentals-remote-sensing | Used for: GE09, GE23, GE62

Table of contents: not checked for this report - see the contents page at the link.

A26 The JORC Code (2012 edition) - Joint Ore Reserves Committee. *Free PDF (Chinese translation also free).*

Link: https://jorc.org/docs/jorc_code2012.pdf | Used for: GE10, GE11

Table of contents: not checked for this report - see the contents page at the link.

A27 Critical Mineral Resources of the United States (USGS Professional Paper 1802) - K.J. Schulz, J.H. DeYoung, R.R. Seal, D.C. Bradley (eds), USGS 2017. *Free PDF (U.S. Government); one chapter per commodity.*

Link: https://pubs.usgs.gov/publication/pp1802 | Used for: GE14

Table of contents: not checked for this report - see the contents page at the link.

A28 Mathematics for Machine Learning - M.P. Deisenroth, A.A. Faisal, C.S. Ong (Cambridge University Press). *Free PDF from the authors' site (published by CUP).*

Link: https://mml-book.github.io/book/mml-book.pdf | Used for: GE15

Table of contents: **1** Introduction and Motivation **2** Linear Algebra **3** Analytic Geometry **4** Matrix Decompositions **5** Vector Calculus **6** Probability and Distributions **7** Continuous Optimization **8** When Models Meet Data **9** Linear Regression **10** Dimensionality Reduction with Principal Component Analysis **11** Density Estimation with Gaussian Mixture Models **12** Classification with Support Vector Machines

A29 Dive into Deep Learning - Zhang, Lipton, Li, Smola. *Free online book and PDF (arXiv).*

Link: https://arxiv.org/abs/2106.11342 | Used for: GE15

Table of contents: not checked for this report - see the contents page at the link.

A30 Deep Learning - I. Goodfellow, Y. Bengio, A. Courville (MIT Press). *Free online edition from the authors.*

Link: https://www.deeplearningbook.org | Used for: GE15

Table of contents: not checked for this report - see the contents page at the link.

A31 An Introduction to Soil Mechanics - A. Verruijt (Springer, 2018). *Open access PDF from Springer.*

Link: https://link.springer.com/book/10.1007/978-3-319-61185-3 | Used for: GE16, GE20, GE28

Table of contents: not checked for this report - see the contents page at the link.

A32 Matrix Structural Analysis (2nd ed.) - W. McGuire, R.H. Gallagher, R.D. Ziemian. *Free PDF from Bucknell Digital Commons.*

Link: https://digitalcommons.bucknell.edu/books/7/ | Used for: GE17

Table of contents: not checked for this report - see the contents page at the link.

A33 Earthquake-Resistant Design Concepts (FEMA P-749) - FEMA / NIBS Building Seismic Safety Council. *Free PDF (U.S. Government).*

Link: https://www.fema.gov/sites/default/files/2020-10/earthquake-resistant-design-concepts_nehrp-seismic-provisions.pdf | Used for: GE19, GE29

Table of contents: not checked for this report - see the contents page at the link.

A34 Practical Rock Engineering - Evert Hoek (Rocscience). *Free PDF from publisher (author's course notes).*

Link: https://www.rocscience.com/assets/resources/learning/hoek/Practical-Rock-Engineering-Full-Text.pdf | Used for: GE21, GE24, GE25, GE30, GE32, GE34, GE35, GE53

Table of contents: **1** What is rock engineering? **2** Intact rock strength **3** Shear strength of discontinuities **4** Rock mass properties **5** In situ and induced stresses **6** When is a rock engineering design acceptable? **7** Structurally controlled instability in tunnels **8** Factor of safety and probability of failure **9** A slope stability problem in Hong Kong **10** Rock slopes in civil and mining engineering **11** Rock mass classification **12** Tunnels in weak rock **13** Design of large underground caverns **14** The Rio Grande project - Argentina **15** Rockbolts and cables **16** Model to demonstrate how rockbolts work **17** Shotcrete support **18** Tensile failure of spalling in tunnels **19** Analysis of rockfall hazards **20** Blasting damage in rock

A35 The SAR Handbook: Comprehensive Methodologies for Forest Monitoring and Biomass Estimation - A. Flores-Anderson et al. (NASA SERVIR, 2019). *Free PDF (NASA Technical Reports Server).*

Link: https://ntrs.nasa.gov/citations/20190002563 | Used for: GE23

Table of contents: not checked for this report - see the contents page at the link.

A36 Technical Manual for Design and Construction of Road Tunnels - Civil Elements (FHWA-NHI-10-034) - U.S. Federal Highway Administration. *Free PDF (U.S. Government).*

Link: https://www.fhwa.dot.gov/bridge/tunnel/pubs/nhi09010/tunnel_manual.pdf | Used for: GE24

Table of contents: not checked for this report - see the contents page at the link.

A37 The Landslide Handbook - A Guide to Understanding Landslides (USGS Circular 1325) - L.M. Highland, P. Bobrowsky (2008). *Public domain; free PDF.*

Link: https://pubs.usgs.gov/publication/cir1325 | Used for: GE25, GE64

Table of contents: not checked for this report - see the contents page at the link.

A38 Project Management (2nd ed.) - A. Watt (BCcampus Open Education). *CC BY 4.0; free web and PDF.*

Link: https://opentextbc.ca/projectmanagement/ | Used for: GE27

Table of contents: not checked for this report - see the contents page at the link.

A39 New Manual of Seismological Observatory Practice (NMSOP-2) - P. Bormann (ed.), GFZ Potsdam / IASPEI. *Open access (GFZ).*

Link: https://nmsop.gfz-potsdam.de | Used for: GE29, GE39

Table of contents: not checked for this report - see the contents page at the link.

A40 Global Industry Standard on Tailings Management - ICMM, UNEP and PRI (Global Tailings Review, 2020). *Free PDF.*

Link: https://globaltailingsreview.org/global-industry-standard/ | Used for: GE37

Table of contents: **1** Topic I: Affected Communities **2** Topic II: Integrated Knowledge Base **3** Topic III: Design, Construction, Operation and Monitoring of the Tailings Facility **4** Topic IV: Management and Governance **5** Topic V: Emergency Response and Long-Term Recovery **6** Topic VI: Public Disclosure and Access to Information

A41 Inverse Problem Theory and Methods for Model Parameter Estimation - A. Tarantola (SIAM, 2005). *Free PDF made available by the author (listed by e-booksdirectory).*

Link: https://www.ipgp.fr/~tarantola/Files/Professional/Books/InverseProblemTheory.pdf | Used for: GE38

Table of contents: **1** The General Discrete Inverse Problem **2** Monte Carlo Methods **3** The Least-Squares Criterion **4** Least-Absolute-Values Criterion and Minimax Criterion **5** Functional Inverse Problems **6** Appendices **7** Problems

A42 PetroWiki - Society of Petroleum Engineers (based on the SPE Petroleum Engineering Handbook). *Free web resource.*

Link: https://petrowiki.spe.org | Used for: GE41, GE45, GE49, GE50, GE52, GE53, GE54, GE55, GE56, GE57, GE58

Table of contents: not checked for this report - see the contents page at the link.

A43 Seismic Data Analysis: Processing, Inversion, and Interpretation of Seismic Data - Oz Yilmaz (SEG). *Open access on the SEG Wiki (web, not PDF).*

Link: https://wiki.seg.org/wiki/Seismic_Data_Analysis | Used for: GE41, GE42, GE45, GE46

Table of contents: **1** Introduction **2** Fundamentals of signal processing **3** Deconvolution **4** Velocity analysis and statics corrections **5** Migration **6** Dip-moveout correction and prestack migration **7** Noise and multiple attenuation **8** 3-D seismic exploration **9** Earth imaging in depth **10** Earth modeling in depth **11** Structural inversion

A44 Groundwater - R. Allen Freeze, John A. Cherry (1979). *Free PDF from The Groundwater Project (preserved book).*

Link: https://gw-project.org/books/groundwater/ | Used for: GE47

Table of contents: not checked for this report - see the contents page at the link.

A45 Handbook of Best Practices for Geothermal Drilling - J. Finger, D. Blankenship (Sandia National Laboratories). *Free PDF (U.S. DOE).*

Link: https://www1.eere.energy.gov/geothermal/pdfs/drillinghandbook.pdf | Used for: GE52

Table of contents: not checked for this report - see the contents page at the link.

A46 GNSS Data Processing, Vol. I: Fundamentals and Algorithms (ESA TM-23/1) - J. Sanz Subirana, J.M. Juan Zornoza, M. Hernandez-Pajares (ESA). *Free PDF from ESA Navipedia.*

Link: https://gssc.esa.int/navipedia/GNSS_Book/ESA_GNSS-Book_TM-23_Vol_I.pdf | Used for: GE59

Table of contents: not checked for this report - see the contents page at the link.

A47 Basics of Space Flight - NASA Jet Propulsion Laboratory. *Free (U.S. Government work); web tutorial.*

Link: https://science.nasa.gov/learn/basics-of-space-flight/participants-guide | Used for: GE60

Table of contents: not checked for this report - see the contents page at the link.

A48 Database Design (2nd ed.) - Adrienne Watt, Nelson Eng (BCcampus). *CC BY; free PDF.*

Link: https://yc-prod-openlibrary-repo.ecampusontario.ca/jspui/bitstream/123456789/247/16/Database-Design-2nd-Edition-1540829576.pdf | Used for: GE61

Table of contents: **1** Before the Advent of Database Systems **2** Fundamental Concepts **3** Characteristics and Benefits of a Database **4** Types of Data Models **5** Data Modelling **6** Classification of Database Management Systems **7** The Relational Data Model **8** The Entity Relationship Data Model **9** Integrity Rules and Constraints **10** ER Modelling **11** Functional Dependencies **12** Normalization **13** Database Development Process **14** Database Users **15** SQL Structured Query Language **16** SQL Data Manipulation Language

A49 NASA Systems Engineering Handbook (SP-2016-6105 Rev 2) - NASA. *Free PDF (U.S. Government).*

Link: https://www.nasa.gov/wp-content/uploads/2018/09/nasa_systems_engineering_handbook_0.pdf | Used for: GE63

Table of contents: **1** Introduction **2** Fundamentals of Systems Engineering **3** NASA Program/Project Life Cycle **4** System Design Processes **5** Product Realization **6** Crosscutting Technical Management

A50 State-of-the-Art of Small Spacecraft Technology - NASA Small Spacecraft Systems Virtual Institute. *Free (U.S. Government), web and PDF.*

Link: https://www.nasa.gov/smallsat-institute/sst-soa/ | Used for: GE63

Table of contents: not checked for this report - see the contents page at the link.

A51 CubeSat 101: Basic Concepts and Processes for First-Time CubeSat Developers - NASA CubeSat Launch Initiative (2017). *Free PDF (U.S. Government work).*

Link: https://www.nasa.gov/wp-content/uploads/2017/03/nasa_csli_cubesat_101_508.pdf | Used for: GE63

Table of contents: not checked for this report - see the contents page at the link.

A52 Lunar Sourcebook: A User's Guide to the Moon - G. Heiken, D. Vaniman, B. French (eds) (Cambridge, 1991; LPI). *Free PDF from the Lunar and Planetary Institute.*

Link: https://www.lpi.usra.edu/publications/books/lunar_sourcebook/ | Used for: GE65

Table of contents: not checked for this report - see the contents page at the link.

Appendix B. Sources

- Colorado School of Mines - Master of Science and Master of Engineering programmes (general requirements): https://catalog.mines.edu/graduate/general-requirements/masters-programs/
- Colorado School of Mines - course catalogue: Geophysical Engineering (GPGN): https://catalog.mines.edu/coursesaz/geophyseng/
- Colorado School of Mines - course catalogue: Geology (GEOL): https://catalog.mines.edu/coursesaz/geology
- Colorado School of Mines - course catalogue: Geological Engineering (GEGN): https://catalog.mines.edu/coursesaz/geoleng/
- Colorado School of Mines - course catalogue: Civil Engineering (CEEN): https://catalog.mines.edu/coursesaz/civileng/civileng.pdf

- Colorado School of Mines - course catalogue: Mining Engineering (MNGN): https://catalog.mines.edu/coursesaz/miningeng/miningeng.pdf
- Colorado School of Mines - course catalogue: Petroleum Engineering (PEGN): https://catalog.mines.edu/coursesaz/petroleumeng/petroleumeng.pdf
- Colorado School of Mines - course catalogue: Space Resources (SPRS): https://catalog.mines.edu/coursesaz/spaceresources/spaceresources.pdf
- MIT EAPS - SM thesis committee proposal form: https://eaps.mit.edu/wp-content/uploads/2024/05/SMThesisCommitteeProposalForm2024.pdf
- MIT Office of Graduate Education - Master of Science general requirements: https://oge.mit.edu/gpp/advanced-degrees/masters-degree/master-of-science
- MIT Office of Graduate Education - English Language Proficiency test policy: https://oge.mit.edu/graduate-admissions/admissions-policies/elp-test-requirement
- MIT CEE - SM Program Guide (June 2023): https://cee.mit.edu/wp-content/uploads/2023/08/SM_Program_Guide_June2023.pdf
- MIT OpenCourseWare graduate subject pages (links in each subject entry): https://ocw.mit.edu
- Tsinghua University - graduate training regulations: https://www.tsinghua.edu.cn/yjsy/fujian/pygz.pdf
- Tsinghua University Language Center - graduate English course exemption notice: https://www.lc.tsinghua.edu.cn/__local/A/29/E6/AAE40F20742057B0BAECE944106_C1ACEDFA_62BAA.pdf
- China University of Geosciences (Wuhan) - graduate training document: https://graduate.cug.edu.cn/__local/C/BB/44/40D6AD23FA7CDCBB4769049FEAF_14DF9605_12E76F.pdf?e=.pdf
- China University of Petroleum (East China) - master's plan, Geological Resources and Geological Engineering: https://cie.upc.edu.cn/pdf/Master/dizhiziyuanyudizhigongcheng.pdf
- China University of Geosciences (Beijing) - programme page used in the earlier reports: https://sesr.cugb.edu.cn/c/2020/06/17/619374.shtml
- European Commission - ECTS Users' Guide 2015: https://education.ec.europa.eu/sites/default/files/document-library-docs/ects-users-guide_en.pdf
- TU Bergakademie Freiberg - English-taught modules, MSc Applied Geoscience: https://tu-freiberg.de/sites/default/files/2025-12/english-taught-modules_applied_geoscience.pdf

Course titles, credits and rules were taken from the documents above as published in 2022-2026; they change from year to year, so check the current versions before formal adoption. The proposed thresholds (grades, language scores, activity points) are recommendations for the faculty to adapt.